Advantage IV and *EB-Flow2* Product Lines

O&M MANUAL

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foreword

Thank you for choosing EBTRON airflow measurement equipment. EBTRON has been the leader in airflow measurement for monitoring and control since 1983. Our advanced thermal dispersion measurement technology has been copied by others but duplicated by none.

We are supported by an extensive, factory trained, ***local*** *representative network that is available to support all of your needs. We also provide toll-free factory support at 800-2EBTRON (232-8766), Monday through Thursday 8:00 AM to 4:30 PM and Friday 8:00 AM to 2:00 PM eastern time.*

Don't know who your local EBTRON representative is? Visit EBTRON.com and click on the REP FINDER button.

This manual has been developed as a comprehensive guide to our Advantage IV and *EB-Flow2* product lines. It is structured as a complete, single source, reference manual and contains the following information:

- FCC Part 15 Compliance Statement
- Section 1 – Installation
- Section 2 – Startup
- Section 3 – Custom Configuration
- Section 4 – Built-in Tools
- Section 5 – Diagnostics
- Section 6 – Troubleshooting
- Appendix A – Placement Guidelines
- Appendix B – Sensor Probe Installation Guides
- Appendix C – Transmitter Installation Guides
- Appendix D – Wiring Guides
- Appendix E – Startup Guides
- Appendix F – Network Registers and Object Lists
- Appendix G – Mechanical Drawings

The following symbols are used in this manual:

⚠ *IMPORTANT. The comment is very important and should not be ignored.*

ⓘ *INFORMATION. The comment provides additional information.*

SUGGESTION. The comment is a suggestion to the user.

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FCC-Part 15 Compliance

ALL GTx116, GTx116e, GTx108, GTx108e, HTx116 and HTx108 Models contain transmitter module:
FCC ID: T9JRN4020 / IC: 6514A-RN4020 or
FCC ID: Y82-DA14531MOD / IC: 9576A-DA14531MOD

Altering the transmitter module(s) or using an antenna other than an Ebtron approved antenna with the transmitter module(s) could void the users' authority to operate the equipment.

The above transmitter module enabled Ebtron models comply with Part 15 of the FCC rules and Industry Canada license-exempt RSS standards.

Les modèles Ebtron activés pour le module émetteur ci-dessus sont conformes aux normes RSS exemptes de licence d'Industrie Canada.

All above models and the EF-x1000, EF-x2000, HTx104 Ebtron models comply with Part 15 of the FCC rules and Industry Canada ICES-003. Operation of these devices is subject to the following two conditions:

1. This device may not cause harmful interference, and
2. This device must accept any interference received, including interference that may cause undesired operation.

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules and Industry Canada ICES-003. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interferences to radio communications. Operations of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

(All above Ebtron models are Class A digital apparatus and complies with Canadian ICES-003.

Operation of these devices is subject to the following two conditions:

1 This device may not cause harmful interference, and
2. This device must accept any interference received, including interference that may cause undesired operation.

Tous les modèles Ebtron ci-dessus sont des appareils numériques de classe A et sont conformes à la norme ICES-003 canadienne. Le fonctionnement de ces appareils est soumis aux deux conditions suivantes: 1 Cet appareil ne doit pas provoquer d'interférences nuisibles, et 2. Cet appareil doit accepter toute interférence reçue, y compris les interférences susceptibles de provoquer un fonctionnement indésirable.

Under Industry Canada regulations, the noted radio transmitters may only operate using an antenna of a type and maximum (or lesser) gain approved for the transmitter by Industry Canada. To reduce potential radio interference to other users, the antenna type and its gain should be so chosen that the equivalent isotropically radiated power (e.i.r.p) is not more than that necessary for successful communication.

Conformément à la réglementation d'Industrie Canada, le présent émetteur radio peut fonctionner avec
une antenne d'un type et d'un gain maximal (ou inférieur) approuvé pour l'émetteur par Industrie Canada. Dans le but de réduire les risques de brouillage radioélectrique à l'intention des autres utilisateurs, il faut choisir le type d'antenne et son gain de sorte que la puissance isotrope rayonnée équivalente (p.i.r.e.) ne dépasse pas l'intensité nécessaire à l'établissement d'une communication satisfaisante.

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Section 1 – Installation

1. SECTION OVERVIEW

This section provides installation and wiring instructions for Advantage IV and *EB-Flow2* remote transmitters.

1.1. SENSOR PROBES SUPPORTED

Transmitters in this manual support the following sensor probe types:

- -P duct/plenum probe
- -U universal mount probe
- -T terminal unit small duct probe
- -F fan inlet probe
- -B bleed sensor probe

1.2. SENSOR PROBE MOUNTING

1.2.1. Locating Probes

Sensor probe placement guidelines are located in the *Appendix A – Placement Guidelines* section at the end of this document.

ⓘ *If minimum placement guidelines cannot be achieved, installed accuracy may be compromised. Transmitters can be field adjusted to match a third-party measurement. Adjusted field measurements typically result in comparative readings within ±3% of the third-party measurement. Be advised that the third-party measurement may have uncertainties greater than or equal to ±10% and should only be used to adjust the airflow measurement device if the probes do not meet minimum placement requirements and the discrepancy is greater than the uncertainty of the third-party source.*

1.2.2. Installation

Installation guides for each probe type are located in the *Appendix B – Sensor Probe Installation Guides* section at the end of this document.

⚠ *Sensor probes must be mounted and spaced properly to achieve the specified installed accuracy.*

1.3. TRANSMITTER MOUNTING

Installation guides for each transmitter model are located in the *Appendix C – Transmitter Installation Guides* section at the end of this document.

Mount transmitters in a location protected from moisture, rain and snow with an ambient temperature between -20 and 120 °F [-28.9 to 48.9 °C]. Provide a weatherproof enclosure and mount away from direct sunlight when outdoor mounting is required.

Transmitters have an LCD and four-button user interface. Select a mounting location where the LCD and pushbutton interface are accessible during normal operation.

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The cover of the Advantage IV Gold Series and Hybrid Series models HTx116 and HTx108 transmitters hinges up for removal. The cover of the Advantage IV Hybrid HTx104 Series transmitter slides up and out of the enclosure. The cover of the *EB-Flow2* transmitter is secured by four #1 Philips head screws.

⚠ *Make sure the transmitter is located where all of the cables from the sensor probes reach the transmitter.*

⚠ *Provide a minimum of 6.5 inches [165 mm] above Hybrid Series HTx104 transmitters to allow for cover removal when slide out covers are provided and a minimum of 2 inches on the sides.*

⚠ *Provide a minimum of 2 inches on sides and top of Gold Series and Hybrid Series model HTx116 and HTx108 transmitters.*

⚠ *Provide a minimum of 3 inches [76 mm] below all transmitters to connect cable plugs from the sensor probes.*

All transmitters contain electrostatic discharge (ESD) sensitive components. Observe ESD precautions during installation, wiring and startup process to avoid possible extensive damage to these components. Failure to comply can result in equipment damage or loss.

1.4. TRANSMITTER WIRING

Wiring guides for each transmitter model are located in the *Appendix D – Wiring Guides* section at the end of this document.

1.4.1. Probe Connections to Transmitter

Each probe has a tag on the cable plug and probe bracket/tube with a TAG ID set to the location name (NAME) provided at the time of order. If a location name is not provided the TAG ID is set to null (no value).

ⓘ *Multiple connector arrangements do not require probes be installed in each receptacle.*

⚠ *Sensor probe plugs are keyed and NOT twist-lock. Align the key and push the plug onto the transmitter receptacle. Twisting the connector onto the receptacle will cause damage and void warranty.*

Probes are “plug and play” and do not need to be connected to a specific receptacle for proper operation. However, use of the following connection convention simplifies sensor node location identification when using the EB-Link Reader or when viewing individual sensor node data using DIAGNOSTICS. Although not required, it is a good practice to install each probe in the corresponding connector. Probes should also be installed sequentially by the labeled probe number, top to bottom, left to right or vice versa in the duct, plenum or fan array.

Depending on the model, up to 8 sensor nodes can be provided in a single probe. A probe number is labeled on each probe hang tag (Probe x of y). The leftmost receptacle used on the transmitter when viewed from the display side is always designated as *Connector 1 (C1)*. Additional connectors increment sequentially from left to right on transmitters with more than one receptacle. Fan array models with two rows of 4 receptacles, increment left to right, *C1* to *C4* on the topmost row closest to the cover, then left to right, *C5* to *C8* on the lower most row.

⚠ *Fan array models (-F/An) must follow this convention if two probes are located in each fan inlet and the individual fan airflow rates are required and/or the fan alarm is enabled (i.e. put probes for fan 1 in C1 and C2, fan 2 in C3 and C4, etc.)*

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Probe numbers (P1 to Py) are dynamically assigned by the transmitter after initial power up, left to right, based on the connector receptacle used. The sensor node furthest from the cable side of the probe on probes with more than one sensor node, is designated as "*Sensor Node 1"*.

1.4.1.1. SPECIAL CONSIDERATIONS FOR DUAL LOCATION TRANSMITTERS

EB-Flow2 transmitters can be configured for two locations in the field. Use the leftmost connector when viewed from display side as "Location 1" and the rightmost connector as "Location 2".

Probes have the suffix AMD-1 and AMD-2 appended to the TAG ID for "Location 1" and "Location 2" respectively for identification.

1.4.2. B.A.S. and Power Connections

B.A.S. and power connections for each transmitter model are located in the *Appendix D – Wiring Guides* section at the end of this document.

1.4.2.1. SIGNAL ISOLATION

1.4.2.1.1. Advantage IV Models

All Advantage IV transmitter models are provided with isolated analog output signals and network connections.

1.4.2.1.2. *EB-Flow2* Models

All *EB-Flow2* transmitter models are provided with non-isolated analog output signals and network connections.

Isolation is typically required and achieved by providing power with a transformer where the secondary is not connected to earth ground.

EBTRON can provide and optional signal isolator for RS-485 network applications.

1.5. POWER UP

Refer to *Section 2 – Startup* and the individual startup guides located in *Appendix E – Startup Guides* at the end of this document prior to moving the power switch to the "ON" position.

1.6. FOR MORE INFORMATION …

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Section 2 – Startup

2. SECTION OVERVIEW

This section discusses startup and basic operation. Advantage IV and *EB-Flow2* remote transmitters are "plug-and-play." Transmitters are designed to be fully functional at power-up. Field configuration is generally only required when factory default output signals and/or network parameters must be modified for the host building automation system or application controller.

2.1. PRE-POWER CHECKS

2.1.1. Sensor Probe Installation

Inspect and verify the following:

- ☐ The location where the probes are located meet or exceed EBTRON published placement guidelines for the sensor probe model and sensor density provided.

ⓘ *If minimum placement guidelines cannot be achieved, installed accuracy may be compromised. Transmitters can be field adjusted to match a third-party measurement. Adjusted field measurements typically result in comparative readings within ±3% of the third-party measurement. Be advised that the third-party measurement may have uncertainties greater than or equal to ±10% and should only be used to adjust the airflow measurement device if the probes do not meet minimum placement requirements and the discrepancy is greater than the uncertainty of the third-party source.*

Inspect and verify the following:

- ☐ The probes are properly spaced in the fan, duct or opening.
- ☐ The airflow directional arrow is pointing in the direction of airflow and the sensor probes are not twisted in the airstream.
- ☐ All sensor probes are properly connected to the transmitter.

2.1.2. B.A.S. and Other Connections

Advantage IV and EB-Flow2 product lines are available with a wide variety of interface options. The connectivity capability provided is indicated by the model code for the x placeholder in the model code. The base model codes are as follows:

- Advantage IV Product Line
  - Gold Series
    - GTx116
    - GTx116e
    - GTx108
    - GTx108e
  - Hybrid Series
    - HTx116
    - HTx108
    - HTx104
- *EB-Flow2* Product Line
  - EF Series

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- EF-x2000

The connectivity code x = {connectivity capability} is as follows:

- Analog Output Signals: x = A, B, C, M, S, F, and U
- RS-485 Network Connection: x = B, C and N
- Ethernet Network Connection: x = B, M and S
- Lon Network Connection: x = F, L
- USB Datalogger: x = D, U

ⓘ *Some models support more than one connectivity type.*

2.1.2.1 TRANSMITTERS WITH ANALOG OUTPUT SIGNALS

Inspect and verify the following for each analog output used:

☐ Twisted pair signal wire has been run between the host control panel and the transmitter.

☐ Signal wiring is shielded and the shield has been properly terminated and grounded at one end only (typically at the host control panel).

⚠ *If twisted pair wire and/or shielded cable is not used, extraneous electrical noise can be picked up between the transmitter and host control panel. If a problem is suspected, measure and record the signal at the host control panel terminals with a test meter set to VDC (across the +/- terminals) for 0-5/ 0-10/1-5/2-10 output signals or DC current (in series with the + or - terminal) for 4-20mA output signals. Disconnect the signal wires from the transmitter and measure the signal across the appropriate transmitter output terminals. There should be no significant change in the reading.*

2.1.2.2. TRANSMITTERS WITH AN RS-485 NETWORK CONNECTION

Inspect and verify the following if the network connection is used:

2.1.2.2.1. Isolated RS-485 Networks

☐ A 3-conductor network cable meeting the BACnet or Modbus standard has been used and all three connections, NET +, NET - and NET COM are connected.

ⓘ *EBTRON Advantage IV transmitters have an isolated RS-485 connection. Simply connect to the network. EB-Flow2 transmitters have a non-isolated RS-485 connection. Isolation is achieved by powering EB-Flow2 transmitters with a 24V power source that is not connected to earth ground (i.e. the secondary on the 24 VAC transformer is not grounded).*

⚠ *If a 2-conductor network cable or other non-conforming cable is used, network speed, length and reliability may be compromised or network failure may occur. This is NOT a limitation of EBTRON network devices.*

2.1.2.2.2. Non-isolated RS-485 Networks

☐ A 2-conductor network cable meeting the BACnet or Modbus standard has been used and both connections, NET +, NET - are connected.

⚠ *If a non-conforming cable is used, network speed, length and reliability may be compromised or network failure may occur. This is NOT a limitation of EBTRON network devices.*

☐ NET COM on Advantage IV transmitters is connected to earth ground.

⚠ *Do not connect NET COM on EB-Flow2 transmitters to earth ground.*

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2.1.2.2.3. All RS-485 Networks

- ☐ All network devices are on a single “daisy-chain”.
- ☐ The first device on the chain is properly terminated for “fail safe bias”, preferred, or “end of line”.

ⓘ *The first device on the chain is often the B.A.S. control panel. If the first device on the chain is not the B.A.S. controller and fail-safe bias is required, EBTRON Advantage IV transmitters can be configured for the fail-safe bias termination. See appropriate wiring diagram for the transmitter provided in Section 1.*

- ☐ The last device on the chain is properly terminated for “end of line”.

ⓘ *If an EBTRON transmitter is the last device on the chain, set the end of line termination. See appropriate wiring diagram for the transmitter provided in Section 1.*

⚠ *Failure to properly terminate the network will result in network instability or network failure.*

2.1.2.3. TRANSMITTERS WITH AN ETHERNET NETWORK CONNECTION

Inspect and verify the following if the network connection is used:

- ☐ CAT5 or higher cable has been used for the Ethernet connection.
- ☐ The cables have been properly terminated with RJ-45 connectors and the connections have been tested prior to connection to the transmitter.

⚠ *Use of an improperly wired RJ-45 connector may cause damage to the Ethernet output circuit of the transmitter.*

2.1.2.4. TRANSMITTERS WITH A LON NETWORK CONNECTION

Lon transmitters use the “free topology” network.

Inspect and verify the following if the network connection is used:

- ☐ The network wire is as specified by Echelon (typically Belden 8471 cable or equivalent).
- ☐ The network wire pair has been properly terminated at the transmitter (polarity insensitive).

2.1.2.5. TRANSMITTERS WITH AN EBTRON RF-LINK CONNECTION

Transmitters with an EBTRON RF-Link card can communicate with EBTRON *Commissioner* or other RF-Link compatible EBTRON master transceivers.

Inspect and verify the following if the RF-Link card is provided:

- ☐ The antenna has been field installed and pointing upward. Install by screwing it into the SMA connector located on the top left of the transmitter enclosure.

ⓘ *Although the wireless radio does not require a clear line of sight between the transmitter and receiving device, a large mass of metal or another RF interference source could degrade signal strength and adversely affect the RF transmission. Note any obstructions between the transmitter and receiving device and avoid placing the transmitter and receiving device between obstructions, whenever possible.*

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2.1.2.6. TRANSMITTERS WITH A USB DATA LOGGER CARD

Transmitters with a USB datalogger card write sensor data to a thumb-drive memory device. Inspect and verify the following:

- ☐ A USB thumb drive memory device is properly seated in the USB connector on the data logger card.

⚠ *To avoid data loss and/or damage do not insert or remove a thumb drive memory device into the USB connector on transmitter with the power switch in the "ON" position.*

2.1.3. Power Connections

Inspect and verify the following:

- ☐ 24 VAC power is connected to both power terminals of the transmitter.
- ☐ Multiple transmitters wired on a single transformer are wired "in-phase" between transmitters (L1 to L1 and L2 to L2).
- ☐ The power transformer has been sized for the total load of all of the devices connected.

2.1.3.1. *EB-FLOW2* TRANSMITTERS

Inspect and verify the following:

- ☐ The secondary of the 24 VAC transformer is not connected to ground.

⚠ *Grounding the power on the secondary side of the transmitter results in non-isolated output signal(s) which may result in ground loops, signal error, and/or damage to the transmitter. Do not use a grounded 24 VAC power source.*

ⓘ *Grounding the primary side of the transformer is not a problem as long as one leg of the secondary or a center tap of the secondary is not connected to ground.*

If devices requiring 24 VAC require grounding, use a separate transformer to power the transmitter(s).

2.2. POWER-UP

Move the power switch to the "ON" position. The transmitter will display the firmware version and conduct power-up diagnostics prior to normal operation. Power-up faults, if detected, are displayed on the display.

ⓘ *EF-x2000 transmitters do not have a power switch. The transmitter is energized when 24 V power is provided to the device.*

⚠ *If "live" power is connected to the EF-x2000 power terminals, take extreme caution not to contact other terminals or the any component on the circuit board. Connecting power to terminals other than the power terminals may damage the transmitter and void warranty. It is advised that the 24 V power source is not "live" when the power connections are made.*

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Inspect and verify the following:

- ☐ The supply voltage to the transmitters under full load (all power switches to “ON” with probes connected and any other devices on the same transformer at full power) is between 22.8 and 26.4 VAC.

After initial power-up, the transmitter is fully functional as an airflow and temperature measurement device. With the exception of network configuration and alarm settings, transmitters are “plug and play” and do not require additional setup unless modifications to the factory default settings are desired by the user.

2.2.1. Display Function

All transmitters are provided with an LCD display.

2.2.1.1. SINGLE LINE DISPLAYS

Single line, nonbacklit, 16-character liquid crystal displays (LCD).

ⓘ *All transmitter models except the GTx116e, GTx108e, HTx116 and HTx108 have a single line display.*

The display contrast is set at the factory. The factory setting can be adjusted on all single line transmitters by turning the contrast potentiometer. See the appropriate wiring diagram in Section 1 for the location of the contrast potentiometer for the model provided.

Single line displays can be configured for one or more views, depending on what information needs to be displayed. In addition to the measured sensor data, the transmitter name, location service(s), active alarms and/or system trouble conditions can be displayed. Factory defaults can be modified by the user (Section 3). Multiple views toggle every 3 seconds.

2.2.1.1.1. Single Location Transmitter Normal Operation

View is active when LCD NAME = ON (Default = OFF)

<	N	A	M	E	>									

View is active when LCD SERV = ON (Default = OFF)

<	S	E	R	V	>									

View is always active

#	#	#	#	#		C	F	M		#	#	.	#		F

Auto space.

View is active when LOW Airflow Alarm = Active

A	L	A	R	M	:	L	O	W		F	L	O	W		

View is active when HIGH Airflow Alarm = Active

A	L	A	R	M	:	H	I	G	H		F	L	O	W	

View is active when Fan Alarm = Active

A	L	A	R	M	:	F	A	N		#					

First fan to go into alarm.

View is active when TRBL = Active and LCD TRBL = ON

#	#		[	E	R	R	O	R		D	E	S	C	R	]

= Trouble Code.
Toggle through additional errors.

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2.2.1.1.2. Dual Location Transmitter Normal Operation

View is active when LCD NAME = ON (Default = OFF)

<	N	A	M	E	>									

View is always active

<	S	E	R	V	1	>								

View is always active

#	#	#	#	#		C	F	M		#	#	.	#		F

Auto space.

View is active when LOW Airflow Alarm = Active

A	L	A	R	M	:	L	O	W		F	L	O	W		

View is active when HIGH Airflow Alarm = Active

A	L	A	R	M	:	H	I	G	H		F	L	O	W	

View is always active

<	S	E	R	V	2	>								

View is always active

#	#	#	#	#		C	F	M		#	#	.	#		F

Auto space.

Note: There is no airflow alarm for location 2.

View is active when TRBL = Active and LCD TRBL = ON

#	#		[	E	R	R	O	R		D	E	S	C	R	]

= Trouble Code.

Toggle through additional errors.

2.2.1.2. DUAL LINE DISPLAYS

Dual line, backlit, 16-character x 2 row, liquid crystal displays (LCD).

ⓘ *Only the GTx116e, GTx108e, HTx116 and HTx108 have a dual line display.*

The display contrast is set at the factory. The factory setting can be adjusted by simultaneously pressing the ESC and ↓ pushbuttons. Change the contrast by pressing the ↑ or ↓ arrows. Press ENT to accept the changes or ESC to ignore the changes. The display will toggle through two views every 3 seconds.

ⓘ *The contrast will return to the factory default setting if the RESET ALL function is performed on the transmitter. See section 5 for more information.*

View 1

S	E	T		C	O	N	T	R	A	S	T		↑	↓
E	N	T		A	C	C	E	P	T		N	E	W	

View 2

S	E	T		C	O	N	T	R	A	S	T		↑	↓
E	S	C		N	O		C	H	A	N	G	E	S	

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2.2.1.2.1. Transmitters without /H humidity Sensor Option Normal Operation

View is active when LCD NAME = ON (Default = OFF)

<	N	A	M	E	>									

View is always active

<	S	E	R	V	>										Not visible when LCD SERV = OFF.
#	#	#	#	#		C	F	M		#	#	.	#		F	Auto space.

View is active when LOW or HIGH Airflow Alarm = Active

A	L	A	R	M	:	L	O	W		F	L	O	W			Not visible when low flow alarm is inactive.
A	L	A	R	M	:	H	I	G	H		F	L	O	W		Not visible when high flow alarm is inactive.

View is active when Fan Alarm = Active

A	L	A	R	M	:	F	A	N		#						First fan to go into alarm.

View is active when TRBL = Active and LCD TRBL = ON

#	#		[	E	R	R	O	R		D	E	S	C	R	]	## = First Trouble Code.
#	#		[	E	R	R	O	R		D	E	S	C	R	]	## = Second Trouble Code, if applicable.
																Toggle through additional errors.

2.2.1.2.2. Transmitters with the /H humidity sensor option

View is active when LCD NAME or LCD SERV = ON

<	N	A	M	E	>											Not visible when LCD NAME = OFF
<	S	E	R	V	>											Not visible when LCD SERV = OFF

View is always active when RHCONFIG=RH

#	#	#	#	#		C	F	M		#	#	.	#		F	Auto space.
										#	#	#	%	R	H	HCONFIG = RH

View is always active when RHCONFIG=DPT

#	#	#	#	#		C	F	M		#	#	.	#		F	Auto space.
#	#	#		F		D	P	T		#	#	#	%	R	H	HCONFIG = DPT

View is always active when RHCONFIG=ENTH

#	#	#	#	#		C	F	M		#	#	.	#		F	Auto space.
#	#	#		B	t	u	/	l	b	#	#	#	%	R	H	HCONFIG = ENTH

Alarm View: View is active when LOW or HIGH Airflow Alarm = Active

A	L	A	R	M	:	L	O	W		F	L	O	W			Not visible when low flow alarm is inactive.
A	L	A	R	M	:	H	I	G	H		F	L	O	W		Not visible when high flow alarm is inactive.

Alarm View: View is active when Fan Alarm = Active

A	L	A	R	M	:	F	A	N							

View is active when TRBL = Active and LCD TRBL = ON

#	#		[	E	R	R	O	R		D	E	S	C	R	]	## = First Trouble Code.
#	#		[	E	R	R	O	R		D	E	S	C	R	]	## = Second Trouble Code, if applicable.
																Toggle through additional errors.

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2.2.1.3. ALL TRANSMITTERS

Inspect and verify the following:

- ☐ The transmitter is displaying airflow and temperature (GTx116e transmitters with an optional /H sensor probe also display relative humidity).
- ☐ No error codes are displayed on the LCD.

ⓘ *Dual location transmitters toggle between locations. Press ENT to hold the current displayed location and ESC to return to toggle mode. If ESC is not pressed within 60 seconds the display will revert back to toggle mode.*

Use the factory default settings when possible. Factory default settings simplify startup, checkout and verification of the measuring device.

2.2.2. Transmitter Configuration

Transmitters do not require any system configuration for operation unless the following is required:

- ☐ SI (international) system of units: m, m/s, L/s, °C in lieu of I-P (US customary units): ft, FPM, CFM °F
- ☐ An analog output signal of 0-5 or 0-10 VDC is required.
- ☐ Dual location operation (*EB-Flow2* devices only).
- ☐ Enthalpy or dewpoint is the desired output of GTx116e transmitters with the optional /H relative humidity sensor installed in the sensor probe.

2.2.2.1. SYSTEM OF UNITS

The transmitter is factory set to I-P system of units. Set the system of units (SYS) to SI if required. See Section 3 for more information.

Verify the following:

- ☐ The transmitter is displaying airflow and temperature in the proper system of units.

⚠ *Changing the system of units restores factory default settings. Modifications made by the user are lost.*

2.2.2.2. DUAL LOCATION OPERATION

EB-Flow2 remote transmitters support two measurement locations. *EB-Flow2* transmitters serving two measurement locations must be configured for 2 locations for dual location operation. Set the LOCATIONS parameter to 2. The locations parameter is located in the GLOBAL submenu of the SETTINGS menu. See Section 3 for more information.

Verify the following:

- ☐ The LOCATIONS parameter is set to 2.

⚠ *Changing the transmitter from 1 to 2 locations, or vice versa, restores factory default settings for the system of units selected. Modifications made by the user are lost.*

Inspect and verify the following:

- ☐ The probes are connected to the transmitter properly for two locations. See Section 1 for more information.

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⚠ *Dual location applications require that the probes are installed in the proper locations and properly connected to the transmitter.*

2.2.2.3. SETUP WIZARDS

-F, -U and -B probe types are shipped with the area parameter(s) set to {null} (unknown). If volumetric airflow, CFM [L/s], is required, the area parameters must be determined. The -F/An fan array probe type has a fan array wizard (FAN WIZ) to configure the number of fans and facilitate area parameter entry. All other probe types have an area setup wizard (AREA WIZ) to facilitate area parameter determination.

Wizards are accessed from the TOOLS menu. See Section 4 for more information.

2.2.2.4. FREE AREA VERIFICATION

☐ Verify the area parameter, AREA, matches the actual area where the probes are installed.

ⓘ *The free area for -P, -U, -T and -B probes is the duct or opening area where the probes are mounted, less any internal insulation (i.e. the probe blockage does not affect the area calculation).*

ⓘ *The free area for -F fan inlet probes is the area calculated at the leading (upstream) edge of the sensor housing where the probes are mounted (i.e. the probe blockage does not affect the area calculation).*

If factory default settings have not been changed in the field, the area parameter will match the area printed on the hang tag of the sensor probes (exception: -F, -U and -B probe types).

⚠ *Conversion of the velocity to volumetric airflow requires that the proper area of the measurement location is used.*

⚠ *If the actual area is different, modify the area parameter and record the new size and area.*

TABLE 2-1 QUICK AREA CALCULATIONS, sq ft [sq m]
W=Width, H=Height, D=Diameter
W, H and D are in inches [mm]

Opening Shape	I-P	SI
Square or Rectangular	0.006944xWxH	0.000001xWxH
Round	0.005454xDxD	0.0000007854xDxD
Flat Oval	0.006944xWxH - 0.00149xHxH	0.000001xWxH - 0.0000002146xHxH

2.2.3. Transmitters with Analog Output Signals

Airflow measurement devices are typically used to determine the volumetric airflow rate in CFM [L/s]. The following procedure assumes that the desired measurement at the B.A.S. is CFM [L/s].

2.2.3.1. ANALOG OUTPUT SIGNAL TYPE AND RANGE

Verify the following:

☐ Verify that the analog input configuration of the B.A.S. matches the output configuration of the transmitter.

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The AOUT parameter firmware setting can be viewed/modified using the shortcut sequences below or through the Settings Menu. See Section 3 for more information.

Simultaneously press the ESC and ↑ pushbuttons during normal operation to quickly verify the setting for AOUT. Some models have an additional prompt to instruct the user to modify jumpers and/or switches on the circuit board or option card.

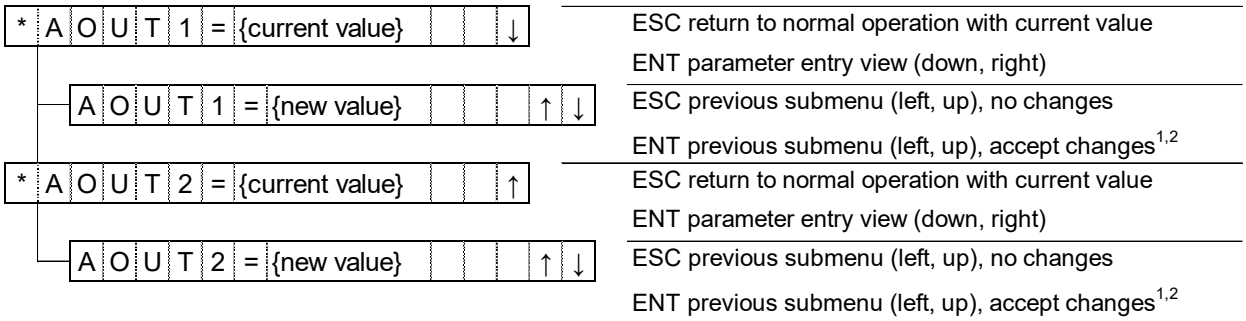
Use the output test tool provided in the TOOLS menu to set a fixed output signal to verify that each of analog output signal is converted properly by the B.A.S. See Section 4 for more information.

Failure to properly configure the output signal will result in significant measurement error at the B.A.S.

2.2.3.1.1. Advantage IV Gold Series Models (Except GTx116e and GTx108e)

The output is set by switches SW1 and SW2 on the output card provided with the transmitter for voltage (VDC) or current (mA). If a switch is set for voltage, only the corresponding voltage options of 0-5V or 0-10V will be available. If a switch is set for current, only 4-20mA will be available.

The AOUT1 and AOUT2 parameter shortcut (ESC and ↑) views are shown below.

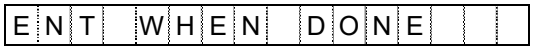


[1]If AOUT# (# = 1 or 2) is a change in output TYPE from voltage to current, toggle through the following views

View 1



View 2

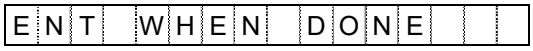


[2]If AOUT# (# = 1 or 2) is a change in output TYPE from current to voltage, toggle through the following views

View 1



View 2



2.2.3.1.2 Advantage IV Gold and Hybrid Series Models (GTx116e, GTx108e, HTx116 and HTx108)

The output is set by the AOUT parameter in firmware. No switches or jumpers are used. The AOUT parameter shortcut (ESC and ↑) view is shown below and applied to all analog output signals.

*	A	O	U	T	=	{current value}		
A	O	U	T	=	{new value}		↑	↓

ESC return to normal operation with current value

ENT select new value and update current value

2.2.3.1.3. Advantage IV Hybrid Series Models HTx104

The output is set by the AOUT parameter in firmware and the position of output jumpers OUT1 and OUT2.

⚠ *Failure to properly set the output jumpers will result in a significant error in the output signal from the transmitter.*

The AOUT parameter shortcut (ESC and ↑) sequence is shown on the next page.

*	A	O	U	T	=	{current value}		
A	O	U	T	=	{new value}		↑	↓

ESC return to normal operation with current value

ENT parameter entry view (down, right)

ESC previous submenu (left, up), no changes

ENT previous submenu (left, up), accept changes[1,2]

[1]If AOUT is changed to current, toggle through the following views:

View 1

S	E	T		J	M	P	R		1	&	2	m	A

View 2

E	N	T		W	H	E	N		D	O	N	E	

[2]If AOUT is changed to voltage, toggle through the following views:

View 1

S	E	T		J	M	P	R		1	&	2	V	D	C

View 2

E	N	T		W	H	E	N		D	O	N	E	

2.2.3.1.4. *EB-Flow2* Models

The output is set by the AOUT parameter in firmware. No switches or jumpers are used. The AOUT parameter shortcut (ESC and ↑) sequence is shown below and applied to all analog output signals.

*	A	O	U	T	=	{current value}		
A	O	U	T	=	{new value}		↑	↓

ESC return to normal operation with current value

ENT parameter entry view (down, right)

ESC previous submenu (left, up), no changes

ENT previous submenu (left, up), accept changes

2.2.3.2. ANALOG OUTPUT SCALING

The analog output units of measure and full-scale readings can be modified in the field. EBTRON measurement devices have percent of reading, not percent of full scale accuracy. There is often no benefit of modifying the factory default settings with today's high-performance application controllers. Multiply the default full scale velocity by the free area of the measurement location to determine the full-scale CFM [L/s] for the B.A.S.

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☐ Verify that the offset or minimum scale used by the B.A.S. is equal to zero.

2.2.3.2.1. Analog Output Units of Measure, AO1 UM = FPM [m/s]

☐ Verify that the span or full scale used by the B.A.S. is equal to: AO1 FS x AREA

2.2.3.2.2. Analog Output Units of Measure, AO1 UM = CFM [L/s]

☐ Verify that the span or full scale used by the B.A.S. is equal to: AO1 FS

2.2.3.2.3. All Analog Output Transmitters

☐ Verify that the airflow reading at the B.A.S. matches the analog output signal converted to CFM [L/s] of the transmitter.

TABLE 2-2 MANUAL VERIFICATION OF AIRFLOW ANALOG OUTPUT SIGNALS TO CFM [L/s]

V_{Out}=measured voltage, VDC, across +/- output terminals of transmitter

I_{Out}=measured current, mA, in series circuit (remove + signal wire)*

**Note: If current is measured in Amps, not mA, multiply reading from meter by 1,000.*

Output Signal	Units of Measure configured to FPM [m/s]	Units of Measure configured to CFM [L/s]
0-5 VDC	V_{Out} / 5 x FS x AREA	V_{Out} / 5 x FS
0-10 VDC	V_{Out} / 10 x FS x AREA	V_{Out} / 10 x FS
1-5 VDC	(V_{Out} - 1) / 4 x FS x AREA	(V_{Out} - 1) / 4 x FS
2-10 VDC	(V_{Out} - 2) / 8 x FS x AREA	(V_{Out} - 2) / 8 x FS
4-20 mA	(I_{Out} - 4) / 16 x FS x AREA	(I_{Out} - 4) / 16 x FS

2.2.4. Transmitters with an RS-485 Network Connection

Verify the following if an RS-485 network connection is used:

☐ The proper network protocol, BACnet or Modbus, is selected for the transmitter.

☐ The transmitter network settings are configured by the network integrator for the protocol used.

☐ The transmitter network communication is enabled by setting RS485 COM to ON.

ⓘ *Consult the BACnet object list or Modbus register map for available variables and variable structure.*

Do not set enable transmitter network communications until the transmitter network settings are properly configured to avoid addressing conflicts.

2.2.5. Transmitters with an Ethernet Network Connection

B.A.S. protocols BACnet IP and Modbus TCP are always enabled on Ethernet transmitters unless BACnet Ethernet is selected, in which case, BACnet Ethernet is enabled and BACnet IP is disabled.

EB-Bus Ethernet is enabled on all Ethernet transmitters and must be connected to an *EB-Bus* Ethernet network with an IAQ-Enforcer® Smart Display Panel SDX-1000, or other EBTRON approved *EB-Bus* host.

ⓘ *EB-Bus Ethernet is available on GTM116e/GTM108e transmitters with output card firmware 4.20 and later is required for EB-Bus Ethernet.*

Verify the following if an Ethernet network connection is used:

- ☐ The proper network protocol, BACnet IP or BACnet Ethernet, is selected if BACnet is used.
- ☐ The transmitter network settings are configured by the network integrator for BACnet IP or BACnet Ethernet, Modbus TCP, and TCP/IP protocols or by the *EB-Bus* host for *EB-Bus* Ethernet protocol.

ⓘ *Consult the BACnet object list or Modbus register map for available variables and variable structure.*

2.2.6. Transmitters with LON Network Connection

LON transmitters are provided with a full featured LonWorks compatible interface.

A “Service” push-button is provided for device commissioning. Once the device is recognized commissioning can be completed by uploading the parameters from the device.

Download the appropriate external interface file (.xif) if required by the installation software.

- ☐ GTF108e Transmitter: https://ebtron.com/wp-content/uploads/software/EBTRON_108e.xif
- ☐ GTF116e Transmitter: https://ebtron.com/wp-content/uploads/software/EBTRON_116e.xif

A “Wink” LED is provided for easy device identification.

An “Activity” LED and separate transmit and receive “TX” and “RX” indicators provide visual indication of transmitter and communication status. The “Activity” LED flashes on for 1 second, off for 1 second when the card is commissioned and online. The “Activity” LED remains illuminated constantly if there is an error.

2.2.7. Transmitters with a USB Data Logger

Transmitters with a USB data logger log sensor node data at 5-minute intervals using Universal Time Coordinated (UTC) based on an onboard real-time clock, whenever power is applied to the transmitter. Data files are automatically appended on power-up. The time zone and interval can be modified by the user using EB-Link software.

2.2.7.1. INSERTING A USB MEMORY DEVICE (“THUMB DRIVE”)

Install the USB memory device into the USB connector on the option card to start logging data.

It is a good practice to set the transmitter power switch to the “OFF” position before inserting the thumb drive memory device.

The USB port must be enabled to log data. Transmitters are shipped with the USB port parameter USB WRITE set to “ON”, which enables the port.

If the transmitter had been previously used to log data, the USB WRITE parameter would most likely have been set to “OFF”. To enable the USB WRITE parameter and start logging data, press the ↑↓ arrow buttons simultaneously to enter the MAIN MENU. The SETTINGS menu is displayed. Press the ENT button to select the top of the SETTINGS submenu category. Press the ↓ button until the USB submenu category is visible. Press the ENT button again to enter the USB submenu. Set the USB WRITE parameter to “ON”.

2.2.7.2. REMOVING A USB MEMORY DEVICE

Remove the USB memory device from the USB connector on the option card after data logging is complete and the USB port has been disabled by setting the USB WRITE parameter to “OFF”.

Disable the USB port to stop logging data. Press the ↑↓ arrow buttons simultaneously to enter the MAIN MENU. The SETTINGS menu is displayed. Press the ENT button to select the top of the SETTINGS

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submenu category. Press the ↓ button until the USB submenu category is visible. Press the ENT button again to enter the USB submenu. Set the USB WRITE parameter to “OFF”.

⚠ *Always set the USB WRITE parameter to “OFF” before removing the USB memory device to avoid data loss/and or damage.*

Remove the USB memory device.

It is a good practice to set the transmitter power switch to the “OFF” position before removing the USB memory device.

2.3. AIRFLOW MEASUREMENT VERIFICATION

In many installations, third-party airflow verification is required. It is recommended that all third-party verifiers have a thorough understanding of the installation requirements and operation of the EBTRON product being verified. The following is EBTRON’s recommended procedure for third-party airflow verification.

2.3.1. Assess the Installation

Third-party verifiers should ensure the following prior to field verification.

- ☐ Verify that the airflow directional arrow is pointing in the direction of airflow.
- ☐ Verify that the sensor node openings are in the direction of airflow and not twisted in the airstream.
- ☐ Verify that the area entered in the transmitter matches the actual area of the duct or opening where the AMD is located.
- ☐ Verify that no active trouble codes are visible on the display.
- ☐ Verify that field adjustment has not been enabled in the transmitter.

Use the EB-Link Reader phone application to verify the area, trouble code status and field adjustment status on Gold Series and Hybrid Series models HTx116 and HTx108 transmitters.

Correct installation issues, enter the proper area (see Section 3), address any active trouble codes and disable field adjustment before proceeding.

- ☐ Determine if probes meet or exceed EBTRON’s minimum placement guidelines.

ⓘ *The airflow rate indicated on the transmitter LCD (and that viewed on the EB-Link Reader) should be within the published installed accuracy of the product provided when probes meet or exceed EBTRON’s minimum placement guidelines.*

⚠ *If the probes do not meet the minimum placement guidelines, the installed accuracy may exceed the published installed accuracy of the product provided and field adjustment may be required.*

2.3.2. Assess the Verification Technique

2.3.2.1. DUCT SYSTEMS AND FAN MEASUREMENT

Ducted field measurements with handheld instruments can yield accuracies of 5% to 10% when an adequate straight run of duct is available. Applications having multiple airflow measuring devices in the same air path can compare one airflow measurement device to the other as part of the verification method.

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Set the system for 100% recirculation and compare the supply airflow measurement device to the return airflow measurement device. Verify the measurement in the best location for the field measurement device.

Set the system for 100% outdoor air and compare the supply airflow measurement device to the outdoor airflow measurement device. Verify the measurement in the best location for the field measurement device.

2.3.2.2. OUTDOOR AIR INTAKES

Direct field measurement of close-coupled outdoor air intakes can approach and exceed 25%. Indirect measurement techniques to determine the outdoor airflow rate using the difference between the supply and return airflow rates or the ratios of the return air, outdoor air and mixed air (or supply air) temperatures should be avoided since uncertainties almost always exceed 25%.

2.3.3. Adjust or Verify?

If the airflow measuring device is within the measurement uncertainty of the verification technique, EBTRON strongly recommends that no field adjustment correction is made. EBTRON airflow measurement devices are factory calibrated to NIST traceable standards. Field adjustment is not required when installed in accordance to published guidelines.

If field adjustment is required, a gain and/or offset adjustment can be made to the output of the device by manual entry (Section 3) or by using the Flow Adjust Wizard (Section 4).

If a one-point adjustment is made, make a gain adjustment rather than an offset adjustment to match the device to the third-party reading.

Do not make an area adjustment in lieu of a gain adjustment since area is used to verify proper setup of the airflow measurement device.

2.4. FOR MORE INFORMATION …

Congratulations! Your EBTRON airflow measurement device is fully operational.

For toll-free factory support call 800-2EBTRON (232-8766), Monday through Thursday 8:00 AM to 4:30 PM and Friday 8:00 AM to 2:00 PM eastern time or contact your local representative.

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O&M MANUAL

Section 3 – Custom Configuration

3. SECTION OVERVIEW

This section discusses access to the settings menu and menu navigation for custom configuration. It includes a table of the factory default settings, optional settings and ranges. A detailed reference and explanation for each setting is also provided.

The settings menu allows customization of the transmitter's factory default parameters.

After initial power-up, the transmitter is fully functional as an airflow and temperature measurement device. Transmitters are "plug and play" and do not require additional setup unless modifications to the factory default settings are desired by the user.

The most common setup parameters modified pertain to analog output, network and alarm settings.

Default and optional settings are shown in this document I-P first, followed by SI in brackets []. The parameter value "as displayed" on the LCD is shown in all capitals followed by a brief descriptive label or units in parenthesis ().

Record any changes to the factory default settings for future reference. All Advantage IV Gold Series and Hybrid Series models HTx116 and HTx108 transmitters are provided with a Bluetooth® Low Energy interface that allows transmitter settings to be quickly downloaded then e-mailed and/or saved using your iOS or Android phone or tablet.

3.1. SETTINGS MENU ACCESS AND NAVIGATION

The SETTINGS menu contains the following submenu categories:

- GLOBAL
- GENERAL
- DISPLAY
- ANALOG OUTPUT
- RS-485
- ETHERNET
- RF-LINK
- USB
- EB-LINK
- ALARM
- FAN ALARM (GTx108e-F/An and HTx108-F/An only)
- RELAY (EF-x2000 transmitters only)
- ADJUSTMENTS

The following symbols are used to describe shortcut paths for menu and submenu navigation:

- > press the enter ENT pushbutton once
- < press the escape ESC pushbutton once
- ↓ press the down arrow pushbutton until the subsequent menu item or parameter appears
- ↑ press the up arrow pushbutton until the subsequent menu item or parameter appears

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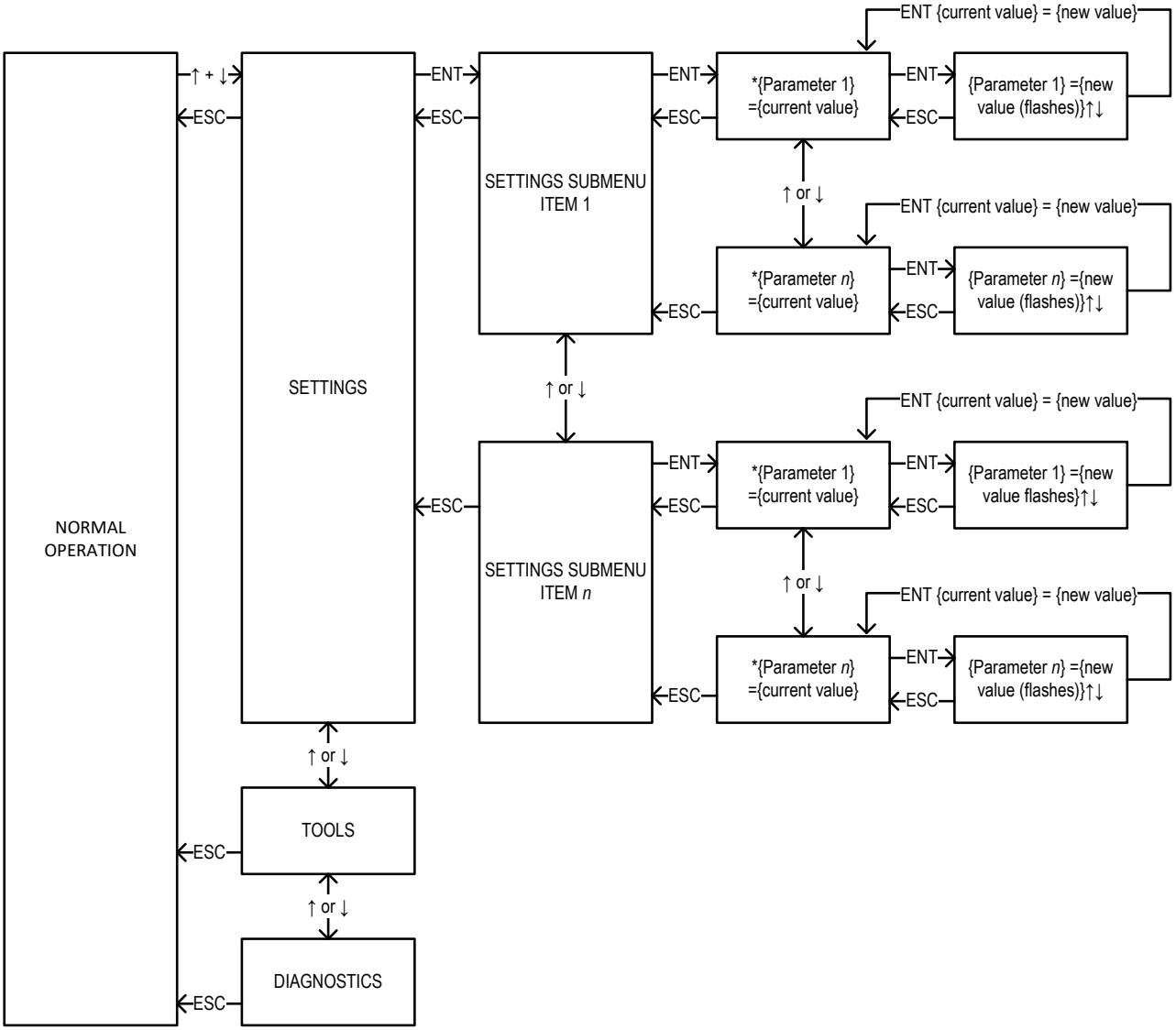
Simultaneously press the ↑ and ↓ pushbuttons during normal operation to enter the main menu. The transmitter continues to operate normally in the background.

The SETTINGS menu is displayed. Press the ENT pushbutton to select the top of the SETTINGS submenu category.

Parameters can be modified unless the transmitter has been locked using the LOCK TOOL (See Section 4). Press the ENT pushbutton to modify a parameter. The parameter is displayed without the "*" prefix and the parameter value flashes. Use the ↑ or ↓ pushbutton to modify the value of the parameter and press the ENT pushbutton to accept the modified value. The modified value is displayed with the "*" prefix. Press the ESC pushbutton instead of the ENT pushbutton to discard changes. After changes are complete, press the ESC pushbutton until the device returns to normal operation.

The SETTINGS menu block diagram is shown below.

FIGURE 3-1 SETTINGS MENU BLOCK DIAGRAM



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TABLE 3-1 Factory Defaults, Optional Settings and Ranges

ⓘ All transmitters are single location transmitters unless otherwise noted. Analog output, network, and Bluetooth capability are model specific (see manual).

	Parameter	Product Line	Probe Type	Special Criteria	Default	Optional Settings/Ranges	Units
GLOBAL	SYS	All	All		I-P	SI	
	LOCATIONS[1]	EB-Flow2	-U		1	2	
	Note 1: Changing the number of locations automatically restores the transmitter to factory default settings.						
GENERAL	NAME	All	All		*{Trans. S.N.}*	Alpha numeric name up to 16 characters at time of order or manually in field.	
	SERV	All	-P, -T, -U, -F		*AMD*	SA, RA, OA, EX, RE	
	SERV1	EB-Flow2	-U	Dual Location 1 of 2	*AMD-1*	SA-1, RA-1, OA-1, EX-1, RE-1	
	SERV2			Dual Location 2 of 2	*AMD-2*	SA-2, RA-2, OA-2, EX-2, RE-2	
	AIRFLOW	All	All		ACT	STD	
	ALT	All	All		0	AUTO[2], 0 to 20000 [0 to 6000]	ft [m]
	Note 2: Auto is only available on GTx116e, GTx108e, HTx116 and HTx108 transmitters.						
	INTG	All	All		30	1 to 999 (1 to 300 for -F/An probe types)	x 300 ms
	INTG1	EB-Flow2	-U	Dual Location 1 of 2		1 to 999	
	INTG2			Dual Location 2 of 2			
	LLIMIT	All	-P, -T, -U, -F		0 FPM	0 to 500 FPM [0.0 to 2.5 m/s]	
			-B	DPCONVERT=NO			
				DPCONVERT=YES	N/A		
	LLIMIT1	EB-Flow2	-U	Dual Location 1 of 2	0 FPM	0 to 500 FPM [0.0 to 2.5 m/s]	
	LLIMIT2			Dual Location 2 of 2			
	AREA	All	-P, -T		*{Order Area}*	0.00 to 9999.99 [0.000 to 999.999]	sq ft [sq m]
			-U, -F, -B		*{Null}*		
	AREA1	EB-Flow2	-U	Dual Location 1 of 2	*{Null}*		
	AREA2			Dual Location 2 of 2	*{Null}*		
	F_AREA*n*	Advantage IV	-F/An	AREA WIZ Saved	*{Null}*	0.00 to 9999.99 [0.000 to 999.999]	sq ft [sq m]
	DPCONVERT	All	-P, -T, -U, -F		N/A		
			-B		NO	YES	
	DIRECTION	All	-P, -T, -U, -F		N/A		
			-B		BI	UNI	
	TEMPMETH (not displayed)	All	All		WGT	None	
	H CONFIG	Advantage IV	-P w/ RH		RH	ENTH, DPT	
	Pb CORR		-P w/ RH		ON	OFF	
	EXT CAB	All	All		0	0 to 40 [0.0 to 12.2]	ft [m]
	EXT CAB1	EB-Flow2	-U	Dual Location 1 of 2			
	EXT CAB2			Dual Location 2 of 2			

TABLE 3-1 Factory Defaults, Optional Settings and Ranges

	Parameter	Product Line	Probe Type	Special Criteria	Default	Optional Settings/Ranges	Units
DISPLAY	LCD DISPL	All	All		ON	OFF[3]	
	Note 3: When LCD DISPL is OFF the LCD will only show "EBTRON" and the transmitter model code.						
	LCD NAME	All	All		OFF	ON	
	LCD SERV	All	-P, -T, -U, -F		OFF	ON	
		EB-Flow2	-U	LOCATIONS=2	ON	N/A	
	LCD TRBL	All	All		ON	OFF	
	LCD UM	All	-P, -T		CFM [L/s][4]	FPM [m/s]	
			-U, -F		FPM [m/s]	CFM [L/s][4]	
			-B	DPCONVERT=NO	FPM [m/s]	CFM [L/s]	
				DPCONVERT=YES	iWG [Pa]	None	
	Note 4: The default and only option for LCD UM is FPM [m/s] when the AREA parameter is set to 0 or {null}.						
	LCD INTG	All	All		100	1 to 999 (1 to 300 for -F/An probe types)	x 300 ms
ANALOG OUTPUT	AOUT[5]	Advantage IV	All		4-20mA	0-5V, 0-10V	
		EB-Flow2			2-10V	0-5V, 0-10V, 1-5V	
	Note 5: Does not apply to GTx116 and GTx108 transmitters without the "e" transmitter model suffix. Set jumpers OUT1 and OUT2 on the main PCB to the "mA" position for 4-20 mA output or "VDC" for 0-5/0-10V output on HTA104 transmitters.						
	AOUT1[6]	Advantage IV	All	SW1 set to mA	4-20mA	N/A	
				SW1 set to V	0-10V	0-5V	
	AOUT2[6]			SW2 set to mA	4-20mA	N/A	
				SW2 set to V	0-10V	0-5V	
	Note 6: Only applies to GTx116 and GTx108 transmitters. Switches are located on the output card.						
	ON FAIL[7]	All	All		HI	LO	
	Note 7: ON FAIL applies to all analog outputs.						
	AO1 ASGN	All	-P, -T, -U, -B, -F		AF	None	
		EB-Flow2	-U	Dual Location 1 of 2	AF1	F1-2, F2-1	
	AO1 UM	All	-P, -T, -U, -F		FPM [m/s]	CFM [L/s][8]	
			-B	DPCONVERT=NO			
				DPCONVERT=YES	iWG [Pa]	None	
	Note 8: The default and only option for AO1 UM is FPM [m/s] when the AREA parameter is set to 0 or {null}.						
	AO1 MS	All	-P, -T, -U, -F		0	None	
			-B	DIRECTION=UNI			
				DIRECTION=BI	*{-AO1 FS}*		
			-P, -T, -U, -F	AO1 ASGN=F1-2 or F2-1	*{-AO1 FS}*		

TABLE 3-1 Factory Defaults, Optional Settings and Ranges

	Parameter	Product Line	Probe Type	Special Criteria	Default	Optional Settings/Ranges	Units
ANALOG OUTPUT	AO1 FS	All	-T, -U, -B	AO1 UM=FPM [m/s]	3000 [15.0]	100 to 15000 [0.5 to 75.0]	FPM [m/s]
			-P		5000 [25.0]		
			-F		10000 [50.0]		
			-T, -U, -B	AO1 UM=CFM [L/s]	3000 [15000] x Area	5 to 999999 [5 to 999999]	CFM [L/s]
			-P		5000 [25000] x Area		
			-F		10000 [50000] x Area	5 to 999999 [5 to 999999]	
			-B	AO1 UM=iWG[Pa]	1.50 [300]	0.05 to 5.00 [10 to 1250]	iWG [Pa]
	AO2 ASGN	All	-P, -T, -U, -B		TEMP	ALRM or TRBL	
		Advantage IV	-F	/SI, /DI	TEMP	ALRM or TRBL	
				/An	TEMP	ALRM, FA or TRBL	
		EB-Flow2	-U	Dual Location 2 of 2	AF2	F1-2, F2-1	
	AO2 UM	All	-P, -T, -U, -F	AO2 ASGN=TEMP	F [C]	None	
		EB-Flow2	-U	Dual Location 2 of 2	FPM [m/s]	CFM [L/s][9]	
	Note 9: The default and only option for AO2 UM is FPM [m/s] when the AREA parameter is set to 0 or {null}.						
	AO2 MS	All	All	AO2 ASGN=TEMP	-20 [-30]	-50 to 160 [-50 to 70]	°F [°C]
		EB-Flow2	-U	AO2 ASGN=AF2	0	None	
				AO2 ASGN=F1-2 or F2-1	*{-AO2 FS}*		
	AO2 FS	All	All	AO2 ASGN=TEMP	160 [70]	-50 to 160 [-50 to 70]	°F [°C]
		EB-Flow2	-U	Dual Location 2 of 2 and AO2 UM=FPM [m/s]	3000 [15.0]	100 to 15000 [0.5 to 75.0]	FPM [m/s]
				Dual Location 2 of 2 and AO2 UM=CFM [L/s]	3000 [15000] x Area	5 to 999999 [5 to 999999]	CFM [L/s]
	NO FAULT	All	All	AO2 ASGN=ALRM, FA or TRBL	HI	LO	
	AO3 ASGN	Advantage IV[10]	-P w/ RH	H CONFIG=RH	RH	None	
				H CONFIG=ENTH	ENTH		
				H CONFIG=DPT	DPT		
	AO3 UM	Advantage IV[10]	-P w/ RH	H CONFIG=RH	% RH	None	
				H CONFIG=ENTH	Btu/lb [kJ/kg]		
				H CONFIG=DPT	F DPT [C DPT]		
	AO3 MS	Advantage IV[10]	-P w/ RH	H CONFIG=RH	0	0 to 100	% RH
				H CONFIG=ENTH		-20 to 400 [-40 to 800]	Btu/lb [kJ/kg]
				H CONFIG=DPT		-50 to 160 [-50 to 70]	°F [°C]

TABLE 3-1 Factory Defaults, Optional Settings and Ranges

	Parameter	Product Line	Probe Type	Special Criteria	Default	Optional Settings/Ranges	Units
ANALOG OUTPUT	AO3 FS	Advantage IV[10]	-P w/ RH	H CONFIG=RH	100	0 to 100	% RH
				H CONFIG=ENTH	200 [400]	-20 to 400 [-40 to 800]	Btu/lb [kJ/kg]
				H CONFIG=DPT	100 [50]	-50 to 160 [-50 to 70]	°F [°C]
	Note 10: AO3 is only applicable to GTx116e transmitters that have the humidity sensor option, /H, provided with the -P probe type.						
RS-485	NETOUT	All	All		BACNET	MODBUS	
	NETADDR	All	All	NETOUT=BACNET	2	0 to 127	
				NETOUT=MODBUS		0 to 247	
	NETBAUD	All	All	NETOUT=BACNET	76800	38400, 19200, 9600	bps
				NETOUT=MODBUS	19200	76800, 38400, 9600	
	PARITY	All	All	NETOUT=MODBUS	EVEN	ODD, NONE1, NONE2	
	NETDI	All	All	NETOUT=BACNET	2	0 to 4194302	
	RS485 COM	All	All		OFF	ON	
ETHERNET	EB-BUS CON[11]	Advantage IV	All	Connected to SDX-1000 (or other EBTRON approved *EB-Bus* host)	T	F	
	Note 11: GTM116e and GTM108e transmitters require output card firmware version 4.32 and later. GTB116e and GTB108e transmitters require output card firmware version 1.02 and later.						
	MODE	Advantage IV[12]	All		DUAL	DAISY CHN	
	Note12: Only applies to GTS116e, GTS108e, HTS116 and HTS108 transmitters						
	DHCP	Advantage IV	All	See Note 13	ON	OFF or set by *EB-Bus* host	
	IP	Advantage IV	All	See Note 13	10.0.5.100	Set by network integrator or *EB-Bus* host	
	MASK	Advantage IV	All	See Note 13	255.255.240.000	Set by network integrator or *EB-Bus* host	
	GATE	Advantage IV	All	See Note 13	10.0.0.1	Set by network integrator or *EB-Bus* host	
	DNS[14]	Advantage IV	All	See Note 13	10.0.0.8	Set by network integrator or *EB-Bus* host	
	BACNET	Advantage IV	All	See Note 13	IP	ETH or set by *EB-Bus* host	
	NETDI	Advantage IV	All	See Note 13	The last octet of the MAC address[15]	0 to 4194302 or set by *EB-Bus* host	
	UDPPORT	Advantage IV[16]	All	See Note 17	47808	0 to 65535 or set by *EB-Bus* host	
	DHCP1	Advantage IV	All	MODE = DUAL	ON	Set by *EB-Bus* host	
	IP1	Advantage IV	All	MODE = DUAL	10.0.5.100	Set by *EB-Bus* host	
	MASK1	Advantage IV	All	MODE = DUAL	255.255.240.000	Set by *EB-Bus* host	
	BACNET1	Advantage IV	All	MODE = DUAL	IP	Set by *EB-Bus* host	
	NETDI1	Advantage IV	All	MODE = DUAL	The last octet of the MAC address[15]	Set by *EB-Bus* host	
	UDPPORT1	Advantage IV[16]	All	MODE = DUAL and BACNET=IP	47808	Set by *EB-Bus* host	
	DHCP2	Advantage IV	All	MODE = DUAL	ON	OFF	
	IP2	Advantage IV	All	MODE = DUAL	10.0.5.100	Set by network integrator	

TABLE 3-1 Factory Defaults, Optional Settings and Ranges

	Parameter	Product Line	Probe Type	Special Criteria	Default	Optional Settings/Ranges	Units
ETHERNET	MASK2	Advantage IV	All	MODE = DUAL	255.255.240.000	Set by network integrator	
	GATE2	Advantage IV	All	MODE = DUAL	10.0.0.1	Set by network integrator	
	DNS2[14]	Advantage IV	All	MODE = DUAL	10.0.0.8	Set by network integrator	
	BACNET2	Advantage IV	All	MODE = DUAL	IP	ETH	
	NETDI2	Advantage IV	All	MODE = DUAL	The last octet of the MAC address[15]	0 to 4194302	
	UDPPORT2	Advantage IV[16]	All	MODE = DUAL and BACNET=IP	47808	0 to 65535	
	Note 13: GTM116, GTM108, GTx116e, GTx108e, HTx116 or HTx108 transmitters (where x = B or M) or MODE = DAISY CHN Note 14: DNS parameter is disabled, and intended for future use. Note 15: The default is 2 for GTM116e and GTM108e transmitters with output card firmware prior to 4.17. Note 16: UDPPORT is only applicable to GTx116e and GTx108e transmitters where x = B, M or S Note 17: GTM116, GTM108, GTx116e, GTx108e, HTx116 or HTx108 transmitters (where x = B or M) or MODE = DAISY CHN and BACNET=IP						
LON	NO SETTINGS REQUIRED FOR LON NETWORK CONFIGURATION						
USB	USB WRITE	Advantage IV	All		ON	OFF	
EB-LINK	EB-LK INTG	Advantage IV	-P, -T, -F		100	1 to 999 (1 to 300 for -F/An probe types)	x 300 ms
	EB-LK STAT	Advantage IV	-P, -T, -F		ON	OFF	
ALARM	LO ALRM	All	All		OFF	ON	
	HI ALRM	All	All		OFF	ON	
	ALRM UM[18]	All	-P, -T, -U, -F		FPM [m/s]	CFM [L/s]	
			-B	DPCONVERT=NO			
				DPCONVERT=YES	iWG [Pa]	None	
	Note 18: The default and only option for ALRM UM is FPM [m/s] when the AREA parameter is set to 0 or {null}.						
	SETPNT	All	-P, -T, -U, -F	ALRM UM=FPM [m/s]	0	0 to 999999 [0.00 to 99.99]	FPM [m/s]
				ALRM UM=CFM [L/s]		0 to 999999 [0 to 999999]	CFM [L/s]
			-B	ALRM UM=FPM [m/s]		0 to 999999 [0.00 to 99.99] (± if DIRECTION=BI)	FPM [m/s]
				ALRM UM=CFM [L/s]		0 to 999999 [0 to 999999] (± if DIRECTION=BI)	CFM [L/s]
				ALRM UM=iWG [Pa]		0.00 to 5.00 [0 to 1250] (± if DIRECTION=BI)	iWG [Pa]
	TOL	All	All		15%	> of 5% or 20 FPM [0.10 m/s] TO 100%	
	DELAY	All	All		2min	0min to 15min	
	ZERO OFF	All	-P, -T, -U, -F		YES	NO	
			-B	DPCONVERT=NO			
				DPCONVERT=YES	N/A	N/A	
	RESET	All	All		AUTO	MANUAL	
FAN ALARM	FAN ALRM	Advantage IV	-F/An		OFF	MIN, DEV, %MAX	
	ON ALRM	Advantage IV	-F/An		KEEP	IGNR	
	ALRM UM	Advantage IV	-F/An	FA TYPE=MIN	FPM [m/s]	CFM [L/s]	

TABLE 3-1 Factory Defaults, Optional Settings and Ranges

	Parameter	Product Line	Probe Type	Special Criteria	Default	Optional Settings/Ranges	Units
FAN ALARM	SETPNT	Advantage IV	-F/An	FA TYPE=DEV or %MAX	25%	10% to 100%	
				FA TYPE=MIN ALRM UM=FPM [m/s]	0	0 to 999999 [0.00 to 99.99]	FPM [m/s]
				FA TYPE=MIN ALRM UM=CFM [L/s]		0 to 999999 [0 to 999999]	CFM [L/s]
	DELAY	Advantage IV	-F/An		2min	0min to 15min	
	ZERO OFF	Advantage IV	-F/An		YES	NO	
	RESET	Advantage IV	-F/An		AUTO	MANUAL	
RELAY	R1 ASGN	EB-Flow2	All		NONE	ALRM or TRBL	
	R1 STATUS	EB-Flow2	All		NO	NC	
ADJUSTMENTS	FLOW ADJ	All	-P, -T, -U, -F		OFF	ON	
			-B	DPCONVERT=NO			
				DPCONVERT=YES	N/A	N/A	
	F_ADJ UM	All	-P, -T, -U, -F		CFM [L/s][19]	FPM [m/s]	
			-B	DPCONVERT=NO			
				DPCONVERT=YES	N/A	N/A	
	Note 19: The default and only option for F_ADJ UM is FPM [m/s] when the AREA parameter is set to 0 or *{null}* .						
	PRESS ADJ	All	-B	DPCONVERT=NO	N/A	N/A	
				DPCONVERT=YES	OFF	ON	
	FAW SMPLS	All	All		100	1 to 999	x 300 ms
	GAIN	All	-P, -T, -U, -F		1.000	0.100 to 9.999	
			-B	DPCONVERT=NO			
				DPCONVERT=YES			
	GAIN1	EB-Flow2	-U	Dual Location 1 of 2			
	GAIN2			Dual Location 2 of 2			
	OFF	All	-P, -T, -U, -F	F_ADJ UM=FPM [m/s]	0	-99999 to 99999 [-99.99 to 99.99]	F_ADJ UM
				F_ADJ UM=CFM [L/s]		-99999 to 99999	
			-B	DPCONVERT=NO F_ADJ UM=FPM [m/s]		-99999 to 99999 [-99.99 to 99.99]	
				DPCONVERT=NO F_ADJ UM=CFM [L/s]		-99999 to 99999	
				DPCONVERT=YES	0.000	-5.000 to 5.000 [-1250 to 1250]	iWG[Pa]
	OFF1	EB-Flow2	-U	Dual Location 1 of 2 F_ADJ UM=FPM [m/s]	0	-99999 to 99999 [-99.99 to 99.99]	F_ADJ UM
				Dual Location 1 of 2 F_ADJ UM=CFM [L/s]		-99999 to 99999	
	OFF2			Dual Location 2 of 2 F_ADJ UM=FPM [m/s]		-99999 to 99999 [-99.99 to 99.99]	
				Dual Location 2 of 2 F_ADJ UM=CFM [L/s]		-99999 to 99999	

FIGURE 3-2 SINGLE LINE DISPLAY MENU STRUCTURE

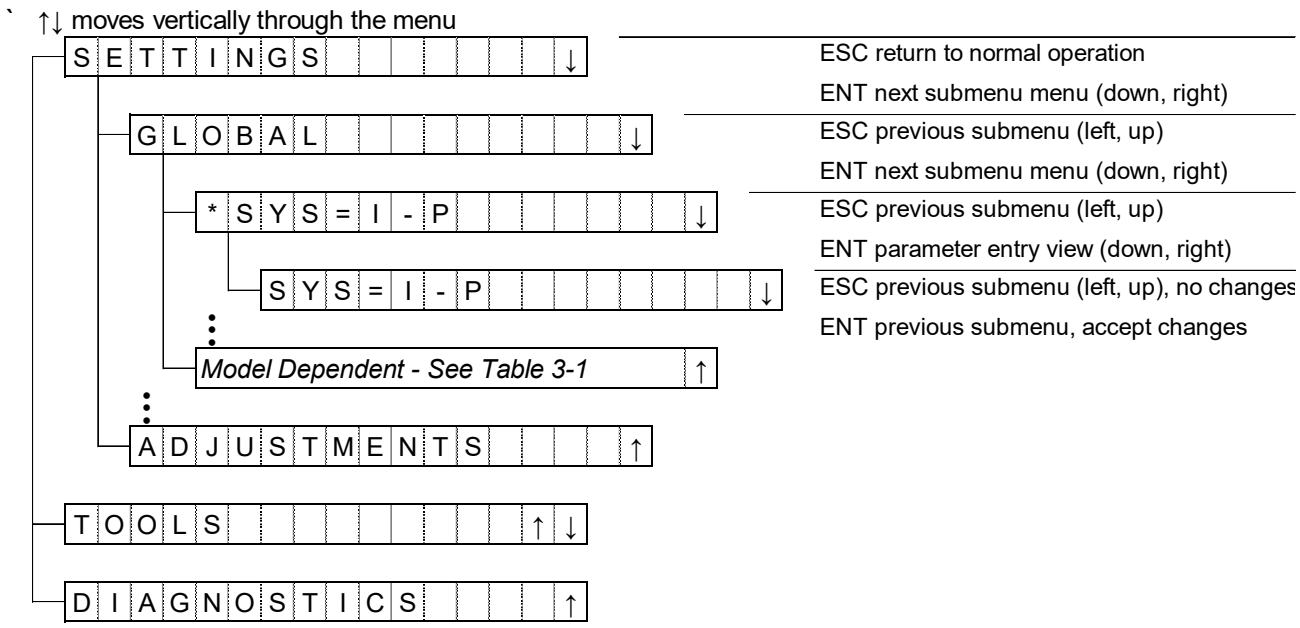
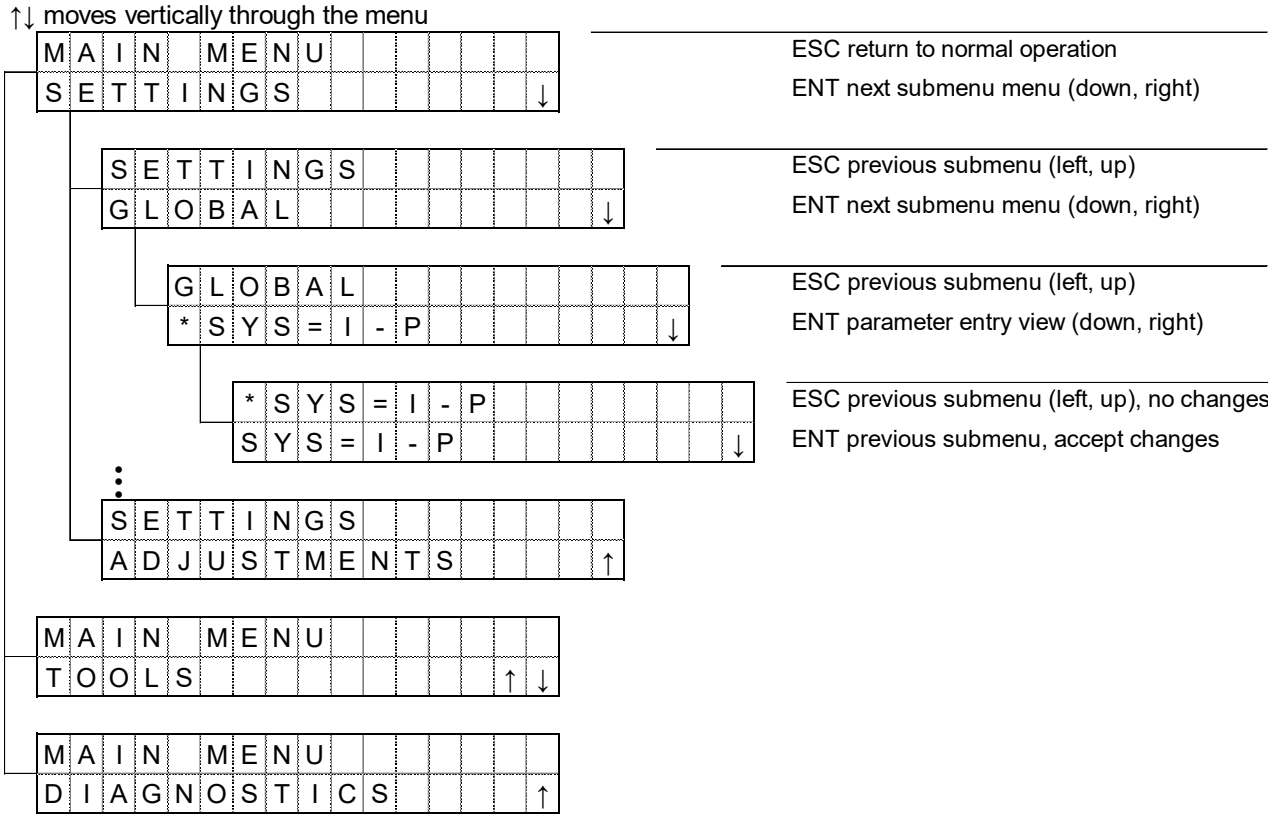


FIGURE 3-3 DUAL LINE DISPLAY MENU STRUCTURE



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3.2. GLOBAL SETTINGS

The GLOBAL submenu contains the following parameters:

- SYS (System Units of Measure)
- LOCATIONS (Number of Measurement Locations)

3.2.1. System Units of Measure [SYS]

The units of measure used by the transmitter.

[Menu path: SETTINGS > GLOBAL > SYS]

Default: I-P (Imperial System of Units)
Optional Settings: SI (International System of Units)

U/M=I-P: The transmitter system of units and abbreviations used in this document are as follows:

- Area: square feet (sq. ft)
- Length: feet (ft)
- Velocity: feet per minute (FPM)
- Volumetric airflow: cubic feet per minute (CFM)
- Pressure: inches of water gauge (iWG)
- Temperature and Dewpoint: degrees Fahrenheit (°F)
- Relative Humidity: percent relative humidity (% RH)
- Enthalpy: British thermal units per pound (Btu/lb)

U/M=SI: The transmitter system of units and abbreviations used in this document are as follows:

- Area: square meters (sq. m)
- Length: meters (m)
- Velocity: meters per second (m/s)
- Volumetric airflow: liters per second (L/s)
- Pressure: Pascals (Pa)
- Temperature and Dewpoint: degrees Celsius/centigrade (°C)
- Relative Humidity: percent relative humidity (% RH)
- Enthalpy: Kilojoules per kilogram (kJ/kg)

⚠ *Changing the system of units restores factory default settings. Modifications made by the user are lost.*

Configure for 2 locations at initial power-up or prior to modifying any factory default settings.

3.2.2. Measurement Locations [LOCATIONS]

The number of measurement locations the transmitter serves.

ⓘ *The measurement location parameter is only visible with EB-Flow2 remote transmitters.*

[Menu path: SETTINGS > GLOBAL >↓ LOCATIONS]

Default: 1
Optional Settings: 2

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Changing the transmitter from 1 to 2 locations, or vice versa, restores factory default settings. Modifications made by the user are lost.

Dual location applications require that the probes are installed in the proper locations and properly connected to the transmitter. See Section 1 for more information.

3.3. GENERAL SETTINGS

The GENERAL submenu contains the following parameters:

- NAME (Location Name)
- SERV (Service Description)
- AIRFLOW (Airflow Calculation Method)
- ALT (Altitude)
- INTG (Airflow Output Integration)
- LLIMIT (Low Limit Airflow Cutoff)
- AREA (Area at the Measurement Location)
- DPCONVERT (Differential Pressure Conversion) -B probe type only
- DIRECTION (Uni or Bidirectional Measurement) -B probe type only
- TEMPMETH (Temperature Averaging Method)
- H CONFIG (Relative Humidity Sensor Configuration) -P/H probe type only
- Pb CORR (Barometric Pressure Correction) -P/H probe type only
- EXT CAB (Extension Cable Length)

3.3.1. Location Name [NAME]

The alpha-numeric name of the location being served.

[Menu path: SETTINGS >↓ GENERAL > NAME]

Default: Transmitter serial number unless a valid location name is provided at the time of order.
Optional Settings: User defined in field using pushbutton interface (up to 16 characters can be entered).

*Enter edit mode by pressing enter when *NAME={parameter name}↑↓ is shown. Use the ↑ and ↓ pushbuttons to select the alpha-numeric character for each character position in the NAME string. Press the ENT pushbutton to move to the next character position to the right. Press the ESC pushbutton to move to the previous character position to the left. Press and hold the ENT pushbutton for 2 seconds to save the name. Press and hold the ESC pushbutton for 2 seconds to exit without accepting changes. The first 8 digits of the name are displayed on the LCD as *NAME={parameter name}↑↓.*

*Press the ENT pushbutton from the *NAME={parameter name}↑↓ and scroll through the name parameter using the ENT pushbutton to view names longer than 8 characters. Press and hold the ESC pushbutton to exit edit mode without making changes to the NAME parameter.*

Use the service description to differentiate the location name of transmitters configured for dual locations.

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3.3.2. Service Description [SERV]

The service description of the measurement location.

[Menu path: SETTINGS >↓ GENERAL >↓ SERV]

Default: AMD
Optional Settings: SA, RA, OA, EX, RE

If the transmitter is configured for two locations, parameters for SERV1 and SERV2 will be displayed in lieu of SERV and the suffix -1 and -2 will be added to the default and optional service descriptions, respectively for each location.

More detailed location naming (up to 16 characters) can be accomplished via BACnet.

The service description can be shown on the LCD to facilitate identification. The service description is particularly useful for display identification on dual location transmitters.

3.3.3. Airflow Calculation Method [AIRFLOW]

The calculation method used by the transmitter to determine the airflow rate.

[Menu path: SETTINGS >↓ GENERAL >↓ AIRFLOW]

Default: ACT (actual airflow)
Optional Settings: STD (standardized mass airflow)

AIRFLOW=ACT. The actual airflow is based on the nominal air density at standardized sea level and the actual air temperature. Modify the altitude parameter, ALT, to adjust the actual air velocity for the nominal altitude (or actual barometric pressure on GTx116e, GTx108e, HTx116 or HTx108 transmitters), if different from standardized sea level.

AIRFLOW=STD. The standardized mass airflow is normalized for 70 °F [21.1 °C] and 29.92” Hg [101.3 kPa].

If AIRFLOW=STD, the letter S will precede the airflow units, FPM [m/s] or CFM [L/s], on the LCD. The letter S is assumed when AIRFLOW is set to STD and intentionally not shown in additional references in this document to simplify reading.

3.3.4. Altitude [ALT]

The altitude used to correct the measured airflow for the nominal air density at the elevation above sea level where the AMD is installed.

[Menu path: SETTINGS >↓ GENERAL >↓ ALT]

Default: 0 (ft) [0 (m)]
Optional Settings: AUTO (GTx116e, GTx108e, HTx116 or HTx108 transmitters only) or 0 to 20000 (ft) [0 to 6000 (m)]

AUTO corrects for the actual barometric pressure rather than the nominal standardized elevation.

The altitude correction only affects the airflow when AIRFLOW=ACT.

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TABLE 3-2 CORRECTION FACTORS AT VARIOUS ALTITUDES Actual Airflow = Measured Airflow x Correction Factor			
ALT Parameter, ft [m]	Correction Factor	ALT, Parameter, ft [m]	Correction Factor
0	1.00	6,000 [1,828]	1.25
1,000 [304.8]	1.04	7,000 [2,134]	1.30
2,000 [609.6]	1.08	8,000 [2,438]	1.35
3,000 [914.4]	1.11	9,000 [2,743]	1.40
4,000 [1,219]	1.16	10,000 [3,048]	1.45
5.000 [1,524]	1.20	15,000 [4,572]	1.75
Note: Correction factors are based on 70 °F [21.1 °C] and 29.92” Hg [101.3 kPa] at sea level.			

3.3.5. Airflow Output Integration [INTG]

The integration buffer size for the running average of the airflow used for analog and network airflow output.

[Menu path: SETTINGS >↓ GENERAL >↓ INTG]

Default: 30
Optional Settings: 1 to 999 (1 to 300 for -F/An probe types and 1 to 750 for HTx202 transmitters)

Transmitters calculate the airflow every 300 milliseconds. The integration buffer is used to smooth real-time airflow data output to AO1 and the network airflow variable AI1. The default integration of 30 is the running average (first in, first out) of the average of all sensor nodes over the preceding 9 second time period.

ⓘ *If the transmitter is configured for two locations, parameters for INTG1 and INTG2 will be displayed in lieu of INTG.*

ⓘ *Airflow output integration is independent of the integration buffer which is used for airflow data displayed on the LCD (which is generally higher). Display integration is discussed in the DISPLAY SETTINGS section.*

Decreasing the integration may improve performance on fast acting control systems such as those found in laboratories and clean rooms.

⚠ *Increasing the integration buffer when installations are mounted in outdoor air intakes may result in control system hunting. Use the default integration and decrease PID output speed if hunting is a problem. EBTRON recommends using the default integration for control and a separate, higher integration, such as that used on the LCD for viewing.*

3.3.6. Low Limit Airflow Cutoff [LLIMIT]

The velocity below which the airflow is forced to 0 FPM [m/s].

[Menu path: SETTINGS >↓ GENERAL >↓ LLIMIT]

Default: 0 FPM [0 m/s]
Optional Settings: 0 FPM to 500 FPM [0 m/s to 2.5 m/s]

ⓘ *If the transmitter is configured for two locations, parameters for LLIMIT1 and LLIMIT2 will be displayed in lieu of LLIMIT.*

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LLIMIT is applied to both directions (i.e. ±) when transmitters with -B probes are configured for bi-directional airflow (DIRECTION=BI). LLIMIT is not supported when transmitters are configured for pressure (DPCONVERT=YES).

Use on outdoor air intakes to eliminate false airflow readings that may occur from transient wind gusts when the intake damper is closed. The low limit airflow cutoff does not affect airflow accuracy or modify any field adjustment coefficients.

3.3.7. Area [AREA]

The area used to convert the average velocity to volumetric airflow when CFM [L/s] is selected for the display and/or airflow output. The area parameter represents the total area of the measurement location.

[Menu path: SETTINGS >↓ GENERAL >↓ AREA]

Default and Optional Settings: See specific probe type

Conversion of the velocity to volumetric airflow requires that the proper area of the measurement location is used.

Record the area stored in the transmitter if the default analog airflow output signal units of measure for velocity is maintained and the analog output signal for velocity is used by the B.A.S. The area is used to determine the full-scale reading used by the B.A.S.

3.3.7.1. -P AND -T PROBE TYPES

Default: Area based on order size
Optional Settings: 0.00 to 9999.99 (sq. ft) [0.000 to 999.999 (sq. m)]

The -P and -T probe types have the area stored in a serial memory chip located in the cable plug of each sensor probe. The area is automatically transferred to the non-volatile memory of the transmitter at initial power-up based on order size. The area is printed on each hang tag connected near the terminal plug of each sensor probe.

Verify that the size and area on the hang tag match the actual area where the probes are installed. If the actual area is different, modify the area parameter and record the new size and area.

3.3.7.2. -U PROBE TYPES

Default: *{Null}*
Optional Settings: 0.00 to 9999.99 (sq. ft) [0.000 to 999.999 (sq. m)]

The -U probe type area(s) can be entered manually or by using the AREA WIZARD.

If the transmitter is configured for two locations, parameters for AREA1 and AREA2 will be displayed in lieu of AREA.

Probe area(s) can be modified directly through SETTINGS or by using the AREA WIZARD.

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3.3.7.3. -F PROBE TYPES

Default: *{Null}*
Optional Settings: 0.00 to 9999.99 (sq. ft) [0.000 to 999.999 (sq. m)]

3.3.7.3.1. -F/SI and -F/DI probe types

The area can be entered manually or by using the FAN WIZARD.

ⓘ *DWDI fans indicate a single AREA parameter equal to the total area of both inlets.*

3.3.7.3.2. -F/An probe types

The area can be entered manually or by using the FAN WIZARD.

ⓘ *Transmitters provided can be configured for up to eight individual fans in a fan array. Transmitters indicate a single AREA parameter equal to the total area of all of the fans in the array. Parameters for F_AREA1, F_AREA2 … F_AREAn are visible after the ARRAY WIZARD is run. Individual fan areas are used to determine individual fan airflow rates and to enable fan alarming features. Once the ARRAY WIZARD has been run the AREA parameter is locked and can only be modified by changing individual fan areas (i.e. the AREA is calculated based on individual fan areas).*

3.3.7.4. -B PROBE TYPES

Default: *{Null}*
Optional Settings: 0.00 to 9999.99 (sq. ft) [0.000 to 999.999 (sq. m)]

The area where the bleed sensor is located is generally not a concern. However, there are applications where the bleed sensor is used to determine the airflow rate through an opening, such as a hallway or corridor. It can also be used to estimate the airflow across a louver. In all cases, a reference airflow is required. The area can be entered manually or by using the AREA WIZARD.

Use the area of the opening to "rough out" the CFM [L/s] and a reference airflow to adjust offset/gain values to "fine tune" the measurement.

3.3.8. Differential Pressure Conversion [DPCONVERT]

Differential pressure conversion applies to the -B probe type only. The air velocity is converted to the equivalent pressure based on a nominal flow coefficient for the bleed sensor housing without any pipe or tube connected, is calculated using the measured air velocity.

[Menu path: SETTINGS >↓ GENERAL >↓ DPCONVERT]

Default: NO
Optional Settings: YES

DPCONVERT=NO. The measured air velocity is not converted to an estimated differential pressure.

DPCONVERT=YES. An estimated differential pressure, based on a nominal flow coefficient for the bleed sensor housing.

⚠ *The bleed airflow sensor is not a pressure sensor. The converted pressure is simply an estimate and should not be used for critical pressure measurement without field adjustment to a known pressure reference.*

⚠ *The addition of pipe or tubing to the bleed sensor housing will increase pressure losses and add additional error to the pressure measurement.*

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3.3.9. Uni or Bidirectional Measurement [DIRECTION]

Uni or bidirectional measurement applies to the -B probe type only.

[Menu path: SETTINGS >↓ GENERAL >↓ DIRECTION]

Default: BI
Optional Settings: UNI

DIRECTION=BI. The sensor measures airflow or equivalent pressure in both directions. The positive direction is in the direction of the airflow arrow.

DIRECTION=UNI. The sensor measures airflow or equivalent pressure in the direction of the airflow arrow only.

3.3.10. Temperature Averaging Method [TEMPMETH]

The method used to determine the average temperature of multiple sensor nodes.

[Menu path: no menu item]

Default: WGT (velocity weighted average)
Optional Settings: None

TEMPMETH=WGT. The velocity weighted average of each sensor node is used.

In a moving airstream, the actual temperature of air with a velocity and temperature profile is not equal to the arithmetic average of the individual sensor node measurements. The true average temperature should be weighted by the velocity at each sensor node.

The velocity-weighted temperature is used to calculate velocity-weighted relative humidity and enthalpy, thus compensating for velocity and temperature profiles and improved relative humidity and enthalpy measurement.

An averaging temperature probe will not match the weighted temperature output if there is both a velocity and temperature profile. Use the non-weighted temperature readings provided in the diagnostics menu to compare temperature readings when verification is required.

3.3.11. Relative Humidity Sensor Configuration [H CONFIG]

The configuration of the relative humidity sensor.

The relative humidity sensor configuration parameter is only visible with Advantage IV GTx116e remote transmitters that have the humidity sensor option, /H, provided with the -P sensor probe type.

[Menu path: SETTINGS >↓ GENERAL >↓ H CONFIG]

Default: RH (relative humidity)
Optional Settings: ENTH (enthalpy), DPT (dew point)

H CONFIG=RH. The calculated velocity-weighted relative humidity based on the relative humidity sensor and velocity-weighted temperature.

H CONFIG=ENTH. The calculated velocity-weighted enthalpy based on the relative humidity sensor dewpoint calculation, velocity-weighted temperature and barometric pressure (see Pb CORR).

H CONFIG=DPT. The calculated dewpoint based on the relative humidity sensor and adjacent sensor node temperature.

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3.3.12. Barometric Pressure Correction [Pb CORR]

Apply the measured barometric pressure to the enthalpy calculation.

ⓘ *The barometric pressure correction parameter is only visible with Advantage IV GTx116e remote transmitters that have the humidity sensor option, /H, provided with the -P sensor probe type.*

[Menu path: SETTINGS >↓ GENERAL >↓ Pb CORR]

Default: ON (Barometric pressure is measured using the onboard barometric pressure sensor)
Optional Settings: OFF (Barometric pressure is calculated based on the value of the altitude parameter, ALT)

3.3.13. Extension Cable Length [EXT CAB]

The length of a single extension cable if extension cables are added in the field.

[Menu path: SETTINGS >↓ GENERAL >↓ EXT CAB]

Default: 0 (ft) [0 (m)]
Optional Settings: 0 to 40 (ft) [0.0 to 12.2 (m)]

Extension cables can be added in the field if the cable length ordered does not reach the location where the transmitter is located. The additional cable length must be entered in the transmitter to compensate for the voltage drop to the heated sensor and ensure sensor accuracy.

⚠ *Extension cables of equal length must be added to each sensor probe.*

⚠ *The total cable length of the original cable and the extension cable must not exceed 50 ft [15.2 m]*

ⓘ *If the transmitter is configured for two locations, parameters for EXT CAB1 and EXT CAB2 will be displayed in lieu of EXT CAB.*

3.4. DISPLAY SETTINGS

An alpha-numeric LCD is provided with all models covered in this manual. The DISPLAY submenu contains the following parameters:

- LCD DSPL (LCD Display Mode)
- LCD NAME (LCD Location Name Indication)
- LCD SERV (LCD Service Indication)
- LCD TRBL (LCD System Trouble Indication)
- LCD UM (LCD Airflow Units of Measure)
- LCD INTG (LCD Airflow Integration)

3.4.1. Display Operation

The following Advantage IV and EB-Flow2 transmitters are provided with a single line non-backlit, 16-character LCD:

- Advantage IV Product Line
    - Gold Series
        - GTx116
        - GTx108

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    - Hybrid Series
        - HTx104
- *EB-Flow2* Product Line
    - EF Series
        - EF-x2000

The following Advantage IV transmitters are provided with a dual line backlit, 16-character by 2-row LCD:

- Advantage IV Product Line
    - Gold Series
        - GTx116e
        - GTx108e
    - Hybrid Series
        - HTx116
        - HTx108

Transmitters can be configured to show or hide specific views based on parameters described in the following sections. Additional views are dependent on alarm settings and trouble code status.

3.4.1.1. SINGLE LINE DISPLAY OPERATION

3.4.1.1.1. Single Location Transmitters

View is active when LCD NAME = ON (Default = OFF)

<	N	A	M	E	>									

View is active when LCD SERV = ON (Default = OFF)

<	S	E	R	V	>									

View is always active

#	#	#	#	#		C	F	M		#	#	.	#		F

Auto space.

View is active when LOW Airflow Alarm = Active

A	L	A	R	M	:	L	O	W		F	L	O	W		

View is active when HIGH Airflow Alarm = Active

A	L	A	R	M	:	H	I	G	H		F	L	O	W	

View is active when Fan Alarm = Active

A	L	A	R	M	:	F	A	N		#					

First fan to go into alarm.

View is active when TRBL = Active and LCD TRBL = ON

#	#		[	E	R	R	O	R		D	E	S	C	R	]

= Trouble Code.
Toggle through additional errors.

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3.4.1.1.2. Dual Location Transmitters

View is active when LCD NAME = ON (Default = OFF)

<	N	A	M	E	>									

View is always active

<	S	E	R	V	1	>								

View is always active

#	#	#	#	#		C	F	M		#	#	.	#		F

Auto space.

View is active when LOW Airflow Alarm = Active

A	L	A	R	M	:	L	O	W		F	L	O	W		

View is active when HIGH Airflow Alarm = Active

A	L	A	R	M	:	H	I	G	H		F	L	O	W	

View is always active

<	S	E	R	V	2	>								

View is always active

#	#	#	#	#		C	F	M		#	#	.	#		F

Auto space.

Note: There is no airflow alarm for location 2.

View is active when TRBL = Active and LCD TRBL = ON

#	#		[	E	R	R	O	R		D	E	S	C	R	]

= Trouble Code.
Toggle through additional errors.

3.4.1.2. DUAL LINE DISPLAY OPERATION

3.4.1.2.1. Transmitters without the /H Humidity Option

View is active when LCD NAME = ON (Default = OFF)

<	N	A	M	E	>									

View is always active

<	S	E	R	V	>									
#	#	#	#	#		C	F	M		#	#	.	#		F

Not visible when LCD SERV = OFF.
Auto space.

View is active when LOW or HIGH Airflow Alarm = Active

A	L	A	R	M	:	L	O	W		F	L	O	W		
A	L	A	R	M	:	H	I	G	H		F	L	O	W	

Not visible when low flow alarm is inactive.
Not visible when high flow alarm is inactive.

View is active when Fan Alarm = Active

A	L	A	R	M	:	F	A	N		#					

First fan to go into alarm.

View is active when TRBL = Active and LCD TRBL = ON

#	#		[	E	R	R	O	R		D	E	S	C	R	]
#	#		[	E	R	R	O	R		D	E	S	C	R	]

= First Trouble Code.
= Second Trouble Code, if applicable.
Toggle through additional errors.

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3.4.1.2.2. Transmitters with the /H Humidity Option

View is active when LCD NAME or LCD SERV = ON

<	N	A	M	E	>									Not visible when LCD NAME = OFF
<	S	E	R	V	>									Not visible when LCD SERV = OFF

View is always active when RHCONFIG=RH

#	#	#	#	#		C	F	M		#	#	.	#		F	Auto space.
										#	#	#	%	R	H	HCONFIG = RH

View is always active when RHCONFIG=DPT

#	#	#	#	#		C	F	M		#	#	.	#		F	Auto space.
#	#	#		F		D	P	T		#	#	#	%	R	H	HCONFIG = DPT

View is always active when RHCONFIG=ENTH

#	#	#	#	#		C	F	M		#	#	.	#		F	Auto space.
#	#	#		B	t	u	/	l	b	#	#	#	%	R	H	HCONFIG = ENTH

Alarm View: View is active when LOW or HIGH Airflow Alarm = Active

A	L	A	R	M	:	L	O	W		F	L	O	W			Not visible when low flow alarm is inactive.
A	L	A	R	M	:	H	I	G	H		F	L	O	W		Not visible when high flow alarm is inactive.

View is active when TRBL = Active and LCD TRBL = ON

#	#		[	E	R	R	O	R		D	E	S	C	R	]	## = First Trouble Code.
#	#		[	E	R	R	O	R		D	E	S	C	R	]	## = Second Trouble Code, if applicable.
																Toggle through additional errors.

3.4.2. LCD Display Mode [LCD DSPL]

Enables or disables LCD option during normal operation.

[Menu path: SETTINGS >↓ DISPLAY >↓ LCD DSPL]

Default: ON (normal display operation)
Optional Settings: OFF (displays EBTRON {transmitter model} only)

3.4.3. LCD Location Name Indication [LCD NAME]

Show or hide the location name, NAME, on the LCD.

[Menu path: SETTINGS >↓ DISPLAY >↓ LCD NAME]

Default: OFF (do not display the location name)
Optional Settings: ON (display the location name)

3.4.4. LCD Service Description Indication [LCD SERV]

Show or hide the service description, SERV, on the LCD.

[Menu path: SETTINGS >↓ DISPLAY >↓ LCD SERV]

3.4.4.1. SINGLE LOCATION ALL PROBE TYPES

Default: OFF (do not display the service location)
Optional Settings: ON (display the service location)

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3.4.4.2. DUAL LOCATION -U PROBE TYPE

Default: ON (display the service location)
Optional Settings: None.

3.4.5. LCD System Trouble Indication [LCD TRBL]

Show or hide active trouble codes and faults on the LCD.

[Menu path: SETTINGS >↓ DISPLAY >↓ LCD TRBL]

Default: ON (display faults and trouble codes)
Optional Settings: OFF (do not display faults and trouble codes)

3.4.6. LCD Airflow Units of Measure [LCD UM]

The airflow units of measure displayed on the LCD.

[Menu path: SETTINGS >↓ DISPLAY >↓ LCD UM]

Default and Optional Settings: See appropriate probe type.

ⓘ *The area parameter(s) [AREA] of -F, -U and -B probe types must be entered (i.e. not equal to the factory default value of {null}) for the LCD UM=CFM [L/s] option to be available. "ENTER AREA <OK>" will be displayed when CFM [L/s] is selected. Press ENT or ESC to return to the "*LCD UM=FPM [m/s]" menu prompt.*

ⓘ *The LCD units of measure is independent of analog and/or network variable units of measure.*

3.4.6.1. -P AND -T PROBE TYPES

Default: CFM [L/s]
Optional Settings: FPM [m/s]

3.4.6.2. -F AND -U PROBE TYPES

Default: FPM [m/s]
Optional Settings: CFM [L/s]

ⓘ *If the ARRAY WIZARD or AREA WIZARD has been run, the setting for LCD UM will be changed to CFM [L/s].*

3.4.6.3. -B PROBE TYPE

3.4.6.3.1. DP CONVERT = NO

Default: FPM [m/s]
Optional Settings: CFM [L/s]

3.4.6.3.2. DP CONVERT = YES

Default: iWG (inwg) [Pa (pascals)]
Optional Settings: None

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3.4.7. LCD Airflow Integration [LCD INTG]

The integration buffer size for the running average of the airflow used for output to the LCD.

[Menu path: SETTINGS >↓ DISPLAY >↓ LCD INTG]

Default: 100
Optional Settings: 1 to 999 (1 to 300 for -F/An probe types)

Transmitters calculate the airflow every 300 milliseconds. The integration buffer is used to smooth real-time airflow data for viewing on the LCD. The default integration of 100 is the running average (first in, first out) of the average of all sensor nodes over the preceding 30 second time period.



Increasing the integration buffer may stabilize the airflow reading indicated on the display in turbulent locations such as outdoor air intakes. The LCD integration buffer has no effect on the output signal or network airflow variable reading.

3.5. ANALOG OUTPUT SETTINGS

Analog output signals are provided with the following transmitter models:

- Advantage IV Product Line
  - Models with the third digit of the model code equal to A, B, C, F, M, S, or U
- *EB-Flow2* Product Line
  - Models with the first digit after EF- equal to A



Failure to properly configure the output signal will result in significant measurement error at the B.A.S.

The ANALOG OUTPUT submenu contains the following parameters:

- AOUT (Analog Output Signal Type and Range)
- ON FAIL (Analog Output on Total Sensor Failure)
- AO1 ASGN (AO1 Assignment)
- AO1 UM (AO1 Unit of Measure)
- AO1 MS (AO1 Minimum Scale)
- AO1 FS (AO1 Full Scale)
- AO2 ASGN (AO2 Assignment)

The following parameters are only available when AO2 ASGN is equal to TEMP, AF2, F1-2 or F2-1

- AO2 UM (AO2 Unit of Measure)
- AO2 MS (AO2 Minimum Scale)
- AO2 FS (AO2 Full Scale)

The following parameter is only available when AO2 ASGN is equal to ALRM, FA or TRBL

- NO FAULT (AO2 Output on Inactive Notification Alarm)

The following parameters are only available with Advantage IV GTx116e remote transmitters that have the humidity sensor option, /H, provided with the -P sensor probe type.

- AO3 ASGN (AO3 Assignment)
- AO3 UM (AO3 Unit of Measure)
- AO3 MS (AO3 Minimum Scale)
- AO3 FS (AO3 Full Scale)

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3.5.1. Analog Output Signals [AOUT]

The analog output type and range for each of the analog outputs.

[Menu path: SETTINGS >↓ ANALOG OUT > AOUT]

3.5.1.1. ADVANTAGE IV MODELS

Default: 4-20mA (current)
Optional Settings: 0-5V, 0-10V (voltage)

⚠ *Current output is "4-wire" type. Do not connect to a loop powered input that provides an excitation voltage.*

ⓘ *The output signal type of Advantage IV GTx108 and GTx116 transmitters (x=C and M) is selected using switches SW1 and SW2 on the output card installed on the main circuit board for analog outputs 1 and 2, respectively. The switch position must be in the VDC position for the optional 0-5V or 0-10V signals to be available. Parameters for AOUT1 and AOUT2 will be displayed in lieu of AOUT.*

ⓘ *Output jumpers, OUT1 (AO1) and OUT2 (AO2) must be configured for the proper output configuration, mA or VDC on Hybrid HTA104/HTA202 transmitters.*

ⓘ *Output signals are isolated from the power supply.*

3.5.1.2. *EB-FLOW2* MODELS

Default: 2-10V (voltage)
Optional Settings: 0-5V, 0-10V, 1-5V (voltage)

⚠ *Output signals are not isolated from the power supply. Provide an individual power transformer, secondary not grounded, if isolation is required.*

ⓘ *The 1-5 VDC and 2-10 VDC outputs can drive a 250 ohm and 500 ohm, respectively, 4-20 mA input circuit.*

3.5.2. Analog Output on Total System Failure [ON FAIL]

The analog output set if a total system failure is detected.

[Menu path: SETTINGS > ANALOG OUT >↓ ON FAIL]

Default: HI (set analog outputs to 100%)
Optional Settings: LO (set analog outputs to 0%)

ONFAIL=LO. The output of AO1 is set to 0%. The output of AO2, if assigned to temperature, is set to 0%. The output of AO3 on GTx116e transmitters is set to 0%.

ONFAIL=HI. The output of AO1 is set to 100%. The output of AO2, if assigned to temperature, is set to 100%. The output of AO3 on GTx116e transmitters is set to 100%.

3.5.3. AO1 Assignment [AO1 ASGN]

The object assigned to analog output AO1.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO1 ASGN]

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3.5.3.1. SINGLE LOCATION ALL PROBE TYPES

Default: AF (average airflow of all sensor nodes)
Optional Settings: None

3.5.3.2. DUAL LOCATION -U PROBE TYPE

Default: AF1 (average airflow of all sensor nodes connected to Location 1)
Optional Settings: F1-2 (Location - Location 2), F2-1 (Location 2 - Location 1)

Dual location transmitters with two connections use the leftmost connector when viewed from the display side as "Location 1".

Differential airflow is generally used in volumetric airflow units (i.e. ∆CFM [L/s]. Set the units of measure for AO1 UM and AO2 UM to CFM [L/s] to output the differential airflow as volumetric airflow.

3.5.4. AO1 Unit of Measure [AO1 UM]

The unit of measure of AO1.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO1 UM]

*The area parameter(s) [AREA] of -F, -U and -B probe types must be entered (i.e. not equal to the factory default value of {null}) for the LCD UM=CFM [L/s] option to be available. "ENTER AREA <OK>" will be displayed when CFM [L/s] is selected. Press ENT or ESC to return to the "*AO1 UM=FPM [L/s]" menu prompt.*

3.5.4.1. ALL PROBE TYPES EXCEPT -B

Default: FPM [m/s]
Optional Settings: CFM [L/s]

Use the default output unit of measure and scaling and avoid field modification. Multiply the default full-scale velocity reading by the area to determine the full-scale CFM [L/s] for the B.A.S.

3.5.4.2. -B PROBE TYPE

3.5.4.2.1. DP CONVERT = NO

Default: FPM [m/s]
Optional Settings: CFM [L/s]

3.5.4.2.2. DP CONVERT = YES

Default: iWG (inwg) [Pa (pascals)]
Optional Settings: None

3.5.5. AO1 Minimum Scale Reading [AO1 MS]

The airflow rate that results in an output signal of 0%.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO1 MS]

3.5.5.1. ALL PROBE TYPES EXCEPT THE -B PROBE TYPE

3.5.5.1.1. AO1 ASGN is set to AF (AF1 on dual location transmitters)

Default: 0
Optional Settings: None

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3.5.5.1.2. AO1 ASGN is set to F1-2 or F2-1 (Dual location transmitters only)

Default: -AO1 FS (negative value of AO1 FS)
Optional Settings: None

3.5.5.2. -B PROBE TYPE

3.5.5.2.1. DIRECTION = UNI

Default: 0
Optional Settings: None

3.5.5.2.2. DIRECTION = BI

Default: -AO1 FS (negative value of AO1 FS)
Optional Settings: None

3.5.6. AO1 Full Scale Reading [AO1 FS]

The airflow rate that results in an output signal of 100%.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO1 FS]

Default and Optional Settings: See appropriate probe type

ⓘ *The output signal is linear to airflow (or pressure on -B probe types configured for DPCONVERT=YES).*

3.5.6.1. -P PROBE TYPES

3.5.6.1.1. AO1 UM = FPM [m/s]

Default: 5000 (FPM) [25.0 (m/s)]
Optional Settings: 100 to 15000 (FPM) [0.5 to 75.0 (m/s)]

3.5.6.1.2. AO1 UM = CFM [L/s]

Default: {5000 x AREA} (CFM) [{25000 x AREA} (L/s)]
Optional Settings: 5 to 999999 (CFM) [5 to 999999 (L/s)]

3.5.6.2. -U AND -T PROBE TYPES

3.5.6.2.1. AO1 UM = FPM [m/s]

Default: 3000 (FPM) [15.0 (m/s)]
Optional Settings: 100 to 15000 (FPM) [0.5 to 75.0 (m/s)]

3.5.6.2.2. AO1 UM = CFM [L/s]

Default: {3000 x AREA} (CFM) [{15000 x AREA} (L/s)]
Optional Settings: 5 to 999999 (CFM) [5 to 999999 (L/s)]

3.5.6.3. -F PROBE TYPE

3.5.6.3.1. AO1 UM = FPM [m/s]

Default: 10000 (FPM) [50.0 (m/s)]
Optional Settings: 100 to 15000 (FPM) [0.5 to 75.0 (m/s)]

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3.5.6.3.2. AO1 UM = CFM [L/s]

Default: {10000 x AREA} (CFM) [{50000 x AREA} (L/s)]
Optional Settings: 5 to 999999 (CFM) [5 to 999999 (L/s)]

3.5.6.4. -B PROBE TYPE

3.5.6.4.1. AO1 UM = FPM [m/s]

Default: 3000 (FPM) [15.0 (m/s)]
Optional Settings: 100 to 15000 (FPM) [0.5 to 75.0 (m/s)]

3.5.6.4.2. AO1 UM = CFM [L/s]

Default: {3000 x AREA} (CFM) [{15000 x AREA} (L/s)]
Optional Settings: 5 to 999999 (CFM) [5 to 999999 (L/s)]

3.5.6.4.3. AO1 UM = inWG [Pa]

Default: 1.50 (iWG) [300 (Pa)]
Optional Settings: 0.05 to 5.00 (iWG) [10 to 1250 (Pa)]

3.5.6.5. ALL PROBE TYPES

ⓘ *Some B.A.S. application controllers use "span" rather than minimum- and full-scales. The span is equal to the full-scale – minimum scale and the offset is equal to the minimum scale.*

ⓘ *EBTRON airflow measurement accuracy is percent of reading. Decreasing the full-scale does not improve measurement accuracy. Today's application controllers generally have sufficient accuracy and input resolution to use the default output scaling of the transmitters.*

3.5.7. AO2 Assignment [AO2 ASGN]

The object assigned to analog output AO2.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO2 ASGN]

3.5.7.1. ALL -P, -T, -B PROBE TYPES AND SINGLE LOCATION -U

Default: TEMP (temperature)
Optional Settings: ALRM (airflow alarm) or TRBL (system trouble alarm)

ⓘ *See "Alarm Settings" for more information on configuring built-in notification alarms.*

3.5.7.2. -F PROBE TYPE

Default: TEMP (temperature)
Optional Settings: ALRM (airflow alarm), FA (fan alarm, FAN TYPE=ARRAY) or TRBL (system trouble alarm)

ⓘ *See "Alarm Settings" for more information on configuring built-in notification alarms.*

ⓘ *FAN TYPE is selected using the ARRAY WIZARD. See "Fan Alarm Settings" for more information on configuring the fan alarm.*

3.5.7.3. DUAL LOCATION -U PROBE TYPE

Default: AF2 (Location 2)
Optional Settings: F1-2 (Location 1 - Location 2), F2-1 (Location 2 - Location 1)

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Use the rightmost connector when viewed from the display side as "Location 2".

Differential airflow is generally used in volumetric airflow units (i.e. ΔCFM [L/s]). Set the units of measure for AO1 UM and AO2 UM to CFM [L/s] to output the differential airflow as volumetric airflow.

Note: Sections 3.5.8. to 3.5.10. only apply when AO2 ASGN=TEMP.

3.5.8. AO2 Unit of Measure [AO2 UM]

The temperature unit of measure of AO2.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO2 UM]

Default: F (°F) [C (°C)]
Optional Settings: None

3.5.9. AO2 Minimum Scale Reading [AO2 MS]

The temperature that results in an output signal of 0%.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO2 MS]

Default: -20 (°F) [-30 (°C)]
Optional Settings: -50 to 160 (°F) [-50 to 70 (°C)]

AO2 MS cannot exceed AO2 FS.

3.5.10. AO2 Full Scale Reading [AO2 FS]

The temperature that results in an output signal of 100%.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO2 FS]

Default: 160 (°F) [70 (°C)]
Optional Settings: -50 to 160 (°F) [-50 to 70 (°C)]

AO2 FS cannot be less than AO2 MS.

The output signal is linear to temperature.

Some B.A.S. application controllers use "span" rather than minimum- and full-scales. The span is equal to the full-scale – minimum scale and the offset is equal to the minimum scale.

EBTRON temperature measurement accuracy is a fixed value. Decreasing the full-scale does not improve measurement accuracy. Today's application controllers generally have sufficient accuracy and input resolution to use the default output scaling of the transmitters.

Note: Sections 3.5.11. to 3.5.13. only apply when AO2 ASGN=AF2, F1-2 or F2-1

3.5.11. AO2 Unit of Measure [AO2 UM]

The airflow unit of measure of AO2.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO2 UM]

Default: FPM [m/s]
Optional Settings: CFM [L/s]

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*The area parameter(s) [AREA] of -U probe type must be entered (i.e. not equal to the factory default value of {null}) for the LCD UM=CFM [L/s] option to be available. "ENTER AREA <OK>" will be displayed when CFM [L/s] is selected. Press ENT or ESC to return to the "*AO2 UM=FPM [L/s]" menu prompt.*

Use the default output unit of measure and scaling and avoid field modification. Multiply the default full-scale velocity reading by the area to determine the full-scale CFM [L/s] for the B.A.S.

3.5.12. AO2 Minimum Scale Reading [AO2 MS]

The airflow rate that results in an output signal of 0%.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO2 MS]

3.5.12.1. -U PROBE TYPE

3.5.12.1.1. AO2 ASGN is set to AF2

Default: 0
Optional Settings: None

3.5.12.1.2. AO2 ASGN is set to F1-2 or F2-1

Default: -AO2 FS (negative value of AO2 FS)
Optional Settings: None

3.5.13. AO2 Full Scale Reading [AO2 FS]

The airflow rate that results in an output signal of 100%.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO2 FS]

The output signal is linear to airflow.

3.5.13.1. -U PROBE TYPE

3.5.13.1.1. AO2 UM = FPM [m/s]

Default: 3000 (FPM) [15.0 (m/s)]
Optional Settings: 100 to 15000 (FPM) [0.5 to 75.0 (m/s)]

3.5.13.1.2. AO2 UM = CFM [L/s]

Default: {3000 x AREA} (CFM) [{15000 x AREA} (L/s)]
Optional Settings: 5 to 999999 (CFM) [5 to 999999 (L/s)]

Some B.A.S. application controllers use "span" rather than minimum- and full-scales. The span is equal to the full-scale – minimum scale and the offset is equal to the minimum scale.

EBTRON airflow measurement accuracy is percent of reading. Decreasing the full-scale does not improve measurement accuracy. Today's application controllers generally have sufficient accuracy and input resolution to use the default output scaling of the transmitters.

Note: Sections 3.5.14. only applies when AO2 ASGN=ALRM, FA or TRBL.

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3.5.14. AO2 Output on an Inactive Notification Alarm [NO FAULT]

The output of AO2 when the notification alarm status is inactive.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ NO FAULT]

Default: HI (output is set to 100%)
Optional Settings: LO (output is set to 0%)

Note: Sections 3.5.15. to 3.5.18. only apply to Advantage IV GTx116e remote transmitters that have the humidity sensor option, /H, provided with the -P sensor probe type.

3.5.15. AO3 Assignment [AO3 ASGN]

The object assigned to analog output AO3.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO3 ASGN]

Default: The current parameter setting for H CONFIG
Optional Settings: None

3.5.16. AO3 Unit of Measure [AO3 UM]

The unit of measure of AO3.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO3 UM]

The unit of measure is dependent on the RH sensor configuration.

3.5.16.1. H CONFIG=RH

Default: % RH
Optional Settings: None

3.5.16.2. H CONFIG=ENTH

Default: Btu/lb [kJ/kg]
Optional Settings: None

3.5.16.3. H CONFIG=DPT

Default: F (°F) [C (°C)]
Optional Settings: None

3.5.17. AO3 Minimum Scale Reading [AO3 MS]

The value that results in an output signal of 0%.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO3 MS]

The minimum scale reading is dependent on the RH sensor configuration.

3.5.17.1. H CONFIG=RH

Default: 0 (% RH)
Optional Settings: 0 to 100 (% RH)

3.5.17.2. H CONFIG=ENTH

Default: 0 (Btu/lb) [0 (kJ/kg)]
Optional Settings: -20 to 400 (Btu/lb) [-40 To 800 (kJ/kg)]

3.5.17.3. H CONFIG=DPT

Default: 0 (°F) [0 (°C)]
Optional Settings: -50 to 160 (°F) [-50 to 70 (°C)]

3.5.18. AO3 Full Scale Reading [AO3 FS]

The value that results in an output signal of 100%.

[Menu path: SETTINGS >↓ ANALOG OUT >↓ AO3 FS]

The full-scale reading is dependent on the RH sensor configuration.

3.5.18.1. H CONFIG=RH

Default: 100 (% RH)
Optional Settings: 0 to 100 (% RH)

3.5.18.2. H CONFIG=ENTH

Default: 200 (Btu/lb) [400 (kJ/kg)]
Optional Settings: -20 to 400 (Btu/lb) [-40 To 800 (kJ/kg)]

3.5.18.3. H CONFIG=DPT

Default: 100 (°F) [50 (°C)]
Optional Settings: -50 to 160 (°F) [-50 to 70 (°C)]

ⓘ *Some B.A.S. application controllers use "span" rather than minimum- and full-scales. The span is equal to the full-scale – minimum scale and the offset is equal to the minimum scale.*

ⓘ *Relative humidity, enthalpy and dew point measurement accuracy is percent of reading. Decreasing the full-scale does not improve measurement accuracy. Today's application controllers generally have sufficient accuracy and input resolution to use the default output scaling of the transmitter*

3.6. RS-485 NETWORK SETTINGS

One RS-485 network connection is provided with the following transmitter models:

- Advantage IV Product Line
  - Models with the third digit of the model code equal to B, C, or N
- *EB-Flow2* Product Line
  - Models with the first digit after EF- equal to N

The RS-485 submenu contains the following parameters:

- NETOUT (RS-485 Network Protocol)
- NETADDR (Network Address)
- NETBAUD (Baud Rate)
- PARITY (Parity)
- NETDI (Device Instance Number)
- RS485 COM (Enable RS-485 Communications)

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3.6.1. RS-485 Network Protocol [NETOUT]

The network protocol of the RS-485 connection.

[Menu path: SETTINGS >↓ RS485 > NETOUT]

Default: BACNET (BACnet MS/TP)
Optional Settings: MODBUS (Modbus RTU)

3.6.2. Network Address [NETADDR]

The network address of the transmitter.

[Menu path: SETTINGS >↓ RS485 >↓ NETADDR]

3.6.2.1. BACNET

Default: 2
Optional Settings:0 to 127

3.6.2.2. MODBUS

Default: 2
Optional Settings:0 to 247

3.6.3. Baud Rate [NETBAUD]

The baud rate of the network.

[Menu path: SETTINGS >↓ RS485 >↓ NETBAUD]

3.6.3.1. BACNET

Default: 76800 (bps)
Optional Settings: 38400, 19200, 9600

3.6.3.2. MODBUS

Default: 19200 (bps)
Optional Settings: 76800, 38400, 9600

3.6.4. Parity [PARITY]

3.6.4.1. BACNET

Not applicable.

3.6.4.2. MODBUS

Sets the parity bit for network data checking.

[Menu path: SETTINGS >↓ RS485 >↓ PARITY]

Default: EVEN
Optional Settings: ODD, NONE1 (one stop bit), NONE2 (two stop bits)

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3.6.5. Device Instance Number [NETDI]

3.6.5.1. BACNET

Sets the device instance number of the transmitter.

[Menu path: SETTINGS >↓ RS485 >↓ NETDI]

Default: 2
Optional Settings: 0 to 4194302

3.6.5.2. MODBUS

Not applicable.

3.6.6. Enable RS-485 Communications [RS485 COM]

Enable the transmitter for RS-485 network communications.

[Menu path: SETTINGS >↓ RS485 >↓ RS485 COM]

Default: OFF
Optional Settings: ON

Do not enable transmitter network communications until the transmitter network settings are properly configured to avoid addressing conflicts.

3.7. ETHERNET NETWORK SETTINGS

One Ethernet network connection is provided with the following transmitter models:

- Advantage IV Product Line
    - Models with the third digit of the model code equal to B, M, or S

The Ethernet network connection supports simultaneous communications below for B.A.S. connections.

- Modbus TCP is always enabled on port 502.
- BACnet IP, if enabled, is on port 47808.
- BACnet Ethernet may be enabled in lieu of BACnet IP.
- A simple web page showing airflow and temperature is available on port 80.

The Ethernet network connection supports *EB-Bus* Ethernet communication for connections with IAQ Enforcer® Smart Display Panel SDX-1000 or other EBTRON approved *EB-Bus* host.

EB-Bus Ethernet settings are configured by EB-Bus host.

GTM116e and GTM108e transmitters with output card firmware 4.20 and later is required for EB-Bus Ethernet.

The ETHERNET submenu contains the following parameters:

- EB-BUS CON (*EB-Bus* Connection Status)
- MODE (Mode)
- DHCP (Dynamic Host Configuration Protocol)
- IP (IP Address)
- MASK (Subnet Mask)
- GATE (Network Gateway)

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- DNS (Domain Name System)
- BACNET (BACnet Protocol)
- NETDI (Device Instance Number)
- UDPPORT (BACnet UDP Port)

3.7.1. EB-Bus Connection Status [EB-BUS CON]

EB-Bus Ethernet network connection status.

[Menu path: SETTINGS >↓ ETHERNET > EB-BUS CON]

Default: T
Optional Settings: F

EB-BUS CON will only be displayed if transmitter is connected to *EB-Bus* Ethernet network. If transmitter is removed from *EB-Bus* Ethernet network, set the parameter to F to disable any *EB-Bus* Ethernet network connection trouble codes.

3.7.2. Ethernet Network Connection Mode [MODE]

Ethernet network connection mode.

[Menu path: SETTINGS >↓ ETHERNET >↓ MODE]

Default: DUAL
Optional Settings: DAISY CHN

ⓘ *MODE is only available on GTS116e, GTS108e, HTS116 and HTS108 transmitters.*

3.7.3. Dynamic Host Configuration Protocol [DHCP]

Enable or disable DHCP.

[Menu path: SETTINGS >↓ ETHERNET >↓ DHCP]

Default: ON
Optional Settings: OFF

When DHCP is ON the IP address, subnet mask and network gateway are automatically set by the DHCP server and cannot be modified at the transmitter.

ⓘ *If GTS116e, GTS108e, HTS116 or HTS108 transmitter and configured for DUAL mode, parameters for DHCP1 and DHCP2 will be displayed in lieu of DHCP.*

3.7.4. IP Address [IP]

The IP address of the transmitter.

[Menu path: SETTINGS >↓ ETHERNET >↓ IP]

Default: 10.0.5.100
Optional Settings: Any valid IP address can be set by the network integrator.

The IP address is required to access Modbus TCP, BACnet IP or the web page. The IP address is not required for BACnet Ethernet. Use the ↑↓ pushbuttons to set the first octet value. Press ENT to move to the subsequent octet until all four octets are complete. Press ENT to save.

ⓘ *The IP address can only be changed when DHCP is OFF.*

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ⓘ *If GTS116e, GTS108e, HTS116 or HTS108 transmitter and configured for DUAL mode, parameters for IP1 and IP2 will be displayed in lieu of IP.*

3.7.5. Subnet Mask [MASK]

The subnet mask of the network.

[Menu path: SETTINGS >↓ ETHERNET >↓ MASK]

Default: 255.255.240.0
Optional Settings: Any valid subnet mask can be set by the network integrator.

The subnet mask is required to access Modbus TCP, BACnet IP or the web page. The subnet mask is not required for BACnet Ethernet. Use the ↑↓ pushbuttons to set the first octet value. Press ENT to move to the subsequent octet until all four octets are complete. Press ENT to save.

ⓘ *The subnet mask can only be changed when DHCP is OFF.*

ⓘ *If GTS116e, GTS108e, HTS116 or HTS108 transmitter and configured for DUAL mode, parameters for MASK1 and MASK2 will be displayed in lieu of MASK.*

3.7.6. Network Gateway [GATE]

The IP address of the network gateway.

[Menu path: SETTINGS >↓ ETHERNET >↓ GATE]

Default: 10.0.0.1
Optional Settings: Any valid IP address for the network gateway can be set by the network integrator.

The network gateway is required to access Modbus TCP, BACnet IP or the web page. The network gateway is not required for BACnet Ethernet. Use the ↑↓ pushbuttons to set the first octet value. Press ENT to move to the subsequent octet until all four octets are complete. Press ENT to save.

ⓘ *The network gateway IP address can only be changed when DHCP is OFF.*

ⓘ *If GTS116e, GTS108e, HTS116 or HTS108 transmitter and configured for DUAL mode, parameter for GATE2 will be displayed in lieu of GATE. There will be no GATE1 parameter displayed.*

3.7.7. Domain Name System [DNS]

The IP address of the network DNS server.

[Menu path: SETTINGS >↓ ETHERNET >↓ DNS]

Default: 10.0.0.8
Optional Settings: Any valid IP address for the DNS server can be set by the network integrator.

ⓘ *DNS parameter is disabled, and is intended for future use.*

ⓘ *DNS IP address can only be changed when DHCP is OFF.*

ⓘ *If GTS116e, GTS108e, HTS116 or HTS108 transmitter and configured for DUAL mode, parameter for DNS2 will be displayed in lieu of DNS. There will be no DNS1 parameter displayed.*

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3.7.8. BACnet Protocol [BACNET]

3.7.8.1. BACNET

Sets the BACnet protocol if BACnet IP or BACnet Ethernet is required.

[Menu path: SETTINGS >↓ ETHERNET >↓ BACNET]

Default: IP (BACnet IP)
Optional Settings: ETH (BACnet Ethernet)

3.7.8.2. MODBUS

Not applicable.

ⓘ *If GTS116e, GTS108e, HTS116 or HTS108 transmitter and configured for DUAL mode, parameters for BACNET1 and BACNET2 will be displayed in lieu of BACNET.*

3.7.9. Device Instance Number [NETDI]

3.7.9.1. BACNET

Sets the device instance number if BACnet IP or BACnet Ethernet is used.

[Menu path: SETTINGS >↓ ETHERNET > ↓NETDI]

Default: The last octet of the MAC address

ⓘ *The default is 2 for GTM116e or GTM108e output card firmware prior to 4.17.*

Optional Settings: 0 to 4194302

3.7.9.2. MODBUS

Not applicable.

ⓘ *If GTS116e, GTS108e, HTS116 or HTS108 transmitter and configured for DUAL mode, parameters for NETDI1 and NETDI2 will be displayed in lieu of NETDI.*

3.7.10. BACnet UDP Port [UDPPORT]

3.7.10.1. BACNET

Sets the UDP port if BACnet IP is used.

[Menu path: SETTINGS >↓ ETHERNET > ↓UDPPORT]

Default: 47808 (0xBAC0)
Optional Settings: 0 to 65535

The port is used to distinguish the messages to the BACnet service running on the user datagram protocol (UDP). This value should match other BACnet IP devices to allow communication and in most cases this will not need to be modified.

3.7.10.2. MODBUS

Not applicable.

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If GTS116e, HTS116, GTS108e or HTS108 transmitter and configured for DUAL mode, parameters for UDPPORT1 and UDPPORT2 will be displayed in lieu of UDPPORT.

3.8. LON NETWORK SETTINGS

One Lonworks free topology connection is provided with the following transmitter models:

- Advantage IV Product Line
  - Models with the third digit of the model code equal to F or L

There are no transmitter settings for Lon.

3.9. USB SETTINGS

One USB port for connection to a Type A USB memory device "thumb drive" (by others) is provided on the following models:

- Advantage IV Product Line
  - Models with the third digit of the model code equal to D or U
- Transmitters with a USB data logger log sensor node data at 5-minute intervals using Universal Time Coordinated (UTC) based on an onboard real-time clock, whenever power is applied to the transmitter. Data files are automatically appended on power-up.
  - The comma separated values (.CSV) filenames and file details are as follows:
    - DATAF.CSV - Average airflow and individual sensor node airflow
      - Format: Date Time, fw rev, Type, FPM, Sen 1,…,Sen 16
    - DATAT.CSV - Average temperature and individual sensor node temperature
      - Format: Date Time, fw rev, Type, deg F, Sen 1,…,Sen 16
  - If the /H humidity sensor option is provided, the following file is also provided:
    - DATARH.CSV - Relative humidity, enthalpy and dew point
      - Format: Date Time, fw rev, Type, %RH, DewPt, Enth

Date time format is MM/DD/YYYY hh:mm. Firmware revision is the firmware revision of the option card. Type refers to sensor probe type connected and can be "PROBE", "FAN", "FAN ARRAY", "BLEED", "NONE". Header on the DATAF.CSV and DATAT.CSV files will always show sensors 1-16, but actual data populated is based on number of sensors connected to the transmitter.

For standard transmitters equipped with EB-Link Bluetooth® low energy interface, the time zone and interval can be modified by the user using EB-Link software.

3.9.1. Enable or Disable the USB Port [USB WRITE]

Enable or disable the write capability on the USB port.

[Menu path: SETTINGS >↓ USB > USB WRITE]

3.9.1.1. INSERTING A USB MEMORY DEVICE ("THUMB DRIVE")

Install the USB memory device into the USB connector on the option card to start logging data.

It is a good practice to set the transmitter power switch to the "OFF" position before inserting the thumb drive memory device.

The USB port must be enabled to log data. Transmitters are shipped with the USB port parameter USB WRITE set to "ON", which enables the port.

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If the transmitter had been previously used to log data, the USB WRITE parameter would most likely have been set to “OFF”. To enable the USB WRITE parameter and start logging data, press the ↑↓ arrow buttons simultaneously to enter the MAIN MENU. The SETTINGS menu is displayed. Press the ENT button to select the top of the SETTINGS submenu category. Press the ↓ button until the USB submenu category is visible. Press the ENT button again to enter the USB submenu. Set the USB WRITE parameter to “ON”.

3.9.1.2. REMOVING A USB MEMORY DEVICE

Remove the USB memory device from the USB connector on the option card after data logging is complete and the USB port has been disabled by setting the USB WRITE parameter to “OFF”.

Disable the USB port to stop logging data. Press the ↑↓ arrow buttons simultaneously to enter the MAIN MENU. The SETTINGS menu is displayed. Press the ENT button to select the top of the SETTINGS submenu category. Press the ↓ button until the USB submenu category is visible. Press the ENT button again to enter the USB submenu. Set the USB WRITE parameter to “OFF”.



Always set the USB WRITE parameter to “OFF” before removing the USB memory device to avoid data loss/and or damage.

Remove the USB memory device.



It is a good practice to set the transmitter power switch to the “OFF” position before removing the USB memory device.

3.10. EB-LINK SETTINGS

The EB-Link is only available and provided with Advantage IV - Gold Series and Hybrid Series models HTx116 and HTx108 transmitters.



The EB-Link Reader application for smart phone or tablet does not support the -B probe type.

The EB-Link Reader application is available for either smart phone or tablet with Android or iOS operating system or comes standard with IAQ Enforcer® Smart Display Panel SDX-1000. For smart phone or tablet application, go to the appropriate application store for the desired operating system, search for EBTRON, and download the application. The EB-Link Bluetooth® Low Energy interface card is provided with all Advantage IV - Gold Series and Hybrid Series models HTx116 and HTx108 transmitters. The EB-Link Reader application is provided free of charge.

The EB-Link interface uses Bluetooth® Low Energy for smart phone or tablet application or *EB-Bus* Ethernet for the IAQ Enforcer Smart Display Panel SDX-1000 to pass information. The application provides real-time viewing of the average airflow and temperature measured by the transmitter. If the transmitter is provided with the /H humidity sensor option, humidity, dewpoint, and enthalpy is also provided. The reader also provides individual sensor node airflow and temperature data for near instantaneous traverses of duct velocities and temperatures. Individual fan airflow rates are available when the FAN TYPE is set to ARRAY using the ARRAY WIZARD. In addition, the reader provides a complete diagnostic text file output that can be viewed and saved by the user and/or e-mailed to EBTRON customer service for analysis.

The EB-Link Reader application for the smart phone or tablet opens to the “SCAN FOR TRANSMITTERS” screen and displays all nearby transmitters that are detected. Transmitters are continuously scanned while the application is on this screen. Select a transmitter by name. The name is determined by the NAME parameter.



The NAME parameter defaults to the transmitter serial number unless a location name was provided with the order or the NAME parameter was modified by the user in the field.

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Once a transmitter is selected and a connection is made, the application moves to the live data screen that shows the real-time average volumetric airflow rate, average velocity, area, velocity weighted temperature and average temperature. Select the menu button (☰) to navigate to other functions of the application, such as traverse data and diagnostics. Traverse and diagnostic data can be saved and e-mailed directly from the application.

In most installations, the Bluetooth signal can be read up to 20 feet from the transmitter with the transmitter cover installed (line of sight, no obstructions).

Some mechanical environments have excessive extraneous radio signals. Moving your position, or even hand position, in some environments may improve connectivity if difficulty in locating transmitters is experienced.

The EB-LINK submenu contains the following parameters:

- EB-LK INTG (EB-Link Integration)
- EB-LK STAT (EB-Link Status)

3.10.1. EB-Link Integration [EB-LK INTG]

The integration buffer size for the running average of the average and individual sensor node airflow rates output to the EB-Link interface card and network connection.

[Menu path: SETTINGS >↓ EB-LINK >↓ EB-LK INTG]

Default: 100
Optional Settings: 1 to 999 (1 to 300 for -F/An probe types)

Transmitters calculate the airflow every 300 milliseconds. The integration buffer is used to smooth real-time airflow data output to the EB-Link Reader. The default integration of 100 is the running average (first in, first out) of the average of all sensor nodes over the preceding 30 second time period.

Be certain the HVAC system is not changing faster than the integration buffer is set to ensure accurate measurement data.

Note: Section 3.10.2. only applies to GTx116e, GTx108e, HTx116 and HTx108 transmitters transmitters. The Bluetooth® Low Energy transceiver of all other Gold Series transmitters is disabled by physically removing the transceiver card from the main circuit board.

3.10.2. EB-Link Status [EB-LK STAT]

Enables/disables the Bluetooth® Low Energy transceiver in the transmitter.

[Menu path: SETTINGS >↓ EB-LINK >↓ EB-LK STAT]

Default: ON
Optional Settings: OFF

3.11. AIRFLOW ALARM SETTINGS

A high/low airflow notification alarm is provided with all models covered in this manual. An “active” alarm is displayed on the LCD and can be assigned to analog output AO2 or the contact closure relay (relay only available with EF-x2000 transmitters).

The airflow alarm is only applicable to location 1 on dual location transmitters.

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ⓘ *The -B probe type high/low airflow alarm is configured as a high/low pressure alarm when DP CONVERT=YES.*

To assign the alarm to analog output, AO2, see “AO2 Assignment”. AO2 becomes a binary high/low output with the inactive state based on the setting for NO FAULT.

To assign the alarm to relay, R1 (EF-x2000 only), see “R1 Assignment”.

ⓘ *If the alarm is assigned to AO2 the default output on an inactive alarm, NO FAULT, is set to HI (100% output, or positive notification, on an inactive alarm). NO FAULT can be set to LO (0% output) on an inactive alarm, if desired. See “AO2 Output on an Inactive Notification Alarm” for more information.*

The ALARM submenu contains the following parameters:

- LO ALRM (Low Alarm Enable)
- HI ALRM (High Alarm Enable)
- ALRM UM (Alarm Units of Measure)
- SETPOINT (Alarm Setpoint)
- TOL (Alarm Tolerance)
- DELAY (Alarm Delay)
- ZERO OFF (Alarm on Zero Flow)
- RESET (Alarm Reset Method)

3.11.1. Low Alarm Enable [LO ALRM]

Enable the low airflow alarm.

[Menu path: SETTINGS >↓ ALARM >↓ LO ALRM]

Default: OFF
Optional Settings: ON

An enabled low airflow alarm results in an “active” alarm state when the average airflow rate measured by the transmitter falls below the alarm setpoint less tolerance (alarm state = “outside”) for the entire delay period specified. The alarm can be configured to reset automatically or require manual reset when the alarm state returns to “normal”. The alarm can be disabled when the airflow rate output is 0% (either measured or forced to 0% by the low limit cutoff) to disable alarming when the system is in “off” or “unoccupied” modes.

3.11.2. High Alarm Enable [HI ALRM]

Enable the high airflow alarm.

[Menu path: SETTINGS >↓ ALARM >↓ HI ALRM]

Default: OFF
Optional Settings: ON

An enabled high airflow alarm results in an “active” alarm state when the average airflow rate measured by the transmitter exceeds the alarm setpoint plus tolerance (alarm state = “outside”) for the entire delay period specified. The alarm can be configured to reset automatically or require manual reset when the alarm state returns to “normal”. The alarm can be disabled when the airflow rate output is 0% (either measured or forced to 0% by the low limit cutoff) to disable alarming when the system is in “off” or “unoccupied” modes.

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3.11.3. Alarm Units of Measure [ALRM UM]

3.11.3.1. ALL PROBE TYPES EXCEPT THE -B PROBE TYPE CONFIGURED FOR PRESSURE (DPCONVERT=YES)

The airflow units of measure used to activate the airflow alarm.

[Menu path: SETTINGS >↓ ALARM >↓ ALRM UM]

Default: FPM [m/s]
Optional Settings: CFM [L/s]

ⓘ *The area parameter(s) [AREA] of -F and -U and -B probe types must be entered (i.e. not equal to the factory default value of {null}) for the LCD UM=CFM [L/s] option to be available. "ENTER AREA <OK>" will be displayed when CFM [L/s] is selected. Press ENT or ESC to return to the "*ALRM UM=FPM [L/s]" menu prompt.*

3.11.3.2. -B PROBE TYPE CONFIGURED FOR PRESSURE (DPCONVERT=YES)

The pressure units of measure used to activate the pressure alarm.

[Menu path: SETTINGS >↓ ALARM >↓ ALRM UM]

Default: iWG [Pa]
Optional Settings: None

3.11.4. Alarm Setpoint [SETPNT]

The setpoint used to activate the airflow alarm.

[Menu path: SETTINGS >↓ ALARM >↓ SETPOINT]

3.11.4.1. ALL PROBE TYPES EXCEPT THE -B PROBE TYPE

3.11.4.1.1. ALRM UM = FPM [m/s]

Default: 0
Optional Settings: 0 to 999999 (FPM) [0.00 to 99.9 9 (m/s)]

3.11.4.1.2. ALRM UM = CFM [L/s]

Default: 0
Optional Settings: 0 to 999999 (CFM) [0.00 to 999999 (L/s)]

3.11.4.2. -B PROBE TYPE

ⓘ *The optional settings range is ± if DIRECTION=BI.*

3.11.4.2.1. ALRM UM = FPM [m/s]

Default: 0
Optional Settings: 0 to 999999 (FPM) [0.00 to 99.99 (m/s)]

3.11.4.2.2. ALRM UM = CFM [L/s]

Default: 0
Optional Settings: 0 to 999999 (CFM [L/s])

3.11.4.2.3. ALRM UM = iWG [Pa]

Default: 0

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Optional Settings: 0.00 to 5.00 (iWG) [0 to 1250 (Pa)]

3.11.5. Alarm Tolerance [TOL]

The airflow tolerance, as a percentage of the setpoint, used to activate the airflow alarm.

[Menu path: SETTINGS >↓ ALARM >↓ TOL]

Default: Greater of 15% or 20 FPM [0.10 m/s]
Optional Settings: Greater of 5% or 20 FPM [0.10 m/s] to 100% or 20 FPM [0.10 m/s]

20 FPM [0.10 m/s] is less than 0.0001 iWG [0.025 Pa] when a -B probe type is configured for pressure and the setpoint is set to 0.

3.11.6. Alarm Delay [DELAY]

The delay, in minutes, that the alarm state must be “outside”, to activate the airflow alarm.

[Menu path: SETTINGS >↓ ALARM >↓ DELAY]

Default: 2min (minutes)
Optional Settings: 0min to 15min

3.11.7. Alarm on Zero Flow [ZERO OFF]

Disable the airflow alarm when the airflow output of the transmitter is set to 0%.

[Menu path: SETTINGS >↓ ALARM >↓ ZERO OFF]

Default: YES
Optional Settings: NO

Use the low limit cutoff (LLIMIT) to improve functionality of the airflow alarm in systems that have “off or “unoccupied” modes.

3.11.8. Alarm Reset Method [RESET]

Automatically reset (clear) the alarm when the fault is inactive or require manual reset.

[Menu path: SETTINGS >↓ ALARM >↓ RESET]

Default: AUTO
Optional Settings: MANUAL

RESET = AUTO. The alarm is automatically reset when the alarm state returns to “normal” or the alarm is manually reset.

RESET = MANUAL. The alarm must be manually reset by pressing the ESC button.

An active alarm can be reset manually by pressing the ESC pushbutton. The alarm state will return to “normal”. If the airflow is outside of the specified setpoint and tolerance, the alarm state returns to “outside” and the delay timer is reset.

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3.12. FAN AIRFLOW ALARM

The fan alarm is available on Advantage IV transmitters when the FAN TYPE is set to ARRAY using the ARRAY WIZARD. The fan alarm can be used to detect a malfunctioning fan in an array, provide notification of the malfunction and/or ignore the airflow measured from the fan. The latter being particularly useful if a fan has failed without a backdraft damper and there is still indication of airflow.

An “active” alarm is displayed on the LCD for the fan numbers in a fault condition and can be assigned to analog output AO2 as a multi-state analog output based on the first fan number to indicate an active fault.

A probe number is labeled on each probe hang tag (Probe x of y). The leftmost receptacle used on the transmitter when viewed from the display side is always designated as *Connector 1 (C1)*. Additional connectors increment sequentially from left to right on transmitters with more than one receptacle. Fan array models with two rows of 4 receptacles, increment left to right, *C1* to *C4* on the topmost row closest to the cover, then left to right, *C5* to *C8* on the lower most row.

Probe numbers (P1 to Py) are dynamically assigned by the transmitter after initial power up, left to right, based on the connector receptacle used.

⚠ *Fan array models (-F/An) must follow this convention if two probes are located in each fan inlet (i.e. put probes for fan 1 in C1 and C2, fan 2 in C3 and C4, etc.)*

To assign the alarm to analog output, AO2, see “AO2 Assignment”. AO2 becomes a multi-state analog output with the inactive state based on the setting for NO FAULT.

An active alarm will set AO2 in increments of 10% of the full-scale output based on the first fan number to indicate an active fault (ex. Fan 2 fault will output 20% of the full-scale).

The FAN ALARM submenu contains the following parameters:

- FAN ALRM (Fan Airflow Alarm Type)
- ON ALRM (Action on Active Fan Alarm)
- ALRM UM (Fan Alarm Units of Measure)
- SETPNT (Fan Alarm Setpoint)
- DELAY (Fan Alarm Delay)
- ZERO OFF (Fan Alarm on Zero Flow)
- RESET (Fan Alarm Reset Method)

There are three alarm types, MIN, DEV and MAX, available that target specific failures with the array.

3.12.1. Fan Airflow Alarm Type [FAN ALRM]

Enable and select one of the fan airflow alarms.

[Menu path: SETTINGS >↓ FAN ALARM > FAN ALRM]

Default: OFF
Optional Settings: DEV, MIN, %MAX

⚠ *Care should be taken when selecting the appropriate fan alarm. Fans that have completely failed in an array that do not have backdraft or isolation dampers may have significant reverse airflow through the fan. The EBTRON -F probe will read approximately a false high reading when airflow across the sensor is in the reverse direction.*

FAN ALRM = DEV. The alarm becomes active when one fan in the array is deviating by a user specified percentage, high or low, from the other fans.

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The DEV fan alarm is generally the most useful method for detecting fan failures when all fans are the same size and operating at the same speed.

FAN ALRM = MIN. The alarm becomes active when the airflow rate falls below a user defined minimum airflow rate.

FAN ALRM = %MAX. The alarm becomes active when the airflow rate falls below a user defined minimum airflow percentage of the maximum airflow rate stored by the transmitter for each fan in the array.

Set the fan speed to maximum during setup to store the highest airflow rate for the %MAX fan alarm.

3.12.2. Action of Active Fan Alarm [ON ALRM]

Enable and select one of the fan airflow alarms.

[Menu path: SETTINGS >↓ FAN ALARM >↓ ON ALRM]

Default: KEEP
Optional Settings: IGNR

ON ALRM = KEEP. Use the airflow of the fan in active alarm to determine the total airflow rate for the fan array.

ON ALRM = IGNR. Ignore (don't use) the airflow of the fan in active alarm to determine the total airflow rate for the fan array.

Note: Section 3.12.3. only applies when FA TYPE=MIN.

3.12.3. Fan Alarm Units of Measure [ALRM UM]

The airflow units of measure used to activate the fan airflow alarm.

[Menu path: SETTINGS >↓ FAN ALARM >↓ ALRM UM]

Default: FPM [m/s]
Optional Settings: CFM [L/s]

*The area parameter(s) [AREA] of -F probe types must be entered (i.e. not equal to the factory default value of {null}) for the LCD UM=CFM [L/s] option to be available. "ENTER AREA <OK>" will be displayed when CFM [L/s] is selected. Press ENT or ESC to return to the "*ALRM UM=FPM [L/s]" menu prompt.*

3.12.4. Fan Alarm Setpoint [SETPNT]

3.12.4.1. DEV AND %MAX ALARMS

The setpoint, as a percentage of airflow, used to activate the fan airflow alarm.

[Menu path: SETTINGS >↓ FAN ALARM >↓ SETPOINT]

Default: 25%
Optional Settings: 10% to 100%

3.12.4.2. MIN ALARM

The airflow setpoint, in the specified units of measure, used to activate the fan airflow alarm.

[Menu path: SETTINGS >↓ FAN ALARM >↓ SETPOINT]

3.12.4.2.1. ALRM UM = FPM [m/s]

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Default: 0
Optional Settings: 0 to 999999 (FPM) [0.00 to 99.99 (m/s)]

3.12.4.2.2. ALRM UM = CFM [L/s]

Default: 0
Optional Settings: 0 to 999999 (CFM [L/s])

3.12.5. Fan Alarm Delay [DELAY]

The delay, in minutes, that the alarm state must be “outside”, to activate the fan airflow alarm.

[Menu path: SETTINGS >↓ FAN ALARM >↓ DELAY]

Default: 2min (minutes)
Optional Settings: 0min to 15min

3.12.6. Fan Alarm on Zero Flow [ZERO OFF]

Disable the fan airflow alarm when the airflow output of the transmitter is set to 0%.

[Menu path: SETTINGS >↓ FAN ALARM >↓ ZERO OFF]

Default: YES
Optional Settings: NO

Use the low limit cutoff (LLIMIT) to improve functionality of the airflow alarm in systems that have “off or “unoccupied” modes.

3.12.7. Fan Alarm Reset Method [RESET]

Automatically reset (clear) the alarm when the fault is inactive or require manual reset.

[Menu path: SETTINGS >↓ FAN ALARM >↓ RESET]

Default: AUTO
Optional Settings: MANUAL

RESET = AUTO. The alarm is automatically reset when the alarm state returns to “normal” or the alarm is manually reset.

RESET = MANUAL. The alarm must be manually reset by pressing the ESC button.

An active alarm can be reset manually by pressing the ESC pushbutton. The alarm state will return to “normal”. If the airflow is outside of the specified setpoint and tolerance, the alarm state returns to “outside” and the delay timer is reset.

3.13. RELAY

Contact closure relay settings only apply to *EB-Flow2* transmitters.

The RELAY submenu contains the following parameters:

- R1 ASGN (Relay Assignment)
- R1 STATUS (Relay Normal Status)

3.13.1. Relay Assignment [R1 ASGN]

The object assigned to relay R1.

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[Menu path: SETTINGS >↓ RELAY > R1 ASGN]

Default: NONE
Optional Settings: ALRM (airflow alarm) or TRBL (system trouble alarm)

ⓘ *See "Alarm Settings", for more information on configuring built-in notification alarms.*

3.13.2. Relay Normal Status [R1 STATUS]

The normal, inactive, status of relay R1.

[Menu path: SETTINGS >↓ RELAY >↓ R1 STATUS]

Default: NO (normally open, N.O.)
Optional Settings: NC (normally closed, N.C.)

3.14. ADJUSTMENTS

Transmitters can be adjusted in the field when conditions warrant. Adjustments should only be made when there is reasonable certainty that the reference measurements are more accurate than the EBTRON measuring device.

ⓘ *If minimum placement guidelines cannot be achieved, installed accuracy may be compromised. Transmitters can be field adjusted to match a third-party measurement. Adjusted field measurements typically result in comparative readings within ±3% of the third-party measurement. Be advised that the third-party measurement may have uncertainties greater than or equal to ±10% and should only be used to adjust the airflow measurement device if the probes do not meet minimum placement requirements and the discrepancy is greater than the uncertainty of the third-party source.*

The ADJUSTMENTS submenu contains the following parameters:

- FLOW ADJ (Airflow Adjustment Enable/Disable)
- F_ADJ UM (Airflow Adjustment Units of Measure)
- PRESS ADJ (Pressure Adjustment Enable/Disable)
- FAW SMPLS (Flow Adjust Wizard Samples)
- GAIN (Airflow or Pressure Gain Factor)
- OFF (Airflow or Pressure Offset Factor)

3.14.1. Airflow Adjustment Enable/Disable [FLOW ADJ]

Enable or disable third-party adjustments made to modify the factory airflow measurement of the transmitter.

[Menu path: SETTINGS >↓ ADJUSTMENTS > FLOW ADJ]

Default: OFF
Optional Setting: ON

ⓘ *If minimum placement guidelines cannot be achieved, installed accuracy may be compromised. Transmitters can be field adjusted to match a third-party measurement. Adjusted field measurements typically result in comparative readings within ±3% of the third-party measurement. Be advised that the third-party measurement may have uncertainties greater than or equal to ±10% and should only be used to adjust the airflow measurement device if the probes do not meet minimum placement requirements and the discrepancy is greater than the uncertainty of the third-party source.*

ⓘ *Airflow adjustments are not available for -B probe types when DP CONVERT is set to YES.*

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⚠ *Enabling user entered offset and gain values overrides EBTRON's NIST traceable airflow calibration.*

3.14.2. Airflow Adjustment Units of Measure [F_ADJ UM]

3.14.2.1. ALL PROBE TYPES EXCEPT THE -B PROBE TYPE CONFIGURED FOR PRESSURE (DPCONVERT=YES)

The airflow units of measure of the offset parameter(s) adjustment coefficient(s).

[Menu path: SETTINGS >↓ ADJUSTMENTS >↓ F_ADJ UM]

Default: CFM [L/s]
Optional Settings: FPM [m/s]

ⓘ *The area parameter(s) [AREA] of -F, -U and -B probe types must be entered (i.e. not equal to the factory default value of {null}) for the F_ADJ UM=CFM [L/s] default to be available. The optional setting of FPM [m/s] is the default value when the area parameter is not entered. "ENTER AREA <OK>" will be displayed when CFM [L/s] is selected. Press ENT or ESC to return to the "*F_ADJ UM=FPM [L/s]" menu prompt.*

ⓘ *The OFFSET adjustment is modified for CFM [L/s] or FPM [m/s] using the appropriate area parameter(s) if the F_ADJ UM is changed after the coefficient(s) for the offset parameter(s) is entered.*

3.14.2.2. -B PROBE TYPE CONFIGURED FOR PRESSURE (DPCONVERT=YES)

Not applicable.

3.14.3. Pressure Adjustment Enable/Disable [PRESS ADJ]

Pressure adjustment only applies to the -B probe type.

3.14.3.1. DPCONVERT = NO

Not applicable.

3.14.3.2. DPCONVERT = YES

Enable or disable adjustments made to modify the factory pressure measurement of the transmitter.

[Menu path: SETTINGS >↓ ADJUSTMENTS >↓ PRESS ADJ]

Default: OFF
Optional Setting: ON

ⓘ *Pressure adjustments will not be converted to airflow adjustments if DP CONVERT is changed from ON to OFF. Pressure adjustments will be saved but not visible when DP CONVERT is set to OFF.*

Adjusting the -B probe type to a known pressure reference typically improves -B probe type pressure estimation.

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3.14.4. FAW Samples [FAW SMPLS]

The sample size used to determine the average, uncorrected, total airflow rate of the transmitter to calculate GAIN and OFFSET coefficients when the FLOW ADJUST WIZARD (FAW) tool is used (Section 4).

[Menu path: SETTINGS >↓ ADJUSTMENTS >↓ FAW SMPLS]

Default: 100
Optional Settings: 1 to 999

Transmitters calculate the airflow every 300 milliseconds. The default integration of 100 calculates the average of all sensor nodes over the preceding 30 second time period.

⚠ *Be certain the HVAC system is not changing faster than the integration buffer is set to ensure accurate measurement data.*

3.14.5. Airflow or Pressure Gain Adjustment [GAIN]

The third-party gain coefficient used to modify the airflow or pressure measurement of the transmitter.

[Menu path: SETTINGS >↓ ADJUSTMENTS >↓ GAIN]

Default: 1.000
Optional Setting: 0.100 to 9.999

3.14.5.1. ALL PROBE TYPES EXCEPT -B CONFIGURED FOR DPCONVERT = YES

The gain adjustment modifies the airflow output by multiplying the unadjusted average airflow by the gain coefficient. The GAIN coefficient can be entered manually or automatically using the FLOW ADJUST WIZARD (FAW) when a one- or two-point adjustment is desired.

ⓘ *If the transmitter is configured for two locations, parameters for GAIN1 and GAIN2 will be displayed in lieu of GAIN.*

Use a gain factor rather than an offset factor when a single point adjustment is used.

3.14.5.2. -B PROBE TYPE CONFIGURED FOR DPCONVERT = YES

The gain adjustment modifies the pressure output by multiplying the unadjusted average pressure by the gain coefficient. The GAIN coefficient is entered manually.

3.14.6. Airflow or Pressure Offset Adjustment [OFF]

The third-party offset coefficient used to modify the airflow or pressure measurement of the transmitter.

[Menu path: SETTINGS >↓ ADJUSTMENTS >↓ OFF]

3.14.6.1. ALL PROBE TYPES EXCEPT -B CONFIGURED FOR DPCONVERT = YES

The offset adjustment modifies the airflow output by adding or subtracting the unadjusted average airflow by the offset coefficient. The OFFSET coefficient can be entered manually or automatically using the FLOW ADUST WIZARD (FAW) when a two-point adjustment is desired.

ⓘ *If the transmitter is configured for two locations, parameters for OFF1 and OFF2 will be displayed in lieu of OFF.*

⚠ Offset values must be entered in the units of F_ADJ UM.

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3.14.6.2. ALL PROBE TYPES EXCEPT THE -B PROBE TYPE CONFIGURED FOR PRESSURE (DPCONVERT=YES)

3.14.6.2.1. F_ADJ_UM = FPM [m/s]

Default: 0
Optional Settings: -99999 to 99999 (FPM) [-99.99 to 99.99 (m/s)]

3.14.6.2.2. F_ADJ UM = CFM [L/s]

Default: 0
Optional Settings: -99999 to 99999 (CFM [L/s])

3.14.6.3. -B PROBE TYPE CONFIGURED FOR PRESSURE (DPCONVERT=YES)

Default: 0
Optional Setting: -5.000 to 5.000 (iWG) [-1250 to 1250 (Pa)]

3.14.7. Temperature Adjustment

Refer to Section 5 if temperature adjustments are required.

ⓘ *The temperature output of the EBTRON device is velocity-weighted and more accurate than the simple arithmetic average obtained by a field temperature measurement or traverse. A field temperature measurement will not match the velocity-weighted output shown on the display. The arithmetic average of the sensor nodes can be viewed through the DIAGNOSTICS Menu.*

⚠ *Making an adjustment to the temperature overrides EBTRON's NIST traceable temperature calibration.*

Note: Section 3.14.8. only applies to Advantage IV GTx116e remote transmitters that have the humidity sensor option, /H, provided with the -P sensor probe type.

3.14.8. Relative Humidity Adjustment

Refer to Section 5 if relative humidity adjustments are required.

ⓘ *The relative humidity output of the EBTRON device is velocity-weighted and more accurate than the simple arithmetic average obtained by a field relative humidity measurement or traverse. A field relative humidity measurement will not match the velocity-weighted output shown on the display. The arithmetic average of the relative humidity sensor(s) can be viewed through the DIAGNOSTICS Menu.*

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4. SECTION OVERVIEW

This section discusses the built-in tools provided with each transmitter to facilitate setup, startup and verification.

4.1. TOOLS MENU ACCESS AND NAVIGATION

The TOOLS menu contains the following submenu categories:

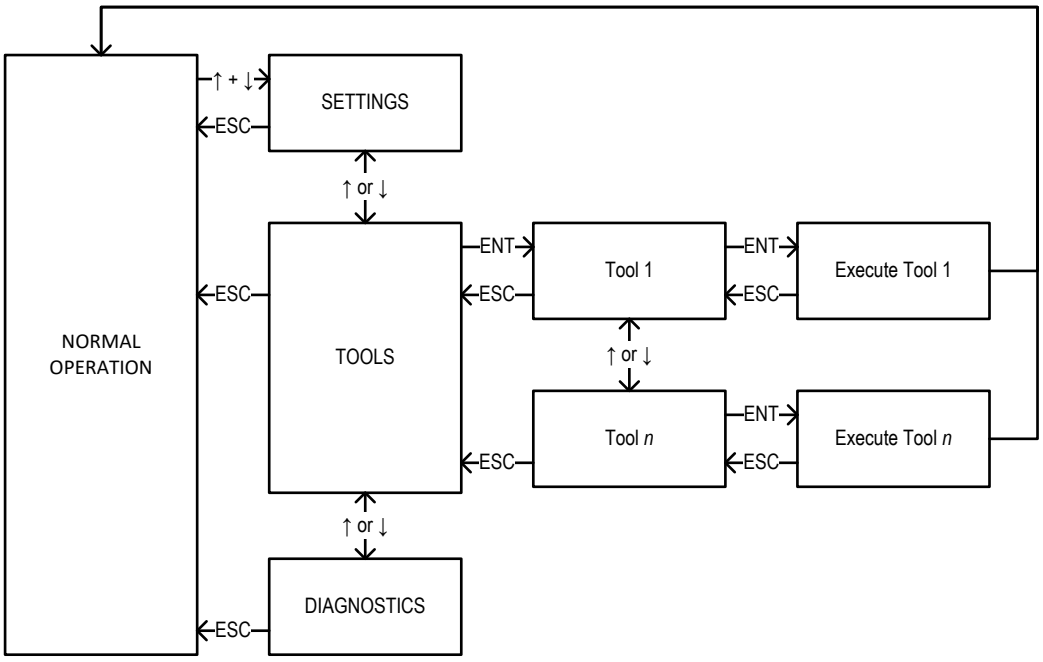
- AREA WIZ (Area Wizard)
- FAN WIZ (Fan Area Wizard)
- FAW (Flow Adjust Wizard)
- TEST OUT (Test Analog Output)
- TEST RELAY (Test Relay)
- LOCK (Lock Parameters and Wizards)

Simultaneously press the ↑↓ pushbuttons during normal operation to enter the main menu. The transmitter will continue to operate normally in the background.

The SETTINGS menu will be displayed. Press ↓ once and the TOOLS menu will be displayed. Press the ENT pushbutton to select the top of the TOOLS submenu category.

The TOOLS menu block diagram is shown below.

FIGURE 4-1 TOOLS MENU BLOCK DIAGRAM



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FIGURE 4-2 SINGLE LINE DISPLAY MENU STRUCTURE

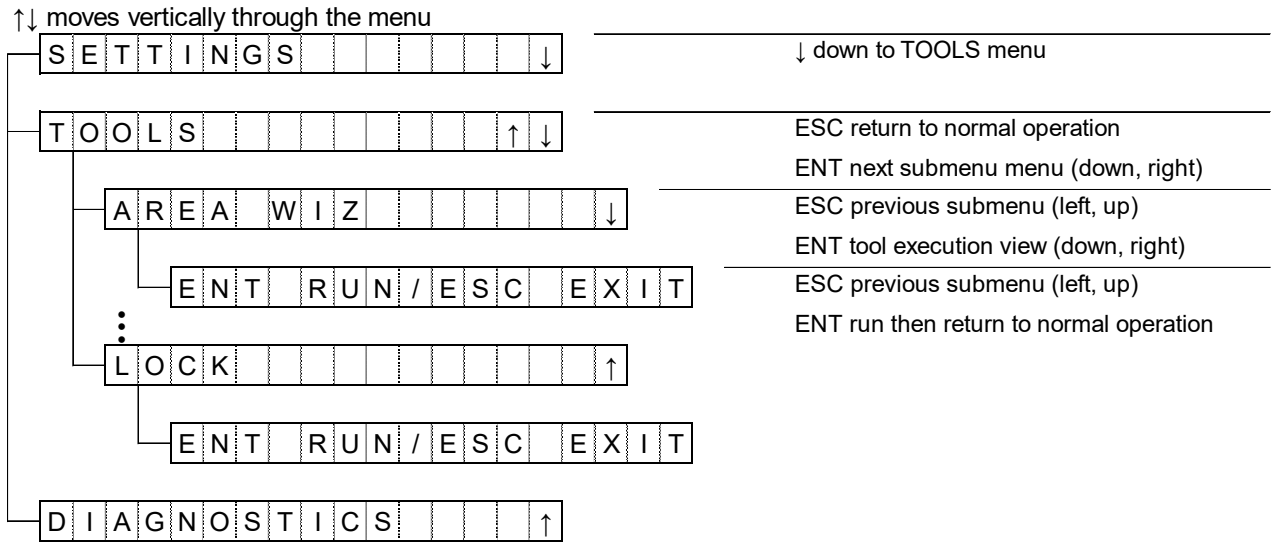
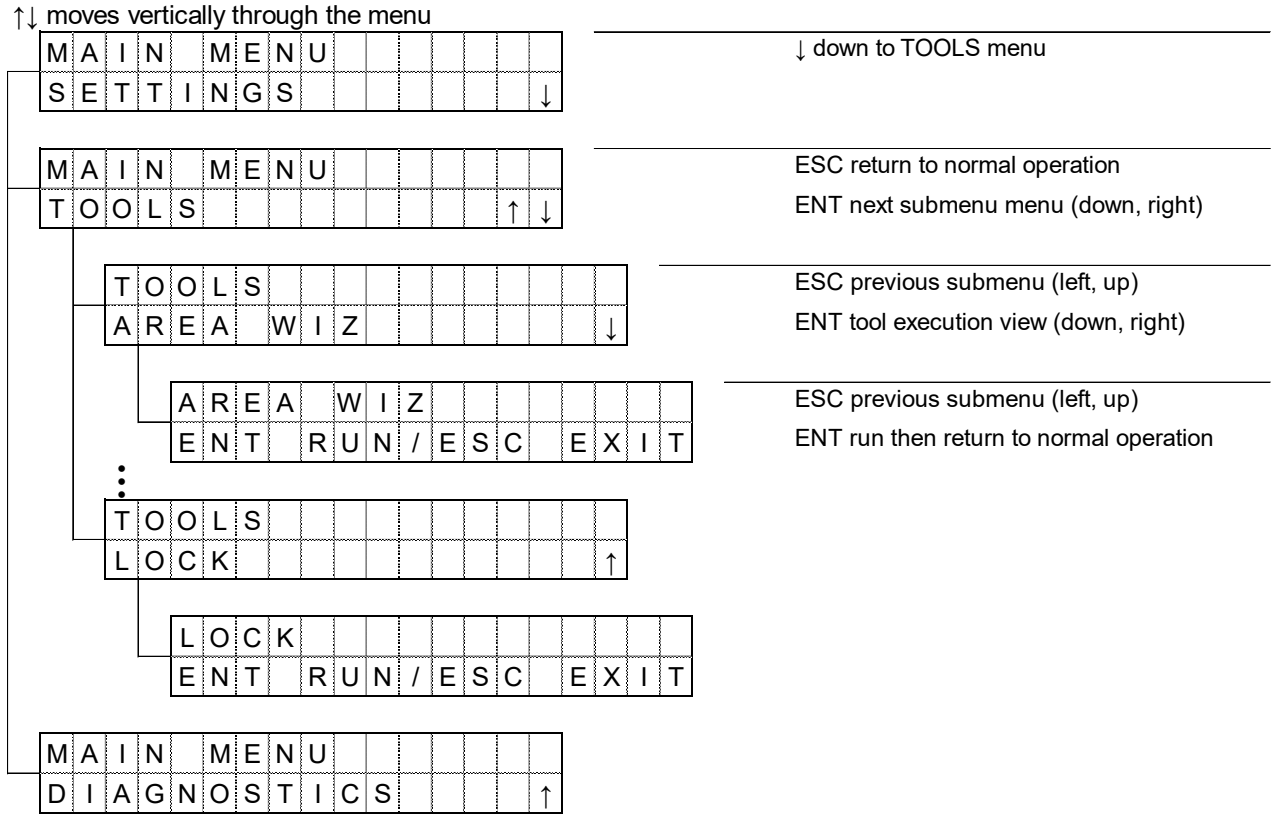


FIGURE 4-3 DUAL LINE DISPLAY MENU STRUCTURE



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4.2. AREA WIZARD [AREA WIZ]

The area of -U and -B probe types generally cannot be determined until the probes are installed. The factory default output is in velocity, FPM [m/s]. In some cases, the area of other probe types does not equal the area based on the original order size. The area is required when volumetric airflow, CFM [L/s], is desired. The area(s) can be entered manually by modifying the area parameter(s) via the SETTINGS menu or the area of each location can be entered using the AREA WIZARD (AREA WIZ) which is accessed via the TOOLS menu.

AREA WIZ (located in the TOOLS menu)

Use ↑↓ buttons to modify, {ENT} to accept and move to next prompt, {ESC} to move to previous prompt.

If the LOCK is enabled, pressing enter will indicate "CONFIG LOCKED" and this tool will not be available.

Navigate entire menu to step 7 to save settings. Press {ESC} at any time to exit without saving changes.

STEP	PROMPT	RESPONSE	DESCRIPTION	SKIP TO
1	LOCATION?	1↓	Location 1 of 2.	
		2↑	Location 2 of 2.	
			Note: This prompt will only be visible on 2 location devices. Single location devices start at prompt #2.	
2	SHAPE?	RECT↓	The duct or opening is square or rectangular.	4
		OVAL↑↓	The duct or opening is a flat oval.	4
		ROUND↑	The duct or opening is round.	
3	DIAM?	{null}↑	The free area duct or opening diameter, in inches [mm]. Value cannot be {null}.	6
4	WIDTH?	{null}↑	The free area duct or opening width, in inches [mm]. Value cannot be {null}.	
5	HEIGHT?	{null}↑	The free area duct or opening height, in inches [mm]. Value cannot be {null}.	
6	AREA	#####.###	Displays calculated AREA. ENT to move to next step. *Note: Dual location devices will indicate AREA1 or AREA2 in lieu of AREA.*	
7	SAVE?	YES↓	Save area to AREA parameter and return to normal operation.	
		NO↑	Do not save AREA parameter and return to normal operation.	

4.3. FAN AREA WIZARD [FAN WIZ]

The area of fan inlet (-F) probe types cannot be determined until the probes are installed. The factory default output is in velocity, FPM [m/s]. The area is required when volumetric airflow, CFM [L/s], is desired. The area can be entered manually by modifying the area parameter or the area of each fan can be entered using the FAN AREA SETUP WIZARD (FAN WIZ) accessed via the TOOLS menu.

The displayed units of measure of transmitters with the -F probe type default to FPM [m/s] since the factory default setting for area is *{null}* (no value). The displayed units of measure are automatically changed to CFM [L/s] when setup wizard changes are saved. The analog output units of measure, if applicable, remain at the factory default value of FPM [L/s].

4.3.1. Fan Arrays

Individual fan areas are used when the airflow of individual fans in a fan array is desired. Individual fan airflow rates are available via network variables and allow the advanced fan alarm capabilities of transmitter to be used. Once entered, individual fan areas can be modified by rerunning the setup wizard or manually via the SETTINGS menu.

Use the default output unit of measure and scaling of FPM [m/s] to determine CFM [L/s] with your B.A.S. Simply multiply the default full-scale velocity reading by the total area to determine the full-scale CFM [L/s] for the B.A.S.

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FAN WIZ (located in the TOOLS menu)

Use ↑↓ buttons to modify, {ENT} to accept and move to next prompt, {ESC} to move to previous prompt.

If the LOCK is enabled, pressing enter will indicate "CONFIG LOCKED" and this tool will not be available.

Navigate entire menu to step 14 to save settings. Press {ESC} at any time to exit without saving changes.

STEP	PROMPT	RESPONSE	DESCRIPTION	SKIP TO
1	CONFIG?	SWSI↓	Single inlet.	
		DWDI↑↓	Dual inlet.	
		ARRAY↑	Fan array.	3
2	DIAM?	{null}↑	The free area duct or opening diameter, in inches [mm]. Value cannot be {null}. *Note: The diameter should be determined at the upstream leading edge where sensor node is installed. The AREA of DWDI fans will be doubled.*	13
3	FANS?	2↑↓	Number of fans in the array, 2 to 8.	
4	DISTR?	EQUAL↓	The number of sensors, 1 or 2, is the same for all fans in the array.	
		VAR↑	The number of sensors, 1 or 2, is not the same for all fans in the array.	
5	SENS/FAN?	1↓	1 sensor node per fan.	
		2↑	2 sensor nodes per fan.	
			Note: If DISTR is set to VAR, there will be a separate prompt for each fan and the SENS/FAN prompt will be followed with the fan number (i.e. SENS/FAN1 … SENS/FANn)	
6	AREA/FAN?	EQUAL↓	The area is equal for each fan in the array.	
		VAR↑	The area is different for at least one fan in the array.	
7	METHOD?	CALC↓	The area is calculated based on dimensions provided by user.	
		AREA↑	The area is directly entered by the user.	12
8	TYPE?	INLET↓	Probes are mounted in the fan inlet cone	
		DMPR↑	Probes are mounted in a backdraft damper	10
9	DIAM?	{null}↑	The free area duct or opening diameter, in inches [mm]. Value cannot be {null}. *Note: The diameter should be determined at the upstream leading edge where sensor node is installed. If AREA/FAN is set to VAR, there will be a separate prompt for each fan and the DIAM prompt will be followed with the fan number (i.e. DIAM1 … DIAMn).*	13
10	WIDTH?	{null}↑	The inside damper width, in inches [mm]. Value cannot be {null}. *If AREA/FAN is set to VAR, there will be a separate prompt for each fan and the WIDTH prompt will be followed with the fan number (i.e. WIDTH1 … WIDTHn).*	
11	HEIGHT?	{null}↑	The inside damper height, in inches [mm]. Value cannot be {null}. *If AREA/FAN is set to VAR, there will be a separate prompt for each fan and the HEIGHT prompt will be followed with the fan number (i.e. HEIGHT1 … HEIGHTn).*	13
12	AREA?	{null}↑	The area of the fan inlet, in sq ft [sq m] where the sensor node is installed. Value cannot be {null} *Note: The area should be determined at the upstream leading edge where sensor node is installed. Enter the area of one inlet on DWDI fans or a single fan in an array if AREA/FAN is set to EQUAL. If AREA/FAN is set to VAR, there will be a separate prompt for each fan and the AREA prompt will be followed with the fan number (i.e. AREA1…AREAn).*	
13	SAVE?	YES↓	Save changes and return to normal operation.	
		NO↑	Do not save changes and return to normal operation.	

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4.4. FLOW ADJUST WIZARD [FAW]

EBTRON airflow measurement devices are factory calibrated to NIST traceable standards. Field adjustment is not required when installed in accordance to published guidelines.

A flow adjust wizard is provided for installations where placement limitations warrant field adjustment of the factory calibration. The wizard can be run with using a one-point (gain only) or two-point (gain and offset). The FAW is not available for -B probe types set to convert airflow to pressure (DPCONVERT=YES).

ⓘ *If minimum placement guidelines cannot be achieved, installed accuracy may be compromised. Transmitters can be field adjusted to match a third-party measurement. Adjusted field measurements typically result in comparative readings within ±3% of the third-party measurement. Be advised that the third-party measurement may have uncertainties greater than or equal to ±10% and should only be used to adjust the airflow measurement device if the probes do not meet minimum placement requirements and the discrepancy is greater than the uncertainty of the third-party source.*

ⓘ *The factory default setting for the number of samples and units of measure can be modified via the SETTINGS MENU under the TOOLS submenu. The factory default number of samples is set to 100 (30 seconds) and the default units of measure is set to CFM [L/s] (as long as a valid area is provided).*

Increasing the number of samples may improve field adjustment in turbulent locations. Make sure that the system is "locked" (i.e. fan speeds, dampers, terminals, etc. are not actively controlling) when the FAW is run.

⚠ *The reference units of measure must be equal to the adjustment units of measure.*

FAW (located in the TOOLS menu)

Use ↑↓ buttons to modify, {ENT} to accept and move to next prompt, {ESC} to move to previous prompt.

If the LOCK is enabled, pressing enter will indicate "CONFIG LOCKED" and this tool will not be available.

Navigate entire menu to step 11 to save settings. Press {ESC} at any time to exit without saving changes.

STEP	PROMPT	RESPONSE	DESCRIPTION	SKIP TO
1	LOCATION?	1↓	Location 1 of 2.	
		2↑	Location 2 of 2.	
			Note: This prompt will only be visible on 2 location devices. Single location devices start at step #3.	
2	# OF FLOWS?	1↓	One flow rate. Only the gain coefficient is calculated.	
		2↑	Two flow rates. The gain and offset coefficients are calculated.	
3	SET FLOW 1		Set FLOW 1. Press ENT to start sample.	
4	REF XXX?	0↑	XXX = FPM [m/s] or CFM [LPS]. Enter reference airflow in appropriate units.	
5	SMPLS=XXX	###%	XXX = number of samples. ### = % complete. Goes to next step at 100%.	
6	Steps 7 to 9 are only visible if # OF FLOWS is set to 2.			
7	SET FLOW 2		Set FLOW 2. Press ENT to start sample.	
8	REF XXX?	0↑	XXX = FPM [m/s] or CFM [LPS]. Enter reference airflow in appropriate units.	
9	SMPLS=XXX	###%	Displays number of integration samples and % complete. Goes to next step at 100%.	
10	G:#.###	O:#####	Displays calculated gain and offset coefficients. ENT to move to next step.	
11	SAVE?	YES↓	Save coeficients, enable adjustments and return to normal operation.	
		NO↑	Do not save coefficents and return to normal operation.	

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4.5. TEST OUTPUT TOOL [TEST OUT]

The test output tool provides a user defined, fixed output signal between 0 and 100% of the full-scale.

TEST OUT (located in the TOOLS menu)

Use ↑↓ buttons to modify, {ENT} to accept and move to next prompt, {ESC} to move to previous prompt.

If the LOCK is enabled, pressing enter will indicate "CONFIG LOCKED" and this tool will not be available.

Navigate entire menu to step 3 to save settings. Press {ESC} at any time to exit and return to normal operation.

STEP	PROMPT	RESPONSE	DESCRIPTION	SKIP TO
1	OUTPUT?	AO1↓	Analog output 1 (AO1).	
		AO2↑↓	Analog output 2 (AO2).	
		AO3↑	Analog output 3 (AO3). [Visible with GTx116e with /H relative humidity sensor installed)	
2	OUTPUT %?	50↑↓	Sets the percent of full-scale output, 0 to 100%.	
3	EXIT?	YES↓	Return to normal operation.	
		NO↑	Return to step 1.	1

4.6. TEST RELAY TOOL [TEST RELAY]

The test relay tool is only available with EF-x2000 transmitters. The test relay tool enables or disables the relay.

- Normally open relays will close when enabled and open when disabled.
- Normally closed relays will open when enabled and close when disabled.

TEST RELAY (located in the TOOLS menu)

Use ↑↓ buttons to modify, {ENT} to accept and move to next prompt, {ESC} to move to previous prompt.

If the LOCK is enabled, pressing enter will indicate "CONFIG LOCKED" and this tool will not be available.

Navigate entire menu to step 2 to save settings. Press {ESC} at any time to exit and return to normal operation.

STEP	PROMPT	RESPONSE	DESCRIPTION	SKIP TO
1	STATUS?	ENABLE↓	Enable relay 1 (R1).	
		DISABLE↑	Disable relay 1 (R1).	
2	EXIT?	YES↓	Return to normal operation.	
		NO↑	Return to step 1.	1

4.7. LOCK TOOL [LOCK]

The lock tool locks transmitter settings and tools. Settings can be viewed but not modified. There are three lock security levels – low, medium and high.

SECURITY = LOW. Low security. The last four digits of the serial number of the transmitter PCB motherboard is used as the lock code. The serial number is located on a label on the PCB.

SECURITY = MED. Medium security. A four-digit code is entered by the user. If the code is lost, a unique "back door" code can be provided by EBTRON customer service to unlock the device.

SECURITY = HIGH. High security. A four-digit code is entered by the user. If the code is lost, the transmitter must be returned to EBTRON to be unlocked.

LOCK Tool (located in the TOOLS menu)

Use ↑↓ buttons to modify, {ENT} to accept and move to next prompt, {ESC} to move to previous prompt.

The LOCK menu item is only visible when the device is UNLOCKED.

Navigate entire menu to step 4 to lock device settings and tools. Press {ESC} twice at any time to exit without locking the device.

STEP	PROMPT	RESPONSE	DESCRIPTION	SKIP TO
1	LOCK?	YES↓	Lock transmitter parameters and tools.	
		NO↑	Do not lock device. Return to normal operation.	
2	SECURITY?	LOW↓	Lock device using PCB serial number as lock/unlock code and return to normal operation.	
		MED↑↓	User defined lock/unlock code with EBTRON "back door" emergency code override.	
		HIGH↑	User defined lock/unlock code. Cannot be unlocked without code or return to factory.	
3	CODE?	0000↑	Enter lock code.	
4	CONFIRM?	0000↑	Reenter lock code. Lock device and return to normal operation. *Note: If an invalid code is entered, the display will indicate "INVALID CODE" for 5 seconds and return to step 3.*	

4.8. UNLOCK TOOL [UNLOCK]

The unlock tool unlocks transmitter settings and tools. Settings can be viewed and modified.

UNLOCK Tool (located in the TOOLS menu)

Use ↑↓ buttons to modify, {ENT} to accept and move to next prompt, {ESC} to move to previous prompt.

The UNLOCK menu item is only visible when the device is LOCKED.

Navigate entire menu to step 2 to unlock settings and tools. Press {ESC} twice at any time to exit without unlocking the device.

STEP	PROMPT	RESPONSE	DESCRIPTION	SKIP TO
1	UNLOCK?	YES↓	Lock transmitter parameters and tools.	
		NO↑	Do not lock device. Return to normal operation.	
2	CODE?	0000↑	Unlock the device and return to normal operation. *Note: If an invalid code is entered, the display will indicate "INVALID CODE" for 5 seconds and return to step 2.*	

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5. SECTION OVERVIEW

This section discusses the built-in diagnostics features of the transmitter.

All Advantage IV Gold Series and Hybrid Series models HTx116 and HTx108 transmitters are provided with a Bluetooth® Low Energy interface that allows transmitter diagnostic information to be quickly downloaded then e-mailed and/or saved using your iOS or Android phone or tablet.

5.1. DIAGNOSTICS MENU ACCESS AND NAVIGATION

The SETTINGS menu contains the following submenu categories:

- DEV INFO
- DIAGNOSTICS
- TRBL CODES

The following symbols are used to describe shortcut paths for menu and submenu navigation:

- > press the enter ENT pushbutton once
- < press the escape ESC pushbutton once
- ↓ press the down arrow pushbutton until the subsequent menu item or parameter appears
- ↑ press the up arrow pushbutton until the subsequent menu item or parameter appears

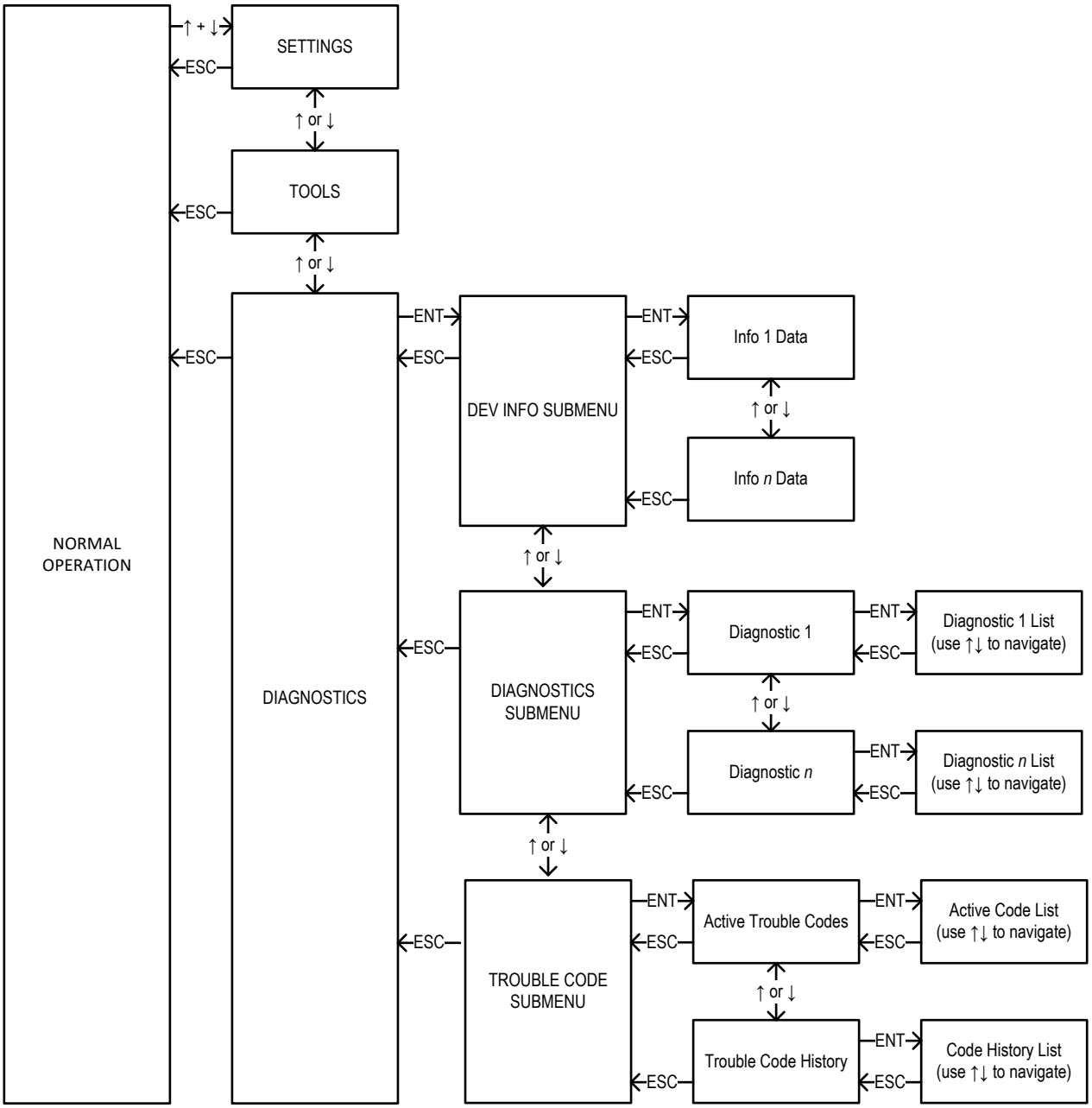
Simultaneously press the ↑ and ↓ pushbuttons during normal operation to enter the main menu. The transmitter continues to operate normally in the background.

The SETTINGS menu is displayed. Press ↓ pushbutton until the DIAGNOSTICS menu is displayed. Press the ENT pushbutton to select the top of the DIAGNOSTICS submenu category.

The DIAGNOSTICS menu block diagram is shown on the next page.

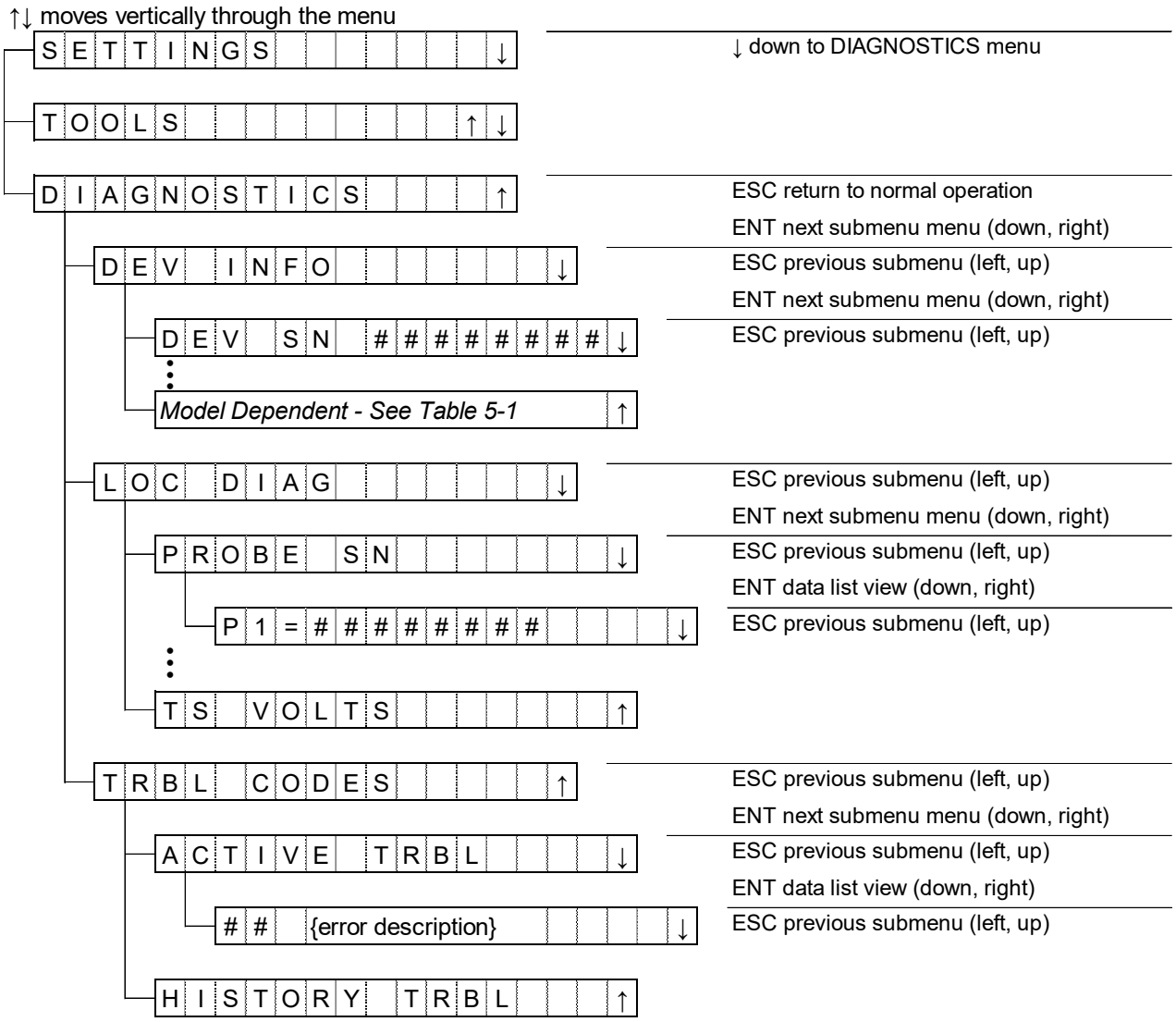
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FIGURE 5-1 DIAGNOSTICS MENU BLOCK DIAGRAM



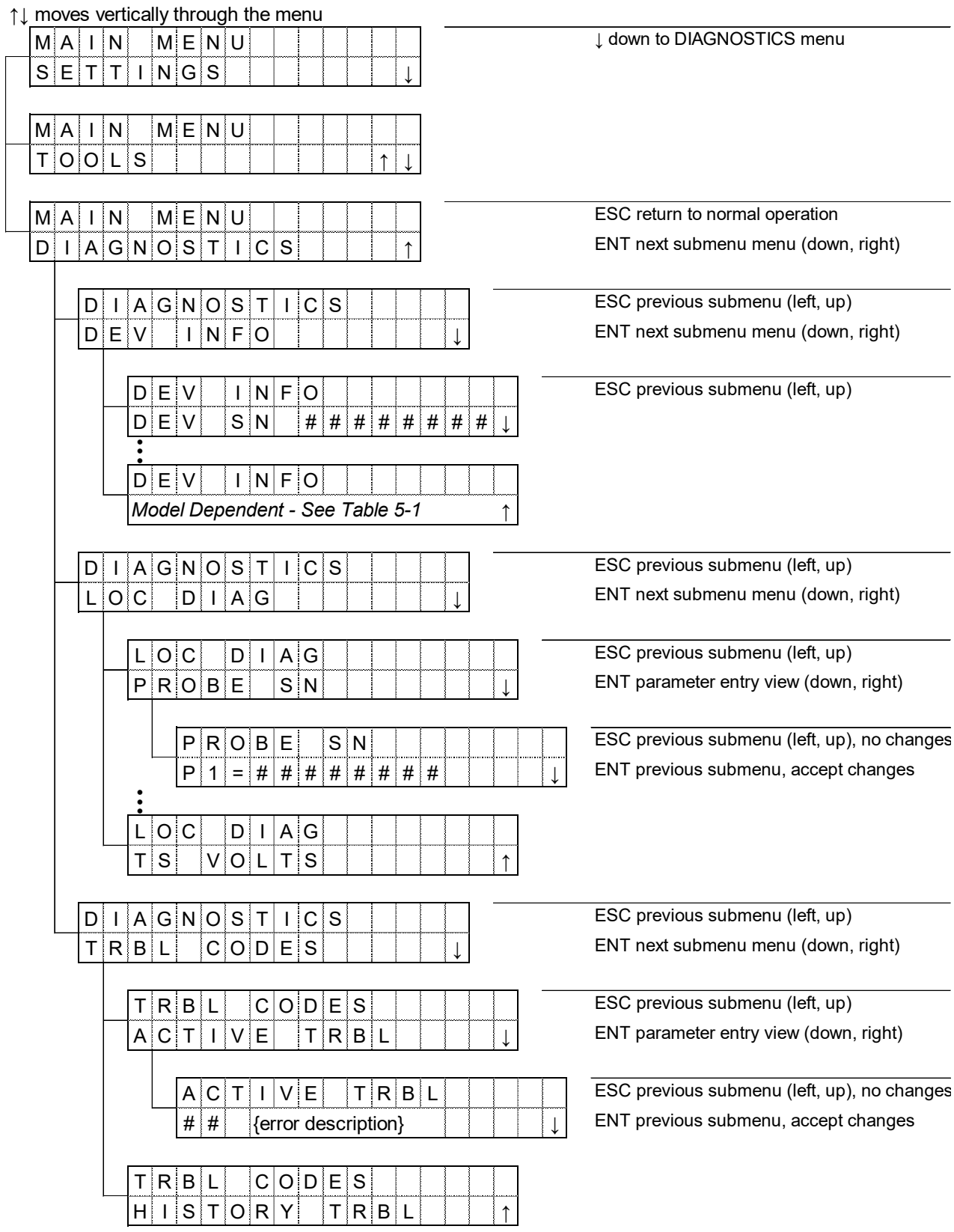
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FIGURE 5-2 SINGLE LINE DISPLAY MENU STRUCTURE



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FIGURE 5-3 DUAL LINE DISPLAY MENU STRUCTURE



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5.2. DEV INFO (DEVICE INFORMATION)

The DEV DIAG submenu contains the following information:

- MODEL (Device Model)
- DEV SN (Device Serial Number)
- PCB SN (Main PCB Serial Number)
- CON SN (Connector Card Serial Number)
- PCB FW (Main Circuit Board Firmware Version)
- CON FW (Connector Card Firmware Version)
- OUT FW (Output/Network Card Firmware Version)
- BLE FW (Bluetooth Low Energy Radio Firmware Version)
- Pb (On-board Barometric Pressure Sensor Reading)

5.2.1. Device Model [MODEL]

The model code of the transmitter system.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO > MODEL]

Applies to: GTx116e and GTx108e models with main circuit board firmware 8.37 and later, and all HTx116 and HTx108 models.

5.2.2. Device Serial Number [DEV SN]

The serial number of the transmitter system.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO >↓ DEV SN]

Applies to: All transmitter models.

ⓘ *The device serial number is identified by a label located on the transmitter enclosure.*

5.2.3. Main PCB Serial Number [PCB SN]

The serial number of the main circuit board.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO >↓ PCB SN]

Applies to: All transmitter models.

5.2.4. Connector Card Serial Number [CON SN]

The serial number of the 8-connector receptacle card.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO >↓ CON SN]

Applies to: GTx108-F/An and GTx108e-F/An models.

5.2.5. Main PCB Firmware Version [PCB FW]

The firmware version of the main transmitter circuit board (i.e. the transmitter operating system)

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO >↓ PCB FW]

Applies to: All transmitter models.

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5.2.6. Connector Card Firmware Version [CON FW]

The firmware version of the 8-connector receptacle card.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO >↓ CON FW]

Applies to: GTx108-F/An and GTx108e-F/An models.

5.2.7. Output/Network Card Firmware Version [OUT FW]

The firmware version of an optional output/network card.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO >↓ OUT FW]

Applies to: All Gold and Hybrid Series transmitters with an optional output/network card installed.

5.2.8. Bluetooth Low Energy Radio Firmware Version [BLE FW]

The firmware version of the BLE transceiver.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO >↓ BLE FW]

Applies to: GTx116e, GTx108e, HTx116 or HTx108 transmitters with an integral BLE transceiver on the main PCB.

5.2.9. On-board Barometric Pressure Sensor Reading [Pb]

The current value reading of the onboard barometric pressure sensor in inHg [Pa].

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO >↓ Pb]

Applies to: GTx116e, GTx108e, HTx116 or HTx108 transmitters only.

5.3. LOC DIAG (LOCATION DIAGNOSTICS)

Provided real-time diagnostic data for the measurement location.

ⓘ *If the transmitter is configured for two locations, submenu tabs for LOC1 DIAG and LOC2 DIAG will be displayed in lieu of LOC DIAG and the information will be replicated for each. See Table 5-1 for more information.*

- PROBE SN (Probe Serial Numbers)
- AVG VELOCITY (Average Velocity of the Sensor Nodes)
- FAN VELOCITY (Average Fan Velocity)
- FAN FLOW (Fan Volumetric Airflow Rate)
- AVG TEMP (Average Temperature of the Sensor Nodes)
- TEMP OFF (Average Temperature Offset Adjustment)
- AVG RH (Average Relative Humidity of the RH Sensors)
- RH OFF (Average Relative Humidity Offset Adjustment)
- NODE VELOCITY (Velocity of each Sensor Node)
- NODE TEMP (Temperature of each Sensor Node)
- RH SENS (Relative Humidity of each RH Sensor)
- HS VOLTS (Self-heated Thermistor Voltage of each Sensor Node)
- TS VOLTS (Zero-power Thermistor Voltage of each Sensor Node)

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5.3.1. Probe Serial Numbers [PROBE SN]

The serial number of each probe connected to the transmitter.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG > PROBE SN > {probe list 1 to n}]

Applies to: All models.

Each probe has a unique serial number that is printed on a label affixed to the probe. The serial number is also stored in a serial memory device located in the cable plug. The serial number identifies where the probe should be located, the probe length and the number of sensors in the probe and is useful information during field diagnostics. Probe numbers (P1 to Py) are dynamically assigned by the transmitter after initial power up, left to right, based on the connector receptacle used. Use the ↑ ↓ pushbuttons to navigate through the list of probes that the transmitter to confirm that all probes for the location are connected. The probe number, serial number and letter "T" is displayed for each probe connected.

ⓘ *If a probe is removed after initial power-up the letter "F" will follow the serial number.*

5.3.2. Average Velocity of the Sensor Nodes [AVG VELOCITY]

The running average velocity of all properly functioning sensor nodes in FPM [m/s]. The running average integration is set to the LCD integration parameter [LCD INTG].

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ AVG VELOCITY]

Applies to: All models.

5.3.3. Average Fan Velocity (FAN VELOCITY)

The running average velocity of each individual fan in a fan array in FPM [m/s]. The running average integration is set to the LCD integration parameter [LCD INTG].

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ FAN VELOCITY > {fan list 1 to n}]

Applies to: GTx108-F/An and GTx108e-F/An models.

5.3.4. Fan Volumetric Airflow Rate (FAN FLOW)

The running average volumetric airflow rate of each individual fan in a fan array in CFM [L/s]. The running average integration is set to the LCD integration parameter [LCD INTG].

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ FAN VELOCITY > {fan list 1 to n}]

Applies to: GTx108-F/An and GTx108e-F/An models.

5.3.5. Average Temperature of the Sensor Nodes [AVG TEMP]

The running average temperature of all properly functioning sensor nodes in °F [°C]. The running average integration is set to 20 and cannot be modified. The average temperature including any user entered offset.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ AVG TEMP]

Applies to: All models.

5.3.6. Average Temperature Offset Adjustment [TEMP OFF]

A user entered temperature offset in °F [°C].

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ TEMP OFF]

Applies to: All models.

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Use the ↑ and ↓ pushbuttons to enter a user defined offset. Press the ENT pushbutton to accept the offset or the ESC pushbutton to keep the previous value.

ⓘ *The temperature output of the EBTRON device is velocity-weighted and more accurate than the simple arithmetic average obtained by a field temperature measurement or traverse. A field temperature measurement will not match the velocity-weighted output shown on the display. The arithmetic average of the sensor nodes is indicated by AVG TEMP in 5.3.5. If an adjustment is made to a field measurement, it should be to AVG TEMP and not the displayed or output temperature of the device.*

⚠ *Making an adjustment to the temperature overrides EBTRON's NIST traceable temperature calibration.*

5.3.7. Average Relative Humidity of the RH Sensors [AVG RH]

The average, non-integrated relative humidity of all properly functioning humidity sensors as %RH.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ AVG RH]

Applies to: GTx116e-P/H models only (i.e. models with the /H humidity sensor option).

5.3.8. Average Relative Humidity Offset Adjustment [RH OFF]

A user entered relative humidity offset in %RH.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ RH OFF]

Applies to: All models.

Use the ↑ and ↓ pushbuttons to enter a user defined offset. Press the ENT pushbutton to accept the offset or the ESC pushbutton to keep the previous value.

ⓘ *The relative humidity output of the EBTRON device is velocity-weighted and more accurate than the simple arithmetic average obtained by a field relative humidity measurement or traverse. A field temperature measurement will not match the velocity-weighted output shown on the display. The arithmetic average of the relative humidity sensor(s) is indicated by AVG RH in 5.3.7. If an adjustment is made to a field measurement, it should be to AVG RH and not the displayed or output relative humidity of the device.*

⚠ *Making an adjustment to either the temperature or the RH sensor modifies the enthalpy and dewpoint calculations.*

5.3.9. Velocity of each Sensor Node [NODE VELOCITY]

The running average velocity of each sensor node in FPM [m/s]. The running average integration is set to 30 and cannot be modified.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ NODE VELOCITY> {sensor node list 1 to n}]

Applies to: All models.

Use the ↑ ↓ pushbuttons to navigate through the sensor nodes.

ⓘ *It is normal for velocity of an individual sensor node to fluctuate. The greater the fluctuation, the greater the turbulence at the measurement location.*

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5.3.10. Temperature of each Sensor Node [NODE TEMP]

The non-integrated temperature of each sensor node in °F [°C] without any offset adjustment.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ NODE TEMP> {sensor node list 1 to n}]

Applies to: All models.

Use the ↑ ↓ pushbuttons to navigate through the sensor nodes.

5.3.11. Relative Humidity of each RH Sensor [RH SENS]

The average, non-integrated relative humidity of each humidity sensors as %RH without any offset adjustment.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ RH SENS]

Applies to: GTx116e-P/H models only (i.e. models with the /H humidity sensor option).

ⓘ *This diagnostic is only visible when more than one humidity sensor has been provided for the measurement location.*

Use the ↑ ↓ pushbuttons to navigate through the individual humidity sensors.

5.3.12. Self-heated Thermistor Voltage of each Sensor Node [HS VOLTS]

The voltage of the self-heated thermistor as measured by the analog to digital (A/D) converter of the transmitter.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ HS VOLTS]

Applies to: All models.

5.3.13. Zero-power Thermistor Voltage of each Sensor Node [TS VOLTS]

The voltage of the zero-power (temperature measuring) thermistor as measured by the analog to digital (A/D) converter of the transmitter.

[Menu path: SETTINGS ↓ DIAGNOSTICS > DEV INFO ↓ LOC DIAG >↓ TS VOLTS]

Applies to: All models.

5.4. TRBL CODES (TROUBLE CODES)

Trouble codes facilitate diagnosing problems with your EBTRON airflow measurement system. Many trouble codes are benign and do not require a remedy for system operation. Other codes require attention.

Both active and historic trouble codes are available. Historic codes are useful to discover intermittent problems and/or failures. Active trouble codes are displayed on the LCD unless the LCD TRBL parameter was set to OFF (Section 3). Active trouble codes also activate a LED flash sequence of 2 seconds on and 2 second off on the activity LED on the main circuit board (Section 6).

Table 5-2 contains a list of the trouble code numbers (Code) and corresponding Error Message by transmitter model.

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TABLE 5-2 TROUBLE CODES

Code	Error Message	GTx116e-P/H	GTx116e-P	HTx116-P	GTx116-P	HTx104-P	HTx104-T	EF-x2000-T	EF-x2000-U	GTx108-F/An	GTx108-F	GTx108e-F/An	GTx108e-F	HTx108-F	HTx104-F	GTx116-B	HTx116-B	HTx104-B	EF-x2000-B
1	NO PROBES	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
2	WRONG SENS	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
3	P1 MISSING	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
4	P2 MISSING	•	•	•	•	•			•	•	•	•	•	•	•	•	•		
5	P3 MISSING	•	•	•	•					•	•	•	•	•	•	•	•		
6	P4 MISSING	•	•	•	•					•	•	•	•	•	•	•	•		
7	P5 MISSING									•	•	•	•	•					
8	P6 MISSING									•	•	•	•	•					
9	P7 MISSING									•	•	•	•	•					
10	P8 MISSING									•	•	•	•	•					
11	P1 MEM ERROR	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
12	P2 MEM ERROR	•	•	•	•	•			•	•	•	•	•	•	•	•	•		
13	P3 MEM ERROR	•	•	•	•					•	•	•	•	•	•	•	•		
14	P4 MEM ERROR	•	•	•	•					•	•	•	•	•	•	•	•		
15	P5 MEM ERROR									•	•	•	•	•					
16	P6 MEM ERROR									•	•	•	•	•					
17	P7 MEM ERROR									•	•	•	•	•					
18	P8 MEM ERROR									•	•	•	•	•					
19	SENSOR 1	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
20	SENSOR 2	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
21	SENSOR 3	•	•	•	•	•				•	•	•	•	•	•	•	•		
22	SENSOR 4	•	•	•	•	•				•	•	•	•	•	•	•	•		
23	SENSOR 5	•	•	•	•					•	•	•	•	•		•	•		
24	SENSOR 6	•	•	•	•					•	•	•	•	•		•	•		
25	SENSOR 7	•	•	•	•					•	•	•	•	•		•	•		
26	SENSOR 8	•	•	•	•					•	•	•	•	•		•	•		
27	SENSOR 9	•	•	•	•														
28	SENSOR 10	•	•	•	•														
29	SENSOR 11	•	•	•	•														
30	SENSOR 12	•	•	•	•														
31	SENSOR 13	•	•	•	•														
32	SENSOR 14	•	•	•	•														
33	SENSOR 15	•	•	•	•														
34	SENSOR 16	•	•	•	•														
35	LOW TEMP	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
36	HIGH TEMP	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
40	RH1 MISSING	•																	
41	RH2 MISSING	•																	
42	RH3 MISSING	•																	
43	RH4 MISSING	•																	

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TABLE 5-2 TROUBLE CODES (Continued)

Code	Error Message	GTx116e-P/H	GTx116e-P	HTx116-P	GTx116-P	HTx104-P	HTx104-T	EF-x2000-T	EF-x2000-U	GTx108-F/An	GTx108-F	GTx108e-F/An	GTx108e-F	HTx108-F	HTx104-F	GTx116-B	HTx116-B	HTx104-B	EF-x2000-B
50	Pb SENS ERROR	•	•	•								•	•	•			•		
60	NUM PROBES								•										
61	FAN COUNT									•		•		•					
70	SYSTEM ERROR	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
71	OUT CARD	•	•	•	•					•	•	•	•	•		•	•		
72	CON CARD									•	•	•	•						
73	BLE COM ERROR	•	•	•								•	•	•			•		
74	AOUT ERR	•	•	•								•	•	•			•		
75	CON BUS	•	•									•	•						
80	EBBUS CON ERR	•	•	•								•	•	•			•		
85	FLOW CALC ERR	•	•	•								•	•	•			•		

Table 5-3 provides a description of each trouble code message. If a trouble code and message is displayed on the LCD, refer to Section 6 – Troubleshooting or contact EBTRON customer service at 1-800-232-8766.

TABLE 5-3 ERROR MESSAGE DESCRIPTION

Error Message	Error Description
NO PROBES	No probes discovered by transmitter.
WRONG SENS	Mismatched sensor probe types on a single transmiter.
Px MISSING	Registered probe missing.
Px MEM ERROR	Serial memory error in probe.
SENSOR x	Sensor node error.
LOW TEMP	Airstream temperature below product's specified minimum (-20°F [-28.9°C])
HIGH TEMP	Airstream temperature above product's specified maximum (160°F [71.1°C])
RHx MISSING	Humidity sensor error.
Pb SENS ERR	Barometric pressure sensor error.
NUM PROBES	Missing probe at Location 2 (dual location EF units).
FAN COUNT	Fan count entry from FAN WIZ does not equal actual sensor probe count.
SYSTEM ERROR	Transmitter memory error.
OUT CARD	Transmitter output card communication error.
CON CARD	-F/An connector card communication error.
BLE COMM ERR	BLE radio error.
AOUT ERR	Analog output error.
CON BUS	"e" transmitter probe serial memory communication error.
EBBUS CON ERR	*EB-Bus* Ethernet network connection error.
FLOW CALC ERR	Flow calculation error (OEM transmitters only)

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6. SECTION OVERVIEW

Each EBTRON product is thoroughly tested prior to shipment. This section contains information on troubleshooting EBTRON Advantage IV and EB-Flow2 measuring devices.

If the product has been physically damaged, skip to Section 6.5. If trouble codes are displayed on the LCD, skip to Section 6.3. If the product is functioning properly and there is a discrepancy in the measurement, skip to Section 6.4.

6.1. INITIAL TROUBLESHOOTING

If the display is blank or no trouble codes are displayed, follow the initial troubleshooting procedure of Table 6-1. If trouble codes are displayed on the LCD, skip to 6.3.

TABLE 6-1 – INITIAL TROUBLESHOOTING

Procedure		Yes	No
1.	Is the activity LED illuminated or flashing?	2.	3.
ⓘ	*The activity LED is located bottom right of the ESC pushbutton of all A4 models and the upper left-hand corner of the printed circuit board of all EB-Flow2 models.*		
2.	Is the flash rate one second on and one second off?	3.	4.
3.	Continue to 6.2. TROUBLESHOOTING (NO TROUBLE CODES)		
4.	Is the flash rate two seconds on and two seconds off?	5.	CALL
5.	Are any trouble codes indicated on the LCD?	7.	6.
6.	The LCD trouble indication parameter, LCD TRBL, may be set to "OFF". Set the LCD TRBL parameter to "ON" (Section 3). Are any trouble codes indicated on the LCD?	7.	CALL
7.	Continue to 6.3. FDS TROUBLESHOOTING		

FDS stands for "Fault Detection System"

6.2. TROUBLESHOOTING (NO TROUBLE CODES)

The most common symptoms and solutions are covered in Table 6-2. If the symptom is not described in Table 6-2, contact customer service (Section 6.5).

Locate the symptom that best describes your issue. Follow the steps outlined and answer the question in the table until you reach “DONE” or “CALL”. In some cases, a yes/no response is not appropriate and instruction is given in the procedure.

6.3. FAULT DETECTION SYSTEM TROUBLESHOOTING

Advantage IV and *EBFlow-2* remote transmitters have a comprehensive fault detection system (FDS) that continuously runs in the background. If a fault is detected, a trouble code is generated and displayed on the LCD along with a short text description of the problem.

Active trouble codes can also be viewed from the DIAGNOSTICS MENU (Section 5) along with a trouble code history that is useful for finding intermittent faults. Identify the active trouble code on the LCD and use Table 6-3 or contact customer service (Section 6.5).

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TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models	Procedure		Yes	No
BASIC OPERATION	The Activity LED flash rate is one second on and one second off and the display is blank.	All Transmitter Models	1.	Increase the LCD contrast to maximum (Section 2). Is the LCD blank?	CALL	DONE
				Set the contrast to the desired level if this corrects the problem.		
	The Activity LED is not illuminated and the LCD is blank	A4 Transmitter Models GTx116, GTx108 and HTx104 where x = {all output types}	1.	Is a 24 VAC power source properly connected to power terminal block with terminal screws secure and no shorts between wires (refer to the appropriate *Wiring Guide* for terminal connections)?	3.	2.
				"Live" wiring to transmitter may damage the transmitter and void warranty.		
			2.	Make sure the 24 VAC power source is not energized. Properly connect the 24 VAC power source to the power terminal block. Energize the 24 VAC power source. Is the LCD blank?	3.	DONE
			3.	Is the power switch in the "ON" position?	5.	4.
			4.	Move the power switch to the "ON" position. Is the LCD blank?	5.	DONE
			5.	Measure the voltage across the power terminal block screws with a VOM configured to measure VAC. Is the voltage between 22.8 to 26.4 VAC?	7.	6.
			6.	Move the power switch to the "OFF" position. Correct input power problem so that the power supplied under load is between 22.8 and 26.4 VAC. Move the power switch to the "ON" position. Is the LCD blank?	7.	DONE
			7.	Move the power switch to the "OFF" position. Remove and inspect the power fuse. Is the fuse blown?	8.	CALL
				Check the fuse with a VOM set to ohms to verify the fuse is good (shorted) or blown (open).		
			8.	Remove all signal wires and probes from transmitter. Replace the glass power fuse with a fast-acting fuse with the power rating indicated on printed circuit board. Move the power switch to the "ON" position. Is the LCD blank?	CALL	9.
			9.	Move the power switch to the "OFF" position. Connect all sensor probes. Move the power switch to the "ON" position. Is the LCD blank?	CALL	10.
			10	Move the power switch to the "OFF" position. Connect all signal wires. Move the power switch to the "ON" position. Is the LCD blank?	11.	DONE
				There was a problem that caused the fuse to blow that is no longer present.		
			11	There is a problem with the signal wiring or the host control system. Move the power switch to the "OFF" position. Disconnect the signal wiring, check for shorts in the signal wiring and/or other problems with the host control system. Is there a control wiring problem or other problem with the host control system?	12.	CALL
			12	Correct the problem with the signal wiring or the host control system. Replace the blown power fuse. Connect all signal wires to the transmitter. Move the power switch to the "ON" position. Is the LCD blank?	CALL	DONE
		A4 Transmitter Models GTx116e, GTx108e, HTx116 and HTx108	1.	Is a 24 VAC power source properly connected to power terminal block with terminal screws secure and no shorts between wires (refer to the appropriate *Wiring Guide* for terminal connections)?	3.	2.
				"Live" wiring to transmitter may damage the transmitter and void warranty.		
				These models have solid state fuses. Allow at least one minute between power cycles to allow the solid state fuses to reset in the event they were in a fault condition.		
			2.	Make sure the 24 VAC power source is not energized. Properly connect the 24 VAC power source to the power terminal block. Energize the 24 VAC power source. Is the LCD blank?	3.	DONE
			3.	Is the power switch in the "ON" position?	5.	4.
			4.	Move the power switch to the "ON" position. Is the LCD blank?	5.	DONE
			5.	Measure the voltage across the power terminal block screws with a VOM configured to measure VAC. Is the voltage between 22.8 to 26.4 VAC?	7.	6.
			6.	Move the power switch to the "OFF" position. Correct input power problem so that the power supplied under load is between 22.8 and 26.4 VAC. Move the power switch to the "ON" position. Is the LCD blank?	7.	DONE

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models	Procedure		Yes	No
BASIC OPERATION			7.	Move the power switch to the "OFF" position. Remove all signal wires and probes from transmitter. Move the power switch to the "ON" position. Is the LCD blank?	CALL	8.
			8.	Move the power switch to the "OFF" position. Connect all sensor probes. Move the power switch to the "ON" position. Is the LCD blank?	CALL	9.
			9.	Move the power switch to the "OFF" position. Connect all signal wires. Move the power switch to the "ON" position. Is the LCD blank?	10.	DONE
			ⓘ	*There was a problem that caused the solid state fuse to trip that is no longer present.*		
			10	There is a problem with the signal wiring or the host control system. Move the power switch to the "OFF" position. Disconnect the signal wiring, check for shorts in the signal wiring and/or other problems with the host control system. Is there a control wiring problem or other problem with the host control system?	11.	CALL
			11	Correct the problem with the signal wiring or the host control system. Connect all signal wires to the transmitter. Move the power switch to the "ON" position. Is the LCD blank?	CALL	DONE
		EB-Flow2 Transmitter Models EF-x2000	1.	Is a 24 VAC power source properly connected to power terminal block with terminal screws secure and no shorts between wires (refer to the appropriate *Wiring Guide* for terminal connections)?	3.	2.
			⚠	*"Live" wiring to transmitter may damage the transmitter and void warranty.*		
				These models have solid state fuses. Allow at least one minute between power cycles to allow the solid state fuses to reset in the event they were in a fault condition.		
			2.	Make sure the 24 VAC power source is not energized. Properly connect the 24 VAC power source to the power terminal block. Energize the 24 VAC power source. Is the LCD blank?	3.	DONE
			3.	Measure the voltage across the power terminal block screws with a VOM configured to measure VAC. Is the voltage between 22.8 to 26.4 VAC?	5.	4.
			4.	Make sure the 24 VAC power source is not energized. Correct input power problem so that the power supplied under load is between 22.8 and 26.4 VAC. Energize the 24 VAC power source. Is the LCD blank?	5.	DONE
			5.	Make sure the 24 VAC power source is not energized Remove all signal wires and probes from transmitter. Energize the 24 VAC power source. Is the LCD blank?	CALL	6.
			6.	Make sure the 24 VAC power source is not energized. Connect all sensor probes. Energize the 24 VAC power source. Is the LCD blank?	CALL	7.
			7.	Make sure the 24 VAC power source is not energized. Connect all signal wires. Energize the 24 VAC power source. Is the LCD blank?	8.	DONE
			ⓘ	*There was a problem that caused the solid state fuse to trip that is no longer present.*		
			8.	There is a problem with the signal wiring or the host control system. Make sure the 24 VAC power source is not energized. Disconnect the signal wiring, check for shorts in the signal wiring and/or other problems with the host control system. Is there a control wiring problem or problem with the host control system?	9.	CALL
			9.	Correct the problem with the signal wiring or the host control system. Connect all signal wires to the transmitter. Energize the 24 VAC power source. Is the LCD blank?	CALL	DONE
OUTPUT	The displayed reading (airflow, temperature or psychrometric value) is significantly fluctuating.	All Transmitter Models	1.	Is the signal of concern the airflow output signal?	2.	CALL
			2.	Is the measurment location in an outdoor air intake?	7.	3.
			3.	Does the measurement location meet or exceed EBTRON's published guidelines?	4.	5.
			4.	Increase the LCD integration parameter, LCD INTG. Is the displayed reading still fluctuating?	CALL	DONE
				Check for fan surge of other fan system instability problems.		
			5.	Can you relocate the AMD to a better location?	CALL	6.

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models		Procedure	Yes	No
OUTPUT			6.	Increase the LCD integration parameter, LCD INTG. Is the displayed reading still fluctuating?	CALL	DONE
			7.	Is the AMD downstream of the outdoor air intake damper?	8.	9.
			8.	Relocate the AMD upstream of the outdoor air intake damper. Call EBTRON for assistance.		
			9.	Is the air velocity less than 150 FPM [0.75 m/s]?	10.	CALL
			10	Transient wind gusts can create false readings on low velocity outdoor air intakes. Install a mechanical dampening filter such as a moisture eliminator, perforated plate, air cleaning filter or increase the airflow setpoint. Increasing the LCD integration, LCD INTG, may improve performance. Call EBTRON for assistance.		
			ⓘ	*Transient wind gusts will also affect analog and network outputs for airflow.*		
	The airflow input measured at the host control system is 0% when the airflow value displayed on the LCD is greater than 0 FPM (CFM) [m/s (L/s)].	A4 Transmitter Models GTx116 and GTx108 where x = C or M	1.	Are the airflow signal wires properly connected to output terminal block of the transmitter with terminal screws secure with no shorts between wires (refer to the appropriate *Wiring Guide* for terminal connections)?	3.	2.
			⚠	*"Live" wiring to transmitter may damage the transmitter and void warranty.*		
			2.	Move the power switch to the "OFF" position. Properly connect the airflow signal wires to the output terminal block. Move the power switch to the "ON" position. Is the airflow input signal at the host control system > 0%?	DONE	3.
			3.	Move the power switch to the "OFF" position. Check that the output card below the LCD is properly seated on the main circuit board. Is the output card properly seated?	6.	4.
			4.	Check pins on the main circuit board for damage. Are the pins damaged?	CALL	5.
			5.	Properly seat the output card on the pins. Move the power switch to the "ON" position. Is the airflow input signal at the host control system > 0%?	DONE	6.
			6.	Move the power switch to the "OFF" position. Remove the signal wire connected to the airflow output terminal. Verify the airflow output signal configuration of the transmitter (Section 2). Is the airflow output signal of the transmitter set to 4-20 mA?	7.	8.
			7.	Configure your VOM to measure mA. Place the leads of the VOM is across the airflow output and common terminals. Move the power switch to the "ON" position and wait for the transmitter to return to normal operation. Measure the output signal current. Is the current greater than 4 mA?	11.	9.
			8.	Configure your VOM to measure VDC. Place the leads of the VOM is across the airflow output and common terminals. Move the power switch to the "ON" position and wait for the transmitter to return to normal operation. Measure the output signal voltage. Is the voltage greater than 0 VDC?	11.	9.
			9.	Move the power switch to the "OFF" position. Remove and inspect the signal fuse, F1, on the output card. Is the fuse blown?	10.	CALL
			⚲	*Check the fuse with a VOM set to ohms to verify the fuse is good (shorted) or blown (open).*		
			10	Replace the glass output signal fuse with a fast-acting 0.125 A fuse. Move the power switch to the "ON" position. Measure the output signal current or voltage as directed in steps 7 or 8 depending on the output configuration of the transmitter. Is the measured output signal greater than 0 VDC (transmitters configured for VDC output) or 4 mA (transmitters configured for mA output)?	11.	CALL
			11	There is most likely a problem with the control wiring or host control system. Check for shorts, extraneous voltages (AC or DC) and/or other problems with the control wiring or the control system input circuit. Resolve the problem before reconnecting the signal wire to the transmitter.		
			⚠	*A problem with the control wiring or host control system may damage the transmitter and void warranty. Do not reconnect the signal wires to the transmitter unless the problem has been resolved.*		

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models	Procedure		Yes	No
OUTPUT		A4 Transmitter Models GTx116e, GTx108e, HTx116 and HTx108	1.	Are the airflow signal wires properly connected to output terminal block of the transmitter with terminal screws secure with no shorts between wires (refer to the appropriate *Wiring Guide* for terminal connections)?	3.	2.
			⚠	*"Live" wiring to transmitter may damage the transmitter and void warranty.*		
			💡	*These models have solid state fuses. Allow at least one minute between power cycles to allow the solid state fuses to reset in the event they were in a fault condition.*		
			2.	Move the power switch to the "OFF" position. Properly connect the airflow signal wires to the output terminal block. Move the power switch to the "ON" position. Is the airflow input signal at the host control system > 0%?	DONE	3.
			3.	Move the power switch to the "OFF" position. Remove the signal wire connected to the airflow output terminal. Verify the airflow output signal configuration of the transmitter (Section 2). Is the airflow output signal of the transmitter set to 4-20 mA?	4.	5.
			4.	Configure your VOM to measure mA. Place the leads of the VOM is across the airflow output and common terminals. Move the power switch to the "ON" position and wait for the transmitter to return to normal operation. Measure the output signal current. Is the current greater than 4 mA?	6.	CALL
			5.	Configure your VOM to measure VDC. Place the leads of the VOM is across the airflow output and common terminals. Move the power switch to the "ON" position and wait for the transmitter to return to normal operation. Measure the output signal voltage. Is the voltage greater than 0 VDC?	6.	CALL
			6.	There is most likely a problem with the control wiring or host control system. Check for shorts, extraneous voltages (AC or DC) and/or other problems with the control wiring or the control system input circuit. Resolve the problem before reconnecting the signal wire to the transmitter.		
			⚠	*A problem with the control wiring or host control system may damage the transmitter and void warranty. Do not reconnect the signal wires to the transmitter unless the problem has been resolved.*		
		A4 Transmitter Models HTA104	1.	Are the airflow signal wires properly connected to output terminal block of the transmitter with terminal screws secure with no shorts between wires (refer to the appropriate *Wiring Guide* for terminal connections)?	3.	2.
			⚠	*"Live" wiring to transmitter may damage the transmitter and void warranty.*		
			2.	Move the power switch to the "OFF" position. Properly connect the airflow signal wires to the output terminal block. Move the power switch to the "ON" position. Is the airflow input signal at the host control system > 0%?	DONE	3.
			3.	Move the power switch to the "OFF" position. Remove the signal wire connected to the airflow output terminal. Verify the airflow output signal configuration of the transmitter (Section 2). Is the airflow output signal of the transmitter set to 4-20 mA?	4.	5.
			4.	Configure your VOM to measure mA. Place the leads of the VOM is across the airflow output and common terminals. Move the power switch to the "ON" position and wait for the transmitter to return to normal operation. Measure the output signal current. Is the current greater than 4 mA?	8.	6.
			5.	Configure your VOM to measure VDC. Place the leads of the VOM is across the airflow output and common terminals. Move the power switch to the "ON" position and wait for the transmitter to return to normal operation. Measure the output signal voltage. Is the voltage greater than 0 VDC?	8.	6.
			6.	Move the power switch to the "OFF" position. Remove and inspect the signal fuse, F1, on the output card. Is the fuse blown?	7.	CALL
			💡	*Check the fuse with a VOM set to ohms to verify the fuse is good (shorted) or blown (open).*		

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models	Procedure		Yes	No
OUTPUT			7.	Replace the glass output signal fuse with a fast-acting 0.125 A fuse. Move the power switch to the "ON" position. Measure the output signal current or voltage as directed in steps 7 or 8 depending on the output configuration of the transmitter. Is the measured output signal greater than 0 VDC (transmitters configured for VDC output) or 4 mA (transmitters configured for mA output)?	8.	CALL
			8.	There is most likely a problem with the control wiring or host control system. Check for shorts, extraneous voltages (AC or DC) and/or other problems with the control wiring or the control system input circuit. Resolve the problem before reconnecting the signal wire to the transmitter.		
			⚠	*A problem with the control wiring or host control system may damage the transmitter and void warranty. Do not reconnect the signal wires to the transmitter unless the problem has been resolved.*		
		EB-Flow2 Transmitter Model EF-A2000	1.	Are the airflow signal wires properly connected to output terminal block of the transmitter with terminal screws secure with no shorts between wires (refer to the appropriate *Wiring Guide* for terminal connections)?	3.	2.
			⚠	*"Live" wiring to transmitter may damage the transmitter and void warranty.*		
			💡	*These models have solid state fuses. Allow at least one minute between power cycles to allow the solid state fuses to reset in the event they were in a fault condition.*		
			2.	Make sure the 24 VAC power source is not energized. Properly connect the airflow signal wires to the output terminal block. Energize the 24 VAC power source. Is the airflow input signal at the host control system > 0%?	DONE	3.
			3.	Make sure the 24 VAC power source is not energized. Remove the signal wire connected to the airflow output terminal. Verify the airflow output signal configuration of the transmitter (Section 2). Configure your VOM to measure VDC. Place the leads of the VOM is across the airflow output and common terminals. Energize the 24 VAC power source. and wait for the transmitter to return to normal operation. Measure the output signal voltage. Is the voltage greater than 0 VDC (2 VDC if the output is set to 2-10 V)?	4.	CALL
			4.	There is most likely a problem with the control wiring or host control system. Check for shorts, extraneous voltages (AC or DC) and/or other problems with the control wiring or the control system input circuit. Resolve the problem before reconnecting the signal wire to the transmitter.		
			⚠	*A problem with the control wiring or host control system may damage the transmitter and void warranty. Do not reconnect the signal wires to the transmitter unless the problem has been resolved.*		
	The host control system is configured for a 0-10 VDC (or 0-5 VDC) input and both of the output signals from the transmitter will not go below 2 VDC (or 1 VDC).	A4 Transmitter Model HTA104	1.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is the AOUT parameter set to 4-20mA?	2.	CALL
			2.	Use the ↑ and ↓ pushbuttons to select 0-10V (or 0-5V). Press the ENT pushbutton. Verify the output jumper positions on the main circuit board for each output when prompted. Press the ESC pushbutton to return to normal operation. Does the output go below 2 VDC (or 1 VDC when set to 0-5 V)?	DONE	CALL
		EB-Flow2 Transmitter Model EF-A2000	1.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is the AOUT parameter set to 2-10V?	2.	CALL
			2.	Use the ↑ and ↓ pushbuttons to select 0-10V (or 0-5V). Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the output go below 2 VDC (or 1 VDC when set to 0-5 V)?	DONE	CALL

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models		Procedure	Yes	No
OUTPUT	The host control system is configured for a 0-10 VDC (or 0-5 VDC) input and one or both of the output signals from the transmitter are greater than 10 VDC.	A4 Transmitter Models GTx116, GTx108 where x = C or M	1.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT1. Use the ↓ pushbutton to determine the setting for AOUT2. Are the parameters set to 4-20mA?	2.	CALL
			2.	Set SW1 (AO1) to VDC and SW2 (AO2) to VDC. Check that AOUT1 and AOUT2 have been modified using the ↑ and ↓ pushbuttons. If a 0-5 VDC signal is required, press the ENT pushbutton when 0-10V is shown and use the ↑ and ↓ pushbuttons to select 0-5V for each output and press the ENT pushbutton to accept each change. Press the ESC pushbutton to return to normal operation. Do the output signals of concern go below 10 VDC?	DONE	CALL
		A4 Transmitter Models GTx116e, GTx108e, HTx116 and HTx108	1.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is the AOUT parameter set to 4-20mA?	2.	CALL
			2.	Use the ↑ and ↓ pushbuttons to select 0-10V (or 0-5V). Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Do the output signals of concern go below 10 VDC?	DONE	CALL
		A4 Transmitter Models HTA104	1.	Check the output jumper positions for OUT1 (AO1) and OUT2 (AO2) on the main circuit board. Are the jumpers configured for mA?	2.	CALL
			2.	Configure the jumpers OUT1 and OUT2 for VDC. Do the output signals of concern go below 10 VDC?	DONE	CALL
	The displayed reading (airflow, temperature or psychrometric value) is relatively stable but the associated analog output signal is significantly fluctuating.	All Transmitter Models	1.	Use the TEST OUT TOOL (refer to Section 4) and set the analog output of concern to 50%. Is the output stable? Press the ESC pushbutton to return to normal operation.	2.	4.
			2.	Is the signal of concern the airflow output signal?	3.	CALL
			3.	Increase the integration parameter, INTG (Section 3). Is the output signal stable?	DONE	CALL
			4.	Move the power switch to the "OFF" position (models with a power switch) or make sure the 24 VAC power source is not energized. Remove all signal wires from the transmitter. Verify the output signal configuration of the output signal of concern (Section 2). Is the output signal of concern set to 4-20 mA?	5.	7.
				⚠ *"Live" wiring to transmitter may damage the transmitter and void warranty*		
			5.	Configure your VOM to measure mA. Place the leads of the VOM is across the airflow output and common terminals. Move the power switch to the "ON" position (models with a power switch) or energize 24 VAC power. Wait for the transmitter to return to normal operation. Measure the output signal current. Is the output signal stable?	6.	CALL
			6.	There is most likely a problem with the control wiring or host control system. Check for intermittent shorts, extraneous voltages (AC or DC) and/or other problems with the control wiring or the control system input circuit. Resolve the problem before reconnecting the signal wire to the transmitter.		
			7.	Configure your VOM to measure VDC. Place the leads of the VOM is across the airflow output and common terminals. Move the power switch to the "ON" position (models with a power switch) or energize 24 VAC power. Wait for the transmitter to return to normal operation. Measure the output signal voltage. Is the output signal stable?	8.	CALL
			8.	Has twisted, shielded wire been provided between the transmitter and host control system with the shield properly terminated at the host control system?	CALL	9.
			9.	Twisted, shielded wire should be provided on transmitters configured for 0-5V, 0-10V or 2-10V to avoid the pickup of extraneous electrical noise. Remedy the problem and properly terminate the shield. Refer to the appropriate Wiring Guide for the transmitter provided.		

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models		Procedure	Yes	No
OUTPUT	The airflow rate, in CFM [L/s], indicated at the host control system does not match the airflow rate shown on the transmitter display.	A4 Transmitter Models GTx116 and GTx108 where x = C or M	1.	Is the transmitter displaying airflow in CFM [L/s]?	3.	2.
			2.	Set the LCD UM parameter to CFM {L/s] (Section 3). Does the airflow indicated by the control system match the transmitter display?	DONE	3.
			3.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AREA (Section 3) parameter. Press the ESC pushbutton to return to normal operation. Is the AREA paramter equal the area where the AMD is installed?	5.	4.
			4.	Modify the AREA parameter so that it matches the area where the AMD is installed. Does the airflow indicated by the control system match the transmitter display?	DONE	5.
				Use the AREA WIZ Tool (or FAN WIZ tool for -F probe types) to determine and automatically save the AREA parameter (Section 4).		
			5.	Is the control system configured for a 4-20 mA input?	6.	8.
			6.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT1. Is AOUT1 set to 4-20mA?	11.	7.
			7.	Set SW1 on the output card to mA. Check that AOUT1 has been modified on the display. Press the ESC pushbutton to return to normal operation. Does the airflow indicated by the control system match the transmitter display?	DONE	11.
			8.	Is the control system configured for a 0-10 VDC (or 0-5 VDC) input?	9.	CALL
			9.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT1. Is AOUT1 set to 0-10V (or 0-5V)?	11.	10.
			10.	Set SW1 to VDC. Check that AOUT1 has been modified.. If a 0-5 VDC signal is required, press the ENT pushbutton when 0-10V is shown and use the ↑ and ↓ pushbuttons to select 0-5V for AO1 and press the ENT pushbutton to accept each change. Press the ESC pushbutton to return to normal operation. Does the airflow indicated by the control system match the transmitter display?	DONE	11.
			11.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO1 UM and AO1 FS (Section 3) parameters. Press the ESC pushbutton to return to normal operation. Is the AO1 UM parameter set to FPM [m/s]?	12.	13.
				This information can easily be retrieved using the EB-Link Reader application for Android® an iOS® phones or tablets (Section 4).		
			12.	The full scale or span of the host control system should equal the AREA paramteter x the AO1 FS parameter. The minimum scale of the host control system should equal 0. Is the host control system setup properly?	CALL	14.
			13.	The full scale or span of the host control system should equal the AO1 FS parameter. The minimum scale of the host control system should equal 0. Is the host control system setup properly?	CALL	14.
			14.	Correct the host control system signal conversion. Does the control system airflow rate match the displayed airflow rate?	DONE	CALL
		A4 Transmitter Models GTx116e, GTx108e, HTx116 and HTx108	1.	Is the transmitter displaying airflow in CFM [L/s]?	3.	2.
			2.	Set the LCD UM parameter to CFM {L/s] (Section 3). Does the airflow indicated by the control system match the transmitter display?	DONE	3.
			3.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AREA (Section 3) parameter. Press the ESC pushbutton to return to normal operation. Is the AREA paramter equal the area where the AMD is installed?	5.	4.
			4.	Modify the AREA parameter so that it matches the area where the AMD is installed. Does the airflow indicated by the control system match the transmitter display?	DONE	5.

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models		Procedure	Yes	No
OUTPUT				*Use the AREA WIZ Tool (or FAN WIZ tool for -F probe types) to determine and automatically save the AREA parameter (Section 4).*		
			5.	Is the control system configured for a 4-20 mA input?	6.	8.
			6.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 4-20mA?	11.	7.
			7.	Use the ↑ and ↓ pushbuttons to select 4-20mA. Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the airflow indicated by the control system match the transmitter display?	DONE	11.
			8.	Is the control system configured for a 0-10 VDC (or 0-5 VDC) input?	9.	CALL
			9.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 0-10V (or 0-5V)?	11.	10.
			10	Use the ↑ and ↓ pushbuttons to select 0-10V (or 0-5V). Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the airflow indicated by the control system match the transmitter display?	DONE	11.
			11	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO1 UM and AO1 FS (Section 3) parameters. Press the ESC pushbutton to return to normal operation. Is the AO1 UM parameter set to FPM [m/s]?	12.	13.
				This information can easily be retrieved using the EB-Link Reader application for Android® an iOS® phones or tablets (Section 4).		
			12	The full scale or span of the host control system should equal the AREA paramteter x the AO1 FS parameter. The minimum scale of the host control system should equal 0. Is the host control system setup properly?	CALL	14.
			13	The full scale or span of the host control system should equal the AO1 FS parameter. The minimum scale of the host control system should equal 0. Is the host control system setup properly?	CALL	14.
			14	Correct the host control system signal conversion. Does the control system airflow rate match the displayed airflow rate?	DONE	CALL
		A4 Transmitter Model HTA104	1.	Is the transmitter displaying airflow in CFM [L/s]?	3.	2.
			2.	Set the LCD UM parameter to CFM {L/s] (Section 3). Does the airflow indicated by the control system match the transmitter display?	DONE	3.
			3.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AREA (Section 3) parameter. Press the ESC pushbutton to return to normal operation. Is the AREA paramter equal the area where the AMD is installed?	5.	4.
			4.	Modify the AREA parameter so that it matches the area where the AMD is installed. Does the airflow indicated by the control system match the transmitter display?	DONE	
				Use the AREA WIZ Tool (or FAN WIZ tool for -F probe types) to determine and automatically save the AREA parameter (Section 4).		
			5.	Is the control system configured for a 4-20 mA input?	6.	10.
			6.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 4-20mA?	8.	7.
			7.	Use the ↑ and ↓ pushbuttons to select 4-20mA. Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the airflow indicated by the control system match the transmitter display?	DONE	8.
			8.	Check the output jumper position for OUT1 (AO1) on the main circuit board. Is the jumper configured for mA?	15.	9.

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models	Procedure		Yes	No
OUTPUT			9.	Configure the jumper OUT1 for mA. Does the airflow indicated by the control system match the transmitter display?	DONE	15.
			10.	Is the control system configured for a 0-10 VDC (or 0-5 VDC) input?	11.	CALL
			11.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 0-10V (or 0-5V)?	13.	12.
			12.	Use the ↑ and ↓ pushbuttons to select 0-10V (or 0-5V). Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the airflow indicated by the control system match the transmitter display?	DONE	13.
			13.	Check the output jumper position for OUT1 (AO1) on the main circuit board. Is the jumper configured for VDC?	15.	14.
			14.	Configure the jumper OUT1 for VDC. Does the airflow indicated by the control system match the transmitter display?	DONE	15.
			15.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO1 UM and AO1 FS (Section 3) parameters. Press the ESC pushbutton untill the transmitter returns to normal operation. Is the AO1 UM parameter set to FPM [m/s]?	16.	17.
			16.	The full scale or span of the host control system should equal the AREA paramteter x the AO1 FS parameter. The minimum scale of the host control system should equal 0. Is the host control system setup properly?	CALL	18.
			17.	The full scale or span of the host control system should equal the AO1 FS parameter. The minimum scale of the host control system should equal 0. Is the host control system setup properly?	CALL	18.
			18.	Correct the host control system signal conversion. Does the control system airflow rate match the displayed airflow rate?	DONE	CALL
		EB-Flow2 Transmitter Model EF-A2000	1.	Is the transmitter displaying airflow in CFM [L/s]?	3.	2.
			2.	Set the LCD UM parameter to CFM {L/s] (Section 3). Does the airflow indicated by the control system match the transmitter display?	DONE	3.
			3.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AREA (Section 3) parameter. Press the ESC pushbutton to return to normal operation. Is the AREA paramter equal the area where the AMD is installed?	5.	4.
			4.	Modify the AREA parameter so that it matches the area where the AMD is installed. Does the airflow indicated by the control system match the transmitter display?	DONE	5.
				Use the AREA WIZ Tool to determine and automatically save the AREA parameter (Section 4).		
			5.	Is the control system configured for a 4-20 mA input?	6.	8.
			6.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 2-10V (or 1-5V)?	11.	7.
				The 1-5 VDC and 2-10 VDC outputs can drive a 250 ohm and 500 ohm, respectively, 4-20 mA input circuit.		
			7.	Use the ↑ and ↓ pushbuttons to select 2-10V (arcross a 500 ohm load) or 1-5V (across a 250 ohm load). Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the airflow indicated by the control system match the transmitter display?	DONE	11.
			8.	Is the control system configured for a 0-10 VDC (or 0-5 VDC) input?	9.	CALL
			9.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 0-10V (or 0-5V)?	11.	10.

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models		Procedure	Yes	No
OUTPUT			10	Use the ↑ and ↓ pushbuttons to select 0-10V (or 0-5V). Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the airflow indicated by the control system match the transmitter display?	DONE	11.
			11	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO1 ASGN, AO1 UM and AO1 FS (Section 3) parameters. Press the ESC pushbutton untill the transmitter returns to normal operation. Is the AO1 UM parameter set to FPM {L/s]?	12.	13.
			⚠	*If the AO1 ASGN parameter is set to F1-2 or F2-1, the AO1 UM paramter must be set to CFM [L/s] unless the area of both locations are identical.*		
			ⓘ	*If the transmitter is configured for two measurement location, repeat steps 12 to 14 for the corresponding AO2 parameters.*		
			12	The full scale or span of the host control system should equal the AREA paramteter x the AO1 FS parameter. The minimum scale of the host control system should equal 0. Is the host control system setup properly?	CALL	14.
			ⓘ	*If the transmitter is configured for two measurement location, AREA1 and AREA2 should be used for Location 1 and Location 2, respectively.*		
			13	The full scale or span of the host control system should equal the AO1 FS parameter. The minimum scale of the host control system should equal 0. Is the host control system setup properly?	CALL	14.
			14	Correct the host control system signal conversion. Does the control system airflow rate match the displayed airflow rate?	DONE	CALL
	The temperature, in °F [°C], indicated at the host control system does not match the temperature shown on the transmitter display.	A4 Transmitter Models GTx116 and GTx108 where x = C or M	1.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO2 ASGN parameter (Section 3). Press the ESC pushbutton until the transmitter returns to normal operation. Is the AO2 ASGN parameter set to TEMP?	3.	2.
			💡	*This information can easily be retrieved using the EB-Link Reader application for Android® an iOS® phones or tablets (Section 4).*		
			2.	The transmitter has been field setup to use AO2 as a binary airflow alarm, fan airflow alarm (-F/An probe types only) or a system status alarm. Verify the desired function of AO2 before making any changes. If temperature output is desired on AO2, change the AO2 ASGN parameter to TEMP and use the values for the AO2 MS and AO2 FS parameters to determine the temperature at the host control system.		
			3.	Is the control system configured for a 4-20 mA input?	4.	6.
			4.	Simultaneously press the ESC and ↑ pushbuttons during normal operation. Use the ↑ and ↓ pushbuttons to determine the setting for AOUT2. Is AOUT2 set to 4-20mA?	9.	5.
			5.	Set SW2 on the output card to mA. Check that AOUT2 has been modified on the display. Press the ESC pushbutton to return to normal operation. Does the temperature indicated by the control system match the transmitter display?	DONE	9.
			6.	Is the control system configured for a 0-10 VDC (or 0-5 VDC) input?	7.	CALL
			7.	Simultaneously press the ESC and ↑ pushbuttons during normal operation. Use the ↑ and ↓ pushbuttons to determine the setting for AOUT2. Is AOUT2 set to 0-10 V (or 0-5V)?	9.	8.
			8.	Set SW2 to VDC. Check that AOUT2 has been modified.. If a 0-5 VDC signal is required, press the ENT pushbutton when 0-10V is shown and use the ↑ and ↓ pushbuttons to select 0-5V for AO2 and press the ENT pushbutton to accept each change. Press the ESC pushbutton to return to normal operation. Does the temperature indicated by the control system match the transmitter display?	DONE	9.

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models	Procedure		Yes	No
OUTPUT			9.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO2 MS and AO2 FS parameters (Section 3). Press the ESC pushbutton to return to normal operation. The full scale of the host control system should be equal to the AO2 FS parameter and the minimum scale should be the AO2 MS parameter. If the controller uses span and offset in lieu of full scale and minimum scale, the span is equal to AO2 FS minus AO2 MS and the offset is equal to AO2 MS. Is the host control system setup properly?	CALL	10.
				This information can easily be retrieved using the EB-Link Reader application for Android® an iOS® phones or tablets (Section 4).		
			10	Correct the host control system signal conversion. Does the temperature indicated by the control system match the transmitter display?	DONE	CALL
		A4 Transmitter Models GTx116e, GTx108e, HTx116 and HTx108	1.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO2 ASGN parameter (Section 3). Press the ESC pushbutton until the transmitter returns to normal operation. Is the AO2 ASGN parameter set to TEMP?	3.	2.
				This information can easily be retrieved using the EB-Link Reader application for Android® an iOS® phones or tablets (Section 4).		
			2.	The transmitter has been field setup to use AO2 as a binary airflow alarm or a system status alarm. Verify the desired function of AO2 before making any changes. If temperature output is desired on AO2, change the AO2 ASGN parameter to TEMP and use the values for the AO2 MS and AO2 FS parameters to determine the temperature at the host control system.		
			3.	Is the control system configured for a 4-20 mA input?	4.	6.
			4.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 4-20mA?	9.	5.
			5.	Use the ↑ and ↓ pushbuttons to select 4-20mA. Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the temperature indicated by the control system match the transmitter display?	DONE	9.
				AOUT affects the output of all analog output signals. If the airflow and/or psychrometric output signal(s) are/is connected to a device with a different input signal configuration, the airflow output signal will no longer be compatible with the host input device.		
			6.	Is the control system configured for a 0-10 VDC (or 0-5 VDC) input?	7.	CALL
			7.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 0-10V (or 0-5V)?	9.	8.
			8.	Use the ↑ and ↓ pushbuttons to select 0-10V (or 0-5V). Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the temperature indicated by the control system match the transmitter display?	DONE	9.
				AOUT affects the output of all analog output signals. If the airflow and/or psychrometric output signal(s) are/is connected to a device with a different input signal configuration, the airflow output signal will no longer be compatible with the host input device.		
			9.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO2 MS and AO2 FS parameters (Section 3). Press the ESC pushbutton to return to normal operation. The full scale of the host control system should be equal to the AO2 FS parameter and the minimum scale should be the AO2 MS parameter. If the controller uses span and offset in lieu of full scale and minimum scale, the span is equal to AO2 FS minus AO2 MS and the offset is equal to AO2 MS. Is the host control system setup properly?	CALL	10.

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models		Procedure	Yes	No
OUTPUT				*This information can easily be retrieved using the EB-Link Reader application for Android® an iOS® phones or tablets (Section 4).*		
			10	Correct the host control system signal conversion. Does the temperature indicated by the control system match the transmitter display?	DONE	CALL
		A4 Transmitter Model HTA104	1.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO2 ASGN parameter (Section 3). Press the ESC pushbutton until the transmitter returns to normal operation. Is the AO2 ASGN parameter set to TEMP?	3.	2.
			2.	The transmitter has been field setup to use AO2 as a binary airflow alarm or a system status alarm. Verify the desired function of AO2 before making any changes. If temperature output is desired on AO2, change the AO2 ASGN parameter to TEMP and use the values for the AO2 MS and AO2 FS parameters to determine the temperature at the host control system.		
			3.	Is the control system configured for a 4-20 mA input?	4.	8.
			4.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 4-20mA?	6.	5.
			5.	Use the ↑ and ↓ pushbuttons to select 4-20mA. Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the temperature indicated by the control system match the transmitter display?	DONE	6.
			⚠	*AOUT affects the output of all analog output signals. If the airflow output signal is connected to a device with a different input signal configuration, the airflow output signal will no longer be compatible with the host input device.*		
			6.	Check the output jumper position for OUT2 (AO2) on the main circuit board. Is the jumper configured for mA?	13.	7.
			7.	Configure the jumper OUT2 for mA. Does the temperature indicated by the control system match the transmitter display?	DONE	13.
			8.	Is the control system configured for a 0-10 VDC (or 0-5 VDC) input?	9.	CALL
			9.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 0-10V (or 0-5V)?	11.	10.
			10.	Use the ↑ and ↓ pushbuttons to select 0-10V (or 0-5V). Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the temperature indicated by the control system match the transmitter display?	DONE	11.
			⚠	*AOUT affects the output of all analog output signals. If the airflow output signal is connected to a device with a different input signal configuration, the airflow output signal will no longer be compatible with the host input device.*		
			11.	Check the output jumper position for OUT2 (AO2) on the main circuit board. Is the jumper configured for VDC?	13.	12.
			12.	Configure the jumper OUT1 for VDC. Does the temperature indicated by the control system match the transmitter display?	DONE	13.
			13.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO2 MS and AO2 FS parameters (Section 3). Press the ESC pushbutton to return to normal operation. The full scale of the host control system should be equal to the AO2 FS parameter and the minimum scale should be the AO2 MS parameter. If the controller uses span and offset in lieu of full scale and minimum scale, the span is equal to AO2 FS minus AO2 MS and the offset is equal to AO2 MS. Is the host control system setup properly?	CALL	14.

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models		Procedure	Yes	No
OUTPUT			14.	Correct the host control system signal conversion. Does the temperature indicated by the control system match the transmitter display?	DONE	CALL
		EB-Flow2 Transmitter Model EF-A2000	1.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO2 ASGN parameter (Section 3). Press the ESC pushbutton until the transmitter returns to normal operation. Is the AO2 ASGN parameter set to TEMP?	3.	2.
			2.	The transmitter has been field setup to use AO2 for a second location or as a binary airflow alarm or a system status alarm. Verify the desired function of AO2 before making any changes. If temperature output is desired on AO2, change the AO2 ASGN parameter to TEMP and use the values for the AO2 MS and AO2 FS parameters to determine the temperature at the host control system.		
			3.	Is the control system configured for a 4-20 mA input?	4.	6.
			4.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 2-10V (or 1-5V)?	9.	5.
			ⓘ	*The 1-5 VDC and 2-10 VDC outputs can drive a 250 ohm and 500 ohm, respectively, 4-20 mA input circuit .*		
			5.	Use the ↑ and ↓ pushbuttons to select 2-10V (arcross a 500 ohm load) or 1-5V (across a 250 ohm load). Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the airflow indicated by the control system match the transmitter display?	DONE	9.
			⚠	*AOUT affects the output of all analog output signals. If the airflow output signal is connected to a device with a different input signal configuration, the airflow output signal will no longer be compatible with the host input device.*		
			6.	Is the control system configured for a 0-10 VDC (or 0-5 VDC) input?	7.	CALL
			7.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 0-10V (or 0-5V)?	9.	8.
			8.	Use the ↑ and ↓ pushbuttons to select 0-10V (or 0-5V). Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the airflow indicated by the control system match the transmitter display?	DONE	9.
			⚠	*AOUT affects the output of all analog output signals. If the airflow output signal is connected to a device with a different input signal configuration, the airflow output signal will no longer be compatible with the host input device.*		
			9.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO2 MS and AO2 FS parameters (Section 3). Press the ESC pushbutton to return to normal operation. The full scale of the host control system should be equal to the AO2 FS parameter and the minimum scale should be the AO2 MS parameter. If the controller uses span and offset in lieu of full scale and minimum scale, the span is equal to AO2 FS minus AO2 MS and the offset is equal to AO2 MS. Is the host control system setup properly?	CALL	10.
			10.	Correct the host control system signal conversion. Does the temperature indicated by the control system match the transmitter display?	DONE	CALL

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models		Procedure	Yes	No
OUTPUT	The psychrometric value, RH in %RH, Enthalpy in Btu/lb [kj/kg] or dew point in °F [°C], indicated at the host control system does not match the psycrhrometric value shown on the transmitter display.	A4 Transmitter Model GTx116e	1.	Is the transmitter displaying the proper psychrometric value (i.e. RH, Enthalpy or Dew Point)?	3.	2.
			2.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and and set H CONFIG for the psychrometric value desired (Section 3). Does the psychrometric value indicated by the control system match the transmitter display?	DONE	3.
			3.	Is the control system configured for a 4-20 mA input?	4.	6.
			4.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 4-20mA?	9.	5.
			5.	Use the ↑ and ↓ pushbuttons to select 4-20mA. Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the psychrometric value indicated by the control system match the transmitter display?	DONE	9.
			⚠	*AOUT affects the output of all analog output signals. If the airflow and/or psychrometric output signal(s) are/is connected to a device with a different input signal configuration, the airflow output signal will no longer be compatible with the host input device.*		
			6.	Is the control system configured for a 0-10 VDC (or 0-5 VDC) input?	7.	CALL
			7.	Simultaneously press the ESC and ↑ pushbuttons during normal operation to determine the setting for AOUT. Is AOUT set to 0-10V (or 0-5V)?	9.	8.
			8.	Use the ↑ and ↓ pushbuttons to select 0-10V (or 0-5V). Press the ENT pushbutton. Press the ESC pushbutton to return to normal operation. Does the psychrometric value indicated by the control system match the transmitter display?	DONE	9.
			⚠	*AOUT affects the output of all analog output signals. If the airflow and/or psychrometric output signal(s) are/is connected to a device with a different input signal configuration, the airflow output signal will no longer be compatible with the host input device.*		
			9.	Press the ↑ and ↓ pushbuttons simlultaneously to enter the setup menu and retrieve the AO3 UM, AO3 MS and AO3 FS parameters (Section 3). Press the ESC pushbutton to return to normal operation. The full scale of the host control system should be equal to the AO3 FS parameter and the minimum scale should be the AO3 MS parameter. If the controller uses span and offset in lieu of full scale and minimum scale, the span is equal to AO3 FS minus AO3 MS and the offset is equal to AO3 MS. The units of measure are equal to the AO3 UM parameter. Is the host control system setup properly?	CALL	10.
			💡	*This information can easily be retrieved using the EB-Link Reader application for Android® an iOS® phones or tablets (Section 4).*		
			10.	Correct the host control system signal conversion. Does the psychrometric value indicated by the control system match the transmitter display?	DONE	CALL
NETWORK	The network is functioning properly but the transmitter is not found.	A4 Transmitter Models GTx116 and GTx108 where x = M, or GTx116e, GTx108e, HTx116 and HTx108 where x = M or S	1.	Is a standard RJ-45 terminated Ethernet cable, CAT5 or greater firmly seated in the output card Ethernet 10/100 connector? (refer to the appropriate Wiring Guide for the Ethernet 10/100 connector location.)	3.	2.
			2.	Connect a standard RJ-45 terminated Ethernet cable, CAT5 or greater to the output card ETHERNET 10/100 connector. Is the transmitter found on the network?	DONE	3.
			3.	Is the activity led on the output card flashing? (refer to the appropriate Wiring Guide for activity led location)	7.	4.
			4.	Move the power switch to the "OFF" position. Check that the output card below the LCD is properly seated on the main circuit board. Is the output card properly seated?	CALL	5.
			5.	Check pins on the main circuit board for damage. Are the pins damaged?	CALL	6.
			6.	Properly seat the output card on the pins. Move the power switch to the "ON" position. Is the activity led on the output card flashing?	7.	CALL

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models		Procedure	Yes	No
NETWORK			7.	Is the link led on the output card illuminated? (refer to the appropriate Wiring Guide for link led location)	8.	CALL
			8.	Verifiy that all Ethernet network settings are correct (Section 3). Are the Ethernet network settings correct?	CALL	9.
			9.	Set the Ethernet settings to proper values. Is the transmitter found on the network?	DONE	CALL
		A4 Transmitter Models GTx116, GTx116e, GTx108, GTx108e, HTx116, and HTx108 where x = C	1.	Is a 3-conductor network cable meeting the corresponding BACnet or Modbus standards connected to the output card terminal block with terminal screws secure with no shorts between wires? (refer to the appropriate Wiring Guide for terminal block location.)	3.	2.
			2.	Move the power switch to the "OFF" position. Properly connect a 3-conductor network cable meeting the corresponding BACnet or Modbus standards to the output card terminal block. Move the power switch to the "ON" position. Is the transmitter found on the network?	DONE	3.
			3.	Move the power switch to the "OFF" position. Check that the output card below the LCD is properly seated on the main circuit board. Is the output card properly seated?	6.	4.
			4.	Check pins on the main circuit board for damage. Are the pins damaged?	CALL	5.
			5.	Properly seat the output card on the pins. Move the power switch to the "ON" position. Is the transmitter found on the network?	DONE	6.
			6.	Is the transmitter the first or last device on the network run?	7.	8.
			7.	Verify the network termination is set to either Fail-Safe Bias for first device or End of Line for last device. (refer to the appropriate Wiring Guide for termination switch location) Is the transmitter found on the network?	DONE	9.
			8.	Verify the network termination is set to No Termination. (refer to the appropriate Wiring Guide for termination switch location) Is the transmitter found on the network?	DONE	9.
			9.	Verifiy that all RS-485 network settings are correct (Section 3). Are the RS-485 network settings correct?	CALL	10.
			10	Set the RS-485 network settings to proper values. Is the transmitter found on the network?	DONE	CALL
		A4 Transmitter Models HTN104	1.	Is a 3-conductor network cable meeting the corresponding BACnet or Modbus standards connected to the output card terminal block with terminal screws secure with no shorts between wires (refer to the appropriate Wiring Guide for terminal block location)?	3.	2.
			2.	Move the power switch to the "OFF" position. Properly connect a 3-conductor network cable meeting the corresponding BACnet or Modbus standards to the output card terminal block. Move the power switch to the "ON" position. Is the transmitter found on the network?	DONE	3.
			3.	Is the transmitter the first or last device on the network run?	4.	5.
			4.	Verify the network termination is set to either Fail-Safe Bias for first device or End-of-Line for last device. (refer to the appropriate Wiring Guide for termination switch location) Is the transmitter found on the network?	DONE	5.
			5.	Verify the network termination is set to No Termination. (refer to the appropriate Wiring Guide for termination switch location) Is the transmitter found on the network?	DONE	6.
			6.	Verifiy that all RS-485 network settings are correct (Section 3). Are the RS-485 network settings correct?	CALL	7.
			7.	Set the RS-485 network settings to proper values. Is the transmitter found on the network?	DONE	CALL

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models		Procedure	Yes	No
NETWORK		EB-Flow2 Transmitter Model EF-N2000	1.	Is a 3-conductor network cable meeting the corresponding BACnet or Modbus standards connected to the terminal block with terminal screws secure with no shorts between wires (refer to the appropriate Wiring Guide for terminal block location)?	3.	2.
			2.	Make sure the 24 VAC power source is not energized. Properly connect a 3-conductor network cable meeting the corresponding BACnet or Modbus standards to the output card terminal block. Energize the 24 VAC power source. Is the transmitter found on the network?	DONE	3.
			⚠	*"Live" wiring to transmitter may damage the transmitter and void warranty.*		
			3.	Is the transmitter the first or last device on the network run?	4.	5.
			4.	Verify the network termination jumper is set to End of Line. (refer to the appropriate Wiring Guide for termination jumper location) Is the transmitter found on the network?	DONE	5.
			5.	Verify the network termination jumper is set to No Termination. (refer to the appropriate Wiring Guide for termination jumper location) Is the transmitter found on the network?	DONE	6.
			6.	Verifiy that all RS-485 network settings are correct (Section 3). Are the RS-485 network settings correct?	CALL	7.
			7.	Set the RS-485 network settings to proper values. Is the transmitter found on the network?	DONE	CALL
		A4 Transmitter Models GTx116e, GTx108e, HTx116 and HTx108 where x = B	1.	Is the issue with the Ethernet network?	2.	11.
			2.	Is a standard RJ-45 terminated Ethernet cable, CAT5 or greater firmly seated in the output card Ethernet 10/100 connector (refer to the appropriate Wiring Guide for Ethernet 10/100 connector location)?	4.	3.
			3.	Connect a standard RJ-45 terminated Ethernet cable, CAT5 or greater to the output card ETHERNET 10/100 connector. Is the transmitter found on the network?	DONE	4.
			4.	Is the activity led on the output card flashing (refer to the appropriate Wiring Guide for activity led location)?	8.	5.
			5.	Move the power switch to the "OFF" position. Check that the output card below the LCD is properly seated on the main circuit board. Is the output card properly seated?	CALL	6.
			6.	Check pins on the main circuit board for damage. Are the pins damaged?	CALL	7.
			7.	Properly seat the output card on the pins. Move the power switch to the "ON" position. Is the activity led on the output card flashing?	8.	CALL
			8.	Is the link LED on the output card illuminated (refer to the appropriate Wiring Guide for link LED location)?	9.	CALL
			9.	Verifiy that all Ethernet network settings are correct (Section 3). Are the Ethernet network settings correct?	CALL	10.
			10	Set the Ethernet settings to proper values. Is the transmitter found on the network?	DONE	CALL
			11	Is a 3-conductor network cable meeting the corresponding BACnet or Modbus standards connected to the output card terminal block with terminal screws secure with no shorts between wires (refer to the appropriate Wiring Guide for terminal block location)?	13.	12.
			12	Move the power switch to the "OFF" position. Properly connect a 3-conductor network cable meeting the corresponding BACnet or Modbus standards to the output card terminal block. Move the power switch to the "ON" position. Is the transmitter found on the network?	DONE	13.
			13	Move the power switch to the "OFF" position. Check that the output card below the LCD is properly seated on the main circuit board. Is the output card properly seated?	16.	14.
			14	Check pins on the main circuit board for damage. Are the pins damaged?	CALL	15.
			15	Properly seat the output card on the pins. Move the power switch to the "ON" position. Is the transmitter found on the network?	DONE	16.
			16	Is the transmitter the first or last device on the network run?	17.	18.
			17	Verify the network termination is set to either Fail-Safe Bias for first device or End of Line for last device. (refer to the appropriate Wiring Guide for termination switch location) Is the transmitter found on the network?	DONE	19.

TABLE 6-2 Non-FDS (Non-Fault Detection System) Troubleshooting

	Symptom	Transmitter Models	Procedure		Yes	No
NETWORK			18.	Verify the network termination is set to No Termination. (refer to the appropriate Wiring Guide for termination switch location) Is the transmitter found on the network?	DONE	19.
			19.	Verifiy that all RS-485 network settings are correct (Section 3). Are the RS-485 network settings correct?	CALL	20.
			20.	Set the RS-485 network settings to proper values. Is the transmitter found on the network?	DONE	CALL
		A4 Transmitter Models GTx116, GTx116e, GTx108 and GTx108e where x = L or x = F	1.	Is a cable specified by Echelon (typically Belden 8471 cable or equivalent) connected to the output terminal block with terminal screws secure with no shorts between wires (refer to the appropriate Wiring Guide for terminal block location)?	3.	2.
			2.	Connect a a cable specified by Echelon (typically Belden 8471 cable or equivalent) to the output terminal block. Is the transmitter found on the network?	DONE	3.
			3.	Is the activity led on the output card flashing (refer to the appropriate Wiring Guide for activity led location)?	7.	4.
			4.	Move the power switch to the "OFF" position. Check that the output card below the LCD is properly seated on the main circuit board. Is the output card properly seated?	CALL	5.
			5.	Check pins on the main circuit board for damage. Are the pins damaged?	CALL	6.
			6.	Properly seat the output card on the pins. Move the power switch to the "ON" position. Is the activity led on the output card flashing?	7.	CALL
			7.	Is the link LED on the output card illuminated (refer to the appropriate Wiring Guide for link LED location)?	8.	CALL
			8.	Has the transmitter been commissioned on the LON network?	CALL	9.
			9.	Commission the transmitter on the LON network. Is the transmitter found on the network?	DONE	CALL

TABLE 6-3 FDS TROUBLESHOOTING

Code(s)	Error Message	Error Description	Effect	Probable Causes	Remedy	Error Priority
1	NO PROBES	No probes are detected by transmitter.	Device will not operate.	No sensor probes are connected to the transmitter.	Connect sensor probes to transmitter.	No other flow or temperature sensor errors will occur while this error exists.
				The serial memory chip of all sensor probe plugs are not communicating with the transmitter.	Inspect sensor probe plugs for damage. Repair/replace damaged plugs. Contact EBTRON Customer Service.	
				Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	
2	WRONG SENS	Mismatched sensor probe types on a single transmiter.	Device will not operate.	Sensor probes of different types are plugged into the transmitter.	Remove the improper sensor probe and reset power to the transmitter or perform RESET SENSORS.	No other flow or temperature sensor errors will occur while this error exists.
3 to 10	Px MISSING, where x=1 to 8	Registered probe missing.	Device will operate, ignore missing sensor nodes and average the remaining sensor nodes.	Sensor probe removed after initial power-up.	Inspect and reinstall removed sensor probe.	Replaces Px MEM ERROR and SENSOR x errors once transmitter detects all sensor node failures.
				Sensor probe plug or cable failed or damaged after initial power-up; transmitter cannot read serial memory chip or measure sensor nodes.	Inspect cable and sensor probe plug for damage. Repair/replace damaged plug/probe. Contact EBTRON Customer Service.	
				Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	
11 to 18	Px MEM ERROR, where x=1 to 8	Serial memory error in probe.	Device will operate normally unless transmitter or probes are reset or a new transmitter is installed.	Serial memory chip in sensor probe plug damaged after initial power-up.	Inspect sensor probe plug for damage. Repair/replace damaged plug. Contact EBTRON Customer Service.	Low priority of sensor related errors. Will not exist with Px MISSING or NO PROBES.
				Communication of serial memory chip in sensor probe plug failed after initial power-up.	Verify ambient temperature of the transmitter is within operationing range of -20 °F to 120 °F. If temperature is outside of operating range adjust ambient temperature or relocate transmitter. If temperature is within operating range contact EBTRON Customer Service.	
				Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	

TABLE 6-3 FDS TROUBLESHOOTING

Code(s)	Error Message	Error Description	Effect	Probable Causes	Remedy	Error Priority
19 to 34	SENSOR x, where x=1 to 16	Sensor node malfunction.	Device will operate, ignore missing sensor nodes and average the remaining sensor nodes.	Sensor node thermistor damaged.	Inspect sensor node for damage. Replace damaged probe. Contact EBTRON Customer Service.	Low priority sensor error. Will not exist with Px MISSING or NO PROBES.
				Connection to sensor node damaged.	Inspect sensor probe plug for damage. Repair/replace damaged plug. Contact EBTRON Customer Service.	
				Sensor node air temperature < -30 °F.	Increase sensor node air temperature.	
				Sensor node air velocity > 15,000 FPM.	Measurement not intended for high flow applications. Reduce sensor node air velocity.	
				Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	
35	LOW TEMP	Airstream temperature below product's specified minimum (-20°F [-28.9°C]).	Device will continue to operate. Error is informational.	Sensor node air temperature < -20 °F.	Increase sensor node air temperature.	Lowest priority of sensor error. Can happen before SENSOR x error if temperature continues to fall.
36	HIGH TEMP	Airstream temperature above product's specified maximum (160°F [71.1°C]).	Device will continue to operate. Error is informational.	Sensor node air tempeture > 160 °F.	Reduce sensor node air temperature.	Lowest priority of sensor error. Can happen before SENSOR x error if temperature continues to rise.
40 to 43	RHx MISSING	Humidity sensor error.	Device will continue to use other connected RH sensors. If all RH sensors are missing devices will operate without psychrometric values.	Sensor probe removed after initial power-up.	Inspect and reinstall removed sensor probe.	NO PROBES is the only error that can prevent RHx MISSING.
				RH sensor damaged.	Inspect sensor for damage. Replace damaged sensor probe. Contact EBTRON Customer Service.	
				Connection to RH sensor damaged.	Inspect cable and sensor probe plug for damage. Repair/replace damaged plug/probe. Contact EBTRON Customer Service.	
				Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	
50	Pb SENS ERROR	Barometric pressure sensor error.	Device will use last valid value or a nominal value if no valid value exists.	Pb sensor damaged.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	Pb sensor error is independent of other errors.
				Connection to Pb sensor damaged; transmitter hardware failture.		

TABLE 6-3 FDS TROUBLESHOOTING

Code(s)	Error Message	Error Description	Effect	Probable Causes	Remedy	Error Priority
60	NUM PROBES	Missing probe at Location 2 (dual location EF units).	Device will continue to operate for the location with connected probes.	No probes are connected to the transmitter for a location.	Verify connected probes are installed in appropriate connector location. Move or install probes to correct location.	Applies to dual location transmitters only. Occurs when no probes are detected for a location after initial power-up.
				The serial memory chip of all sensor probe plugs at a location have failed.	Inspect sensor probe plugs for damage. Repair/replace dampaged plugs. Contact EBTRON Customer Service.	
				EF-x2000-U LOCATIONS setting may be inadvertently set to 2.	Verify LOCATIONS setting is correct in GLOBAL menu.	
				Dual-location transmitter installed in single location application.	Verify transmitter is correct for application.	
				Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	
61	FAN COUNT	Fan count entry from FAN WIZ does not equal actual sensor probe count.	Device will continue to operate for the detected fans.	Incorrect FAN WIZ configuration.	Re-evaluate FAN WIZ settings. FANS x SENS/FAN.	FAN COUNT error is independent of other errors.
				Probes not connected to transmitter.	Connect missing probes.	
				Serial memory chip of sensor probe plug has failed.	Inspect connected sensor probe plugs for damage. Repair/replace damaged plugs. Contact EBTRON Customer Service.	
				Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	
				Additional probes connected to transmitter.	Remove additional probes from transmitter.	
70	SYSTEM ERROR	Transmitter memory error.	Device will not operate.	Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	SYSTEM ERROR is independent of other errors.
71	OUT CARD	GTx116e, GTx108e, HTx116 or HTx108 output card communication error.	Device will operate, outputs for connected card will be invalid.	Output card connection removed from transmitter.	Inspect output card and ensure that it is securely connected.	OUT CARD error is independent of other errors.
				Output card communication to transmitter failed.	Inspect output card for damage. Repair/replace damaged card. Contact EBTRON Customer Service.	
				Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	

TABLE 6-3 FDS TROUBLESHOOTING

Code(s)	Error Message	Error Description	Effect	Probable Causes	Remedy	Error Priority
72	CON CARD	-F/An connector card communication error.	Device may operate, ignore missing sensor nodes and average the remaining sensor nodes.	Connector card damaged or removed after initial power-up.	Inspect connector card for proper connection or damage. Adjust/repair/replace connector. Contact EBTRON Customer Service.	CON CARD error is independent of other errors.
				Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	
73	BLE COM ERROR	BLE radio error.	Device will continue to operate. Data from BLE connection will be invalid.	Communication to BLE module damaged.	Repair/replace damaged transmitter components.	BLE COM ERROR is independent of other errors.
				BLE module failure.		
				Transmitter hardware failure.		
74	AOUT ERR	GTx116e, GTx108e, HTx116 or HTx108 analog output error.	Device will continue to operate. Analog output values may be unreliable.	Communication to analog output circuit failed.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	AOUT ERR is independent of other errors.
				Analog output connection damaged.		
				Transmitter hardware failure.		
75	CON BUS	"e" transmitter probe serial memory communication error.	Device will continue to operate. Additional connected probes will not be discovered. Psychrometric values will not be available.	Sensor probe plug or cable damaged.	Inspect connected cables and sensor probe plugs for damage. Repair/replace damaged plug/probe. Contact EBTRON Customer Service.	CON BUS error is independent of other errors.
				Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	
80	EBBUS CON ERR	GTx116e, GTx108e, HTx116 or HTx108 EB-Bus Ethernet network communication error.	Device will continue to operate. No communication with *EB-Bus* host.	Ethernet network wiring issue, damaged cable or disconnected cable	Inspect network cables for damage.Repair/replace damaged cables Contact EBTRON Customer Service.	EBBUS CON ERR is independent of other errors.
				Transmitter hardware failure.	Repair/replace damaged transmitter components. Contact EBTRON Customer Service.	
				EB-Bus host hardware failure.	Repair/replace damaged components. Contact EBTRON Customer Service.	
85	FLOW CALC ERR	GTx116e, GTx108e, HTx116 or HTx108 flow calculation error (OEM transmitters only).	Device will not operate.	No area set for OEM transmitter model	Enter correct area for installed location. Contact EBTRON Customer Service.	FLOW CALC ERR is independent of other errors.

6.4. DISCREPENCIES

In many installations, third-party airflow verification is required. If there is a discrepancy between third-party measurements and the EBTRON airflow measuring device, the following procedures are recommended.

6.4.1 Assess the Installation

- ☐ Verify that the airflow directional arrow is pointing in the direction of airflow.
- ☐ Verify that the sensor node openings are in the direction of airflow and not twisted in the airstream.
- ☐ Verify that the area entered in the transmitter matches the actual area of the duct or opening where the AMD is located.
- ☐ Verify that no active trouble codes are visible on the display.
- ☐ Verify that field adjustment has not been enabled in the transmitter.
- ☐ Verify readings using the LCD. Do not verify at the host B.A.S. system at this time.



Use the EB-Link Reader phone application to verify the area, trouble code status and field adjustment status on Gold Series and Hybrid Series models HTx116 and HTx108 transmitters.

Correct installation issues, enter the proper area, address any active trouble codes and disable field adjustment before proceeding.

- ☐ Determine if probes meet or exceed EBTRON's minimum placement guidelines.



The airflow rate indicated on the transmitter LCD (and that viewed on the EB-Link Reader) should be within the published installed accuracy of the product provided when probes meet or exceed EBTRON's minimum placement guidelines.



If the probes do not meet the minimum placement guidelines, the installed accuracy may exceed the published installed accuracy of the product provided and field adjustment may be required.

6.4.2. Assess the Verification Technique

6.4.2.1. DUCT SYSTEMS AND FAN MEASUREMENT

Ducted field measurements with handheld instruments can yield accuracies of 5% to 10% when an adequate straight run of duct is available. Applications having multiple airflow measuring devices in the same air path can compare one airflow measurement device to the other as part of the verification method.



Set the system for 100% recirculation and compare the supply airflow measurement device to the return airflow measurement device. Verify the measurement in the best location for the field measurement device.



Set the system for 100% outdoor air and compare the supply airflow measurement device to the outdoor airflow measurement device. Verify the measurement in the best location for the field measurement device.

6.4.2.2. OUTDOOR AIR INTAKES

Direct field measurement of close-coupled outdoor air intakes can approach and exceed 25%. Indirect measurement techniques to determine the outdoor airflow rate using the difference between the supply and return airflow rates or the ratios of the return air, outdoor air and mixed air (or supply air) temperatures should be avoided since uncertainties almost always exceed 25%.

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6.4.2.3. ADJUST OR VERIFY?

If the airflow measuring device is within the measurement uncertainty of the verification technique, EBTRON strongly recommends that no field adjustment correction is made. EBTRON airflow measurement devices are factory calibrated to NIST traceable standards. Field adjustment is not required when installed in accordance to published guidelines.

If field adjustment is required, a gain and/or offset adjustment can be made to the output of the device by manual entry (Section 3) or by using the Flow Adjust Wizard (Section 4).

If a one-point adjustment is made, make a gain adjustment rather than an offset adjustment to match the device to the third-party reading.

Do not make an area adjustment in lieu of a gain adjustment since area is used to verify proper setup of the airflow measurement device.

6.5. RESETTING TRANSMITTER

The SERVICE menu contains the RESET submenu category.

Simultaneously press and hold the ESC and ENT pushbuttons for three seconds during normal operation to enter the service menu. The transmitter continues to operate normally in the background.

The PRESETS submenu will be displayed. Press ↓ once and the RESET submenu will be displayed. Press the ENT pushbutton to select the top of the RESET submenu category.

6.5.1. Reset All

The Reset All function restores all factory default settings.

The Reset All function will restore any preconfigured factory settings entered prior to shipment to factory default settings.

[Menu path: PRESETS ↓ RESETS > ALL > NO ↓ YES]

Applies to: All Advantage IV and EB Flow2 transmitter models with LCD and pushbutton interface.

6.5.2. Reset Sensors

The Reset Sensors function clears only sensor probe data.

[Menu path: PRESETS ↓ RESETS > ALL ↓ SENS > NO ↓ YES]

Applies to: All Advantage IV and EB Flow2 transmitter models with LCD and pushbutton interface.

6.6. CONTACTING CUSTOMER SERVICE

For toll-free factory support call 800-2EBTRON (232-8766), Monday through Thursday 8:00 AM to 4:30 PM and Friday 8:00 AM to 2:00 PM eastern time or contact your local representative.

Getting and keeping you up and running is our number one priority. We have several factory options to get you back on track quickly.

Save time and use the EB-Link Reader application for Android and iOS phones and tablets to retrieve diagnostic data prior to contacting EBTRON Customer Service. Tell the service technician you have downloaded and saved the diagnostic file for information.

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6.6.1. Advance Replacement Service

EBTRON will send out new replacement product(s) to the customer without first examining the defective/damaged product(s) at the EBTRON factory. The Advance Replacement Policy is described below.

Advance Replacement Policy

1. The customer must perform basic field diagnostics as requested by EBTRON Customer Service to determine which product component(s) is/are defective/damaged. Once approved, a return merchandise authorization number (RMA#) will be assigned so that the defective/damaged product(s) can be returned to EBTRON by the customer at the customer's expense.
2. The customer must provide a contingency purchase order for the cost of the replacement product(s) in the event that EBTRON determines the defective/damaged product(s) is/are not covered under warranty.
3. EBTRON will expedite manufacturing and ship the advance replacement product(s) via UPS Ground or similar method, shipping charges deferred.
4. Any expedited shipping method requested by the customer will be charged to the customer.
5. The defective/damaged product(s) must be received by EBTRON within 10 working days after customer receipt of the replacement product(s) or the service will be considered non-warranty.
6. The equipment will be inspected based upon the condition at arrival at the EBTRON factory to determine cause of damage (warranty or non-warranty). The customer should properly package and insure the equipment in case of shipping loss or damage. Product(s) damaged in shipment will void any warranty claim.
7. If the damage is deemed to be warranty, the contingency purchase order will be destroyed and the return shipping charge will be paid by EBTRON. If the damage is deemed to be non-warranty, the customer will be invoiced for the complete cost of all advance replacement product(s) they received and the associated shipping charges regardless of repair required if the product was returned for Standard Factory Repair Service.

6.6.2. Standard Factory Repair Service

The customer returns the defective damaged products to the EBTRON factory for inspection and/or repair. The Standard Factory Repair Service Policy is described below.

Standard Factory Service Policy

1. The customer must perform basic field diagnostics as requested by EBTRON Customer Service to determine which product component(s) is/are defective/damaged. Once approved, a return merchandise authorization number (RMA#) will be assigned so that the defective/damaged product(s) can be returned to EBTRON by the customer at the customer's expense.
2. The equipment will be inspected based upon the condition at arrival at the EBTRON factory to determine cause of damage (warranty or non-warranty). The customer should properly package and insure the equipment in case of shipping loss or damage. Product(s) damaged in shipment will void any warranty claim.
3. If non-warranty repairs are required after inspection, the customer will be contacted with a cost estimate for the repair. The customer must provide a purchase order for the non-warranty repairs to initiate the repair process.
4. Most repairs, warranty and non-warranty will be performed in one week of receipt of the product (warranty) and one week of receipt of the repair purchase order (non-warranty). Exceptions include large or special service requests that will be handled on a case-by-case basis.
5. If the damage is deemed to be warranty, the product(s) will be shipped in the same manner it was received by the customer and the return shipping charge will be paid by EBTRON. If the damage is deemed to be non-warranty, the customer will be invoiced for the repair and the associated shipping charges based on the customer's shipping instructions.

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6.6.3. Other Service Options

If the EBTRON Advance Replacement or Standard Factory Service options do not meet your needs, field service and/or replacement product(s) can be obtained/purchased directly from your local EBTRON representative.

For the name and phone number of your local EBTRON representative visit us online at https://ebtron.com/. Click on the REP FINDER button on the top right side of the page.

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